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Lack of CCDC146, a ubiquitous centriole and microtubule-associated protein, leads to non-syndromic male infertility in human and mouse

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Abstract

Genetic mutations are a recurrent cause of male infertility. Multiple morphological abnormalities of the flagellum (MMAF) syndrome is a heterogeneous genetic disease, with which more than 50 genes have been linked. Nevertheless, for 50% of patients with this condition, no genetic cause is identified. From a study of a cohort of 167 MMAF patients, pathogenic bi-allelic mutations were identified in the *CCDC146* gene in two patients. This gene encodes a poorly characterized centrosomal protein which we studied in detail here. First, protein localization was studied in two cell lines. We confirmed the centrosomal localization in somatic cells and showed that the protein also presents multiple microtubule-related localizations during mitotic division, suggesting that it is a microtubule-associated protein (MAP). To better understand the function of the protein at the sperm level, and the molecular pathogenesis of infertility associated with *CCDC146* mutations, two genetically modified mouse models were created: a *Ccdc146* knock-out (KO) and a knock-in (KI) expressing a HA-tagged CCDC146 protein. KO male mice were completely infertile, and sperm exhibited a phenotype identical to our two MMAF patient's phenotype with *CCDC146* mutations. No other pathology was observed, and the animals were viable. CCDC146 expression starts during late spermiogenesis, at the time of flagellum biogenesis. In the spermatozoon, the protein is conserved but is not localized to centrioles, unlike in somatic cells, rather it is present in the axoneme at the level of microtubule doublets. Expansion microscopy associated with the use of the detergent sarkosyl to solubilize microtubule

doublets, suggest that the protein may be a microtubule inner protein (MIP). At the subcellular level, the absence of CCDC146 affected the formation, localization and morphology of all microtubule-based organelles such as the manchette, the head–tail coupling apparatus (HTCA), and the axoneme. Through this study, we have characterized a new genetic cause of infertility, identified a new factor in the formation and/or structure of the sperm axoneme, and demonstrated that the CCDC146 protein plays several cellular roles, depending on the cell type and the stages in the cell cycle.

eLife assessment

This study presents **valuable** information that demonstrates CCDC146 as a novel cause of male infertility that play key role in microtubule-associated structures. The evidence supporting the claims of the authors is **solid** using combination of human and mouse genetics, biochemical and imaging approaches. This paper would be of interest to cell and developmental biologists working on genes involved in spermatogenesis and male infertility.

Introduction

Infertility is a major health concern, affecting approximately 50 million couples worldwide [1], or 12.5% of women and 10% of men. It is defined by the World Health Organization (WHO) as the “failure to achieve a pregnancy after 12 months or more of regular unprotected sexual intercourse”. In almost all countries, infertile couples have access to assisted reproductive technology (ART) to try to conceive a baby, and there are now 5 million people born as a result of ART. Despite this success, almost half of the couples seeking medical support for infertility fails to successfully conceive and bear a child by the end of their medical care. The main reason for these failures is that one member of the couple produces gametes that are unable to support fertilization and/or embryonic development. Indeed, ART does not specifically treat or even try to elucidate the underlying causes of a couple’s infertility, rather it tries to bypass the observed defects. Consequently, when defects in the gametes cannot be circumvented by the techniques currently proposed, ART fails. To really treat infertility, a first step would be to gain a better understanding of the problems with gametogenesis for each patient. This type of approach should increase the likelihood of adopting the best strategy for affected patients, and if necessary, should guide the development of innovative therapies.

Male infertility has several causes, such as infectious diseases, anatomical defects, or a genetic disorder including chromosomal or single gene deficiencies. Genetic defects play a major role in male infertility, with over 4000 genes thought to be involved in sperm production, of which more than 2000 are testis-enriched and almost exclusively involved in spermatogenesis [2]. Mutations in any of these genes can negatively affect spermatogenesis and produce one of many described sperm disorders. The characterization and identification of the molecular bases of male infertility is thus a real challenge. Nevertheless, thanks to the emergence of massively parallel sequencing technologies, such as whole exome sequencing (WES) and whole genome sequencing (WGS), the identification of genetic defects has been greatly facilitated in recent years. As a consequence, remarkable progress has been made in the characterization of numerous human genetic diseases, including male infertility.

Today, more than 120 genes are associated with all types of male infertility [3], including quantitative and qualitative sperm defects. Qualitative spermatogenesis defects impacting sperm morphology, also known as “teratozoospermia”[4, 5], are a heterogeneous group of abnormalities covering a wide range of sperm phenotypes. Among these phenotypes, some relate to the morphology of the flagellum. These defects are usually not uniform, and patients' sperm show a wide range of flagellar morphologies such as short and/or coiled and/or irregularly sized flagella. Due to this heterogeneity, this phenotype is now referred to as multiple morphological abnormalities of the sperm flagellum (MMAF) [5]. Sperm from these patients are generally immotile, and patients are sterile.

Given the number of proteins present in the flagellum and necessary for its formation and functioning, many genes have already been linked to the MMAF phenotype. Study of the MMAF phenotype in humans has allowed the identification of around 50 genes [6] coding for proteins involved in axonemal organization, present in the structures surrounding the axoneme – such as the outer dense fibers and the fibrous sheath – and involved in intra-flagellar transport (IFT). Moreover, some genes have been identified from mouse models, and their human orthologs are very good gene candidates for MMAF, even if no patient has yet been identified with mutations in these genes. Finally, based on the remarkable structural similarity of the axonemal structure of motile cilia and flagella, some MMAF genes were initially identified in the context of primary ciliary dyskinesia (PCD). However, this structural similarity does not necessarily imply a molecular similarity, and only around half (10 of the 22 PCD-related genes identified so far [7]) are effectively associated with male infertility. However, in most cases, the number of patients is very low and the details of the sperm tail phenotype are unknown [7].

We have recruited 167 patients with MMAF. Following whole exome sequencing, biallelic deleterious variants in 22 genes were identified in 83 subjects. The genes identified are *AK7*[8], *ARMC2* [9], *CFAP206* [10], *CCDC34* [11], *CFAP251* [12], *CFAP43* and *CFAP44* [13], *CFAP47* [14], *CFAP61* [15], *CFAP65* [16], *CFAP69* [17], *CFAP70* [18], *CFAP91* [19], *CFAP206* [10], *DNAH1* [20], *DNAH8* [21], *FSIP2* [22], *IFT74* [23], *QRICH2* [24], *SPEF2* [25], *TTC21A* [26] and *TTC29* [27]. Despite this success, a molecular diagnosis is obtained in half of the patients (49.7%) with this sperm phenotype, suggesting that novel candidate genes remain to be identified. We have pursued our effort with this cohort to identify further mutations that could explain the patient MMAF phenotype. As such, we have identified bi-allelic truncating mutations in *CCDC146* in two unrelated infertile patients displaying MMAF. *CCDC146* is known to code for a centrosomal protein when heterologously expressed in HeLa cells [28, 29], but minimal information is available on its distribution within the cell, or its function when naturally present. Moreover, this gene has never been associated with any human disease.

The centrosome, located adjacent to the nucleus, is a microtubule-based structure composed of a pair of orthogonally-oriented centrioles surrounded by the pericentriolar material (PCM). The centrosome is the major microtubule-organizing center (MTOC) in animal cells, and as such regulates the microtubule organization within the cell. Therefore, it controls intracellular organization and intracellular transport, and consequently regulates cell shape, cell polarity, and cell migration. The centrosome is also crucial for cell division as it controls the assembly of the mitotic/meiotic spindle, ensuring correct segregation of sister chromatids in each of the daughter cells [30]. The importance of this organelle is highlighted by the fact that 3% (579 proteins) of all known human proteins have been experimentally detected in the centrosome (*https://www.proteinatlas.org/humanproteome/subcellular/centrosome*). Centrioles also play essential roles in spermatogenesis and particularly during spermiogenesis. In round spermatids, the centriole pair docks to the cell membrane, whereas the distal centriole serves as the basal body initiating assembly of the axoneme. The proximal centriole then tightly attaches to the sperm nucleus and gradually develops the head-to-tail coupling apparatus (HTCA), linking the sperm head to the flagellum [31]. In human and bovine sperm, the proximal centriole is retained and the distal centriole is remodeled to produce an 'atypical' centriole [32]; in contrast, in rodents, both centrioles are degenerated during epididymal maturation [33]. Despite the number of proteins making up the centrosome, and its importance in sperm differentiation and flagellum formation, very few centrosomal proteins have been linked to MMAF in humans – so far only CEP135 [34], CEP148 [35] or DZIP1 [36]. Moreover, some major axonemal proteins with an accessory location in the centrosome, such as CFAP58 [37] and ODF2 [38, 39], have also been reported to be involved in MMAF syndrome. Other centrosomal proteins lead to MMAF in mice, these include CEP131 [40] and CCDC42 [41]. The discovery that MMAF in humans is linked to CCDC146, known so far as a centrosomal protein, adds to our knowledge of proteins important for axoneme biogenesis.

In this manuscript, we first evaluated the localization of endogenous CCDC146 during the cell cycle in two types of cell cultures, immortalized HEK-293T cells and primary human foreskin fibroblasts. To validate the gene candidate and improve our knowledge of the corresponding protein, we also generated two mouse models. The first one was a *Ccdc146* KO model, with which we studied the impact of lack of the protein on the general phenotype, and in particular on male reproductive function using several optical and electronic microscopy techniques. The second model was a HA-tagged CCDC146 model, with which we studied the localization of the protein in different cell types. Data from these genetically modified mouse models were confirmed in human sperm cells.

Results

1/ WES identifies *CCDC146* as a gene involved in MMAF

We performed whole exome sequencing (WES) to investigate a highly-selected cohort of 167 MMAF patients previously described in [9]. The WES data was analyzed using an open-source bioinformatics pipeline developed in-house, as previously described [42]. From these data, we identified two patients with homozygous truncating variants in the *CCDC146* (coiled-coil domain containing 146) gene, NM_020879.3 (**Figure 1A**), which contains 19 exons in human. Sperm parameters of both patients are presented in **Figure 1—Figure supplement 1**. No other candidate variants reported to be associated with male infertility was detected in these patients. Despite an ubiquitous expression, this gene is highly transcribed in human testes (**Figure 1—Figure supplement 2A**). The first identified mutation is located in exon 9 and corresponds to c.1084C>T, the second is located in exon 15 and corresponds to c.2112Del (**Figure 1A**). The c.1084C>T variant is a nonsense mutation, whereas the single-nucleotide deletion c.2112Del is predicted to induce a translational frameshift. Both mutations were predicted to produce premature stop codons: p.(Arg362Ter) and p.(Arg704serfsTer7), respectively, leading either to the complete absence of the protein, or to the production of a truncated and non-functional protein. These variants are annotated with a high impact on the protein structure (MoBiDiC prioritization algorithm (MPA) score = 10) [43]. The two mutations are therefore most-likely deleterious. Both variants were absent in our control cohort and their minor allele frequencies (MAF), according to the gnomAD v3 database, were 6.984×10^{-5} and 6.5×10^{-06} respectively. The presence of these variants and their homozygous state were verified by Sanger sequencing, as illustrated in **Figure 1B**. Taken together, these elements strongly suggest that mutations in the *CCDC146* gene could be responsible for the infertility of these two patients and the MMAF phenotype.

2/ *Ccdc146* knock-out mouse model confirms that lack of CCDC146 is associated with MMAF

Despite a lower expression in mouse testes compared to human (the level of expression is only medium, **Figure 1—Figure supplement 2B**), we produced by CRISPR/Cas9 two mouse lines carrying each a frameshift mutation in *Ccdc146* (ENSMUST00000115245). One line (line 1) has a deletion of 4- bp (**Figure 2—Figure supplement 1AB**) and the other (line 2) an insertion of 250-bp, both in exon 2. We used these lines to address the hypothesis that CCDC146 deficiency leads to MMAF and male infertility. We analyzed the reproductive phenotype of the gene-edited mice from

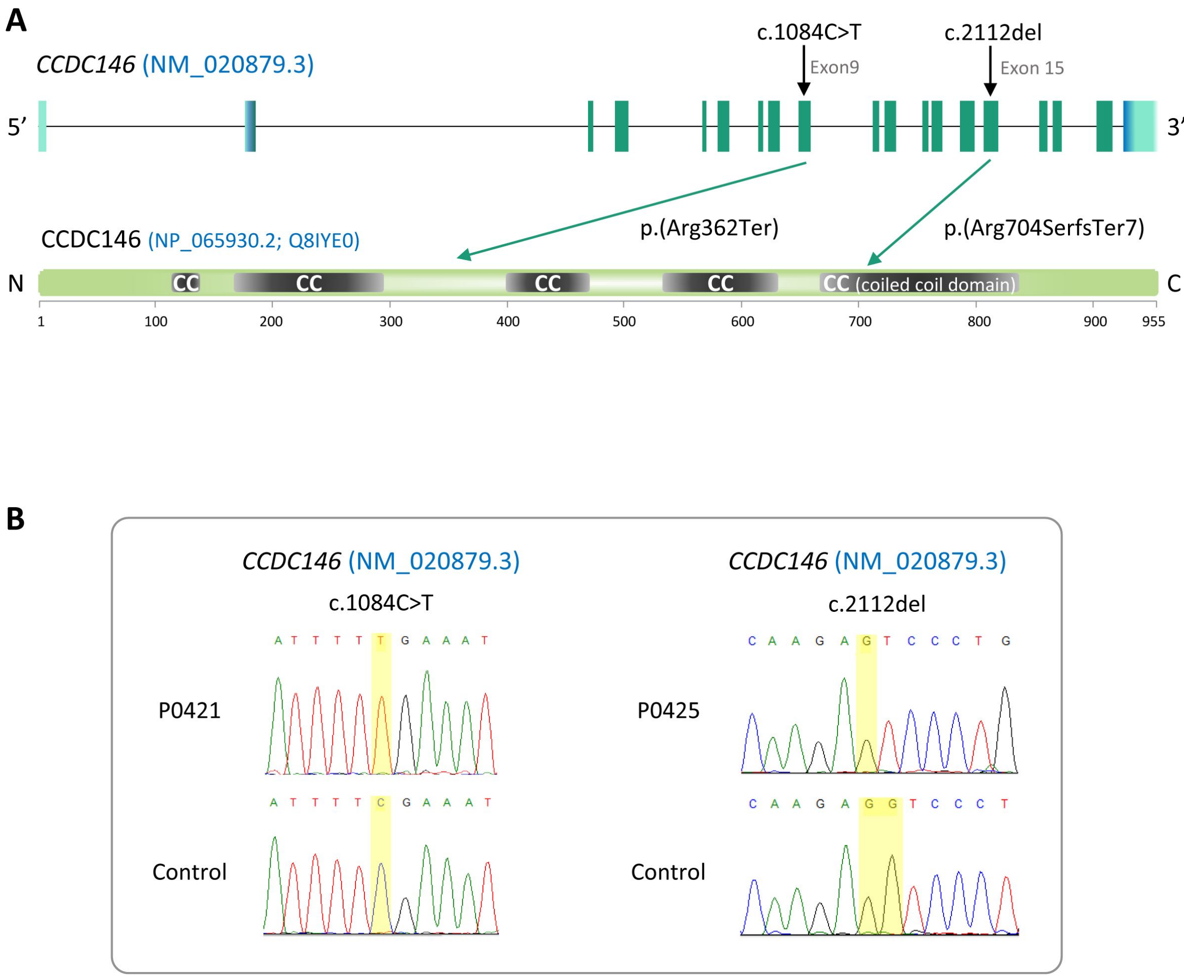

Figure 1

Identification of two *CCDC146* gene variants in MMAF patients

(A) Structure of the canonical *CCDC146* gene transcript showing the position of the variants and their impact on translation. Variants are annotated according to HGVS recommendations. Position of the observed variants in both probands. (B) Electropherograms indicating the homozygous state of the identified variant: variant c.1084C>T is a nonsense mutation, and c.2112Del is a single-nucleotide deletion predicted to induce a translational frameshift.

the F2 generation and found that homozygous males reproduced the MMAF phenotype, like the two patients carrying the homozygous variants in the orthologous gene (**Figure 2** for line 1 and **Figure 2—Figure supplementary 1CD** for line 2). Based on these findings, we restricted our study to a strain with a 4-bp deletion in exon 2 (c.164_167delTTCG).

The *Ccdc146* KO mice were viable without apparent defects. The reproductive phenotypes of male and female mice were explored. WT or heterozygous animals and KO females were fertile, whereas KO males were completely infertile (**Figure 2A**). This infertility is associated with a 90% decrease in epididymal sperm concentration (from ~30- to ~3-million) (**Figure 2B**) and a significant decrease of testicle weight relative to whole body weight (**Figure 2C**), suggesting a germ cell rarefaction in the seminiferous epithelia due to high apoptosis level. A study of spermatogenic cell viability by TUNEL assay confirmed this hypothesis, with a significant increase in the number of fluorescent cells in *Ccdc146* KO animals (**Figure 2—Figure supplement 2**). Closer examination revealed sperm morphology to be strongly altered, with a typical MMAF phenotype and marked defects in head morphology (**Figure 2D**) indicative of significantly impaired spermiogenesis. The percentage of abnormal form, flagellum and head anomalies is 100% (**Figure 2E**). Moreover, an almost complete absence of motility was observed (**Figure 2F**), Comparative histological studies (**Figure 3**) showed that on sections of spermatogenic tubules, structural and shape defects were present from the elongating spermatid stage in *Ccdc146* KO mice, with almost complete disappearance of the flagella in the lumen and very long spermatid nuclei (**Figure 3A**). At the epididymal level, transverse sections of the epididymal tubules from KO males contained almost no spermatozoa, and the tubules were filled with an acellular substance (**Figure 3B**).

3/ *CCDC146* codes for a centriolar protein

CCDC146 has been described as a centriolar protein in immortalized HeLa cells [28, 29]. To confirm this localization, we performed immunofluorescence experiments (IF) (**Figure 4**). First, we validated the specificity of the anti-CCDC146 antibody (Ab) in HEK-293T cells by expressing DDK-tagged CCDC146. We observed a nice colocalization of the DDK and CCDC146 signals in both interphase and mitotic cells, showing that both Abs target the same protein (**Figure 4—Figure supplement 1**), strongly suggesting that the anti-CCDC146 Ab is specific. Next, we focused on the centrosome (**Figure 4A**). In HEK-293T cells, using an antibody recognizing centrin (anti-centrin Ab) as a centriole marker and the anti-CCDC146 Ab, CCDC146 was shown to colocalize with centrioles. This colocalization strongly suggests that CCDC146 is a centriolar protein. However, the signal was not strictly localized to centrioles, as peri-centriolar labeling was clearly visible (**Figure 4A1d**, overlay). As this labeling pattern suggests the presence of centriolar satellite proteins, we next performed co-labeling with an antibody recognizing PCM1, a canonical centriolar satellite marker [44]. Once again, the colocalization was only partial (**Figure 4—Figure supplement 2**). Finally, the presence of CCDC146 was assessed at the basal body of primary cilia in cells cultured under serum-deprived conditions (**Figure 4B**). Similarly to centrosome staining, CCDC146 was observed on the basal body by also around it (**Figure 4B1d**). Based on these observations, CCDC146 has a unique localization profile in somatic cells that may indicate specific functions.

4/ CCDC146 co-localizes with multiple tubulin-based organelles

We next evaluated the presence of CCDC146 on centrioles during their duplication (**Figure 5A**). For this purpose, we synchronized HEK-293T cells with thymidine and nocodazole, and CCDC146 labelling was observed by IF. Both centrioles pairs were stained by CCDC146 Ab in cells blocked in the G2 phase (**Figure 5A**). Because our immunofluorescence experiments revealed that CCDC146 labeling was not strictly limited to centrosomes, we next assessed the presence of CCDC146 on other tubulin-containing cellular substructures, particularly structures emerging during cell division (**Figure 5B-E**). In non-synchronized cells, the mitotic spindle was labeled at its base and

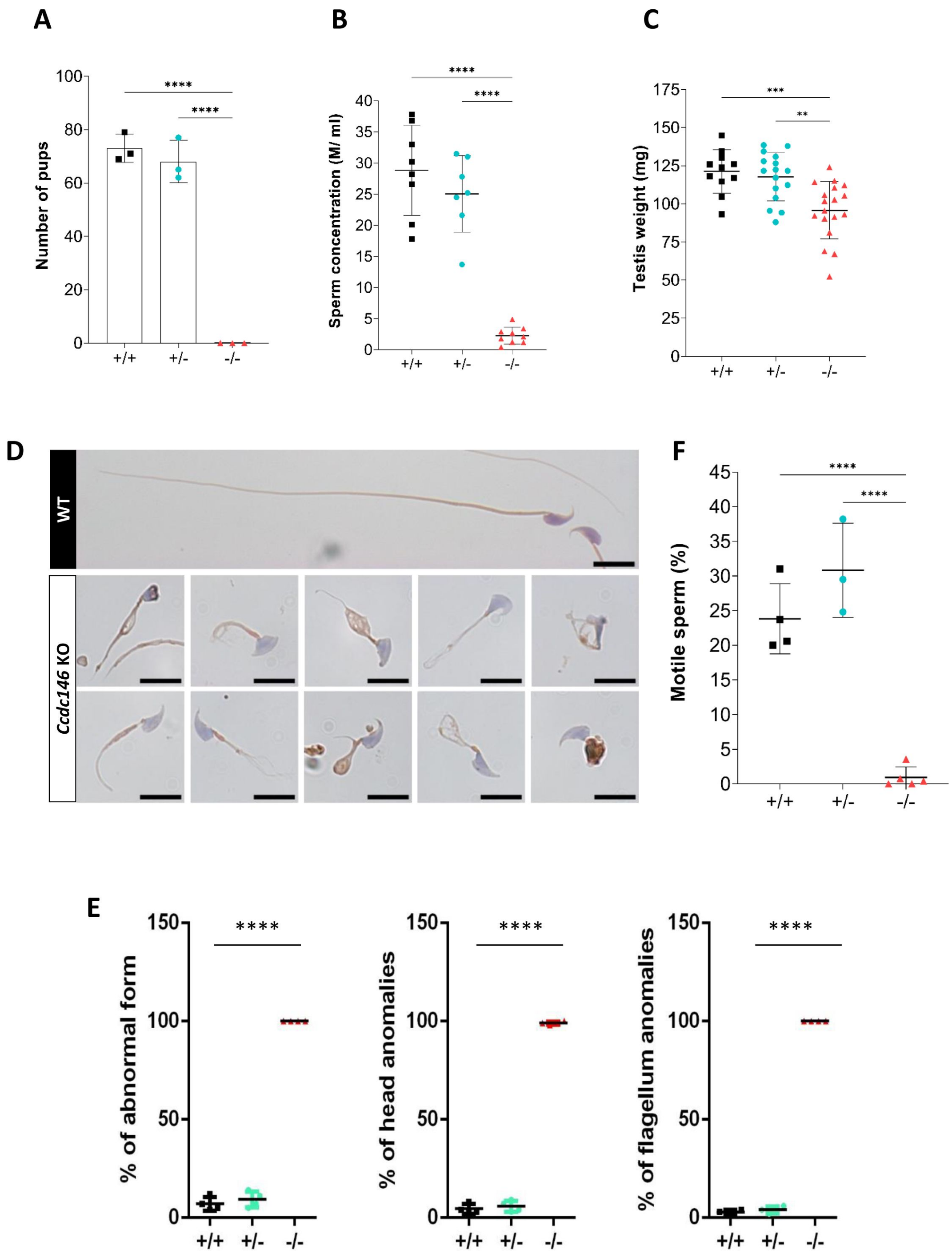

Figure 2

Ccdc146 KO mice are infertile and KO sperm exhibit a typical "multiple morphological abnormalities of the flagellum (MMAF)" phenotype.

(A) Number of pups produced by wild-type (+/+, WT), *CCDC146* heterozygote (+/-) and *Ccdc146* knock-out (-/-, KO) males (3 males per genotype) after mating with fertile WT females (2 females per male) over a period of three months. (B) Sperm concentration and (C) Comparison of testis weights (mg). (D) Illustration of WT and KO sperm morphologies stained with Papanicolaou and observed under optic microscopy. Scale bars of images represents 10 µm. (E) Histograms showing proportions of total, head and flagella morphological anomalies (mean ± SD) for each *Ccdc146* genotype (n=5). (F) sperm mobility. (A-D, F) Statistical comparisons were based on ordinary one-way ANOVA tests and (E) statistical significance was assessed by applying an unpaired t-test; p-values: **** p<0.0001; *** p<0.001, **p<0.01, *p<0.05.

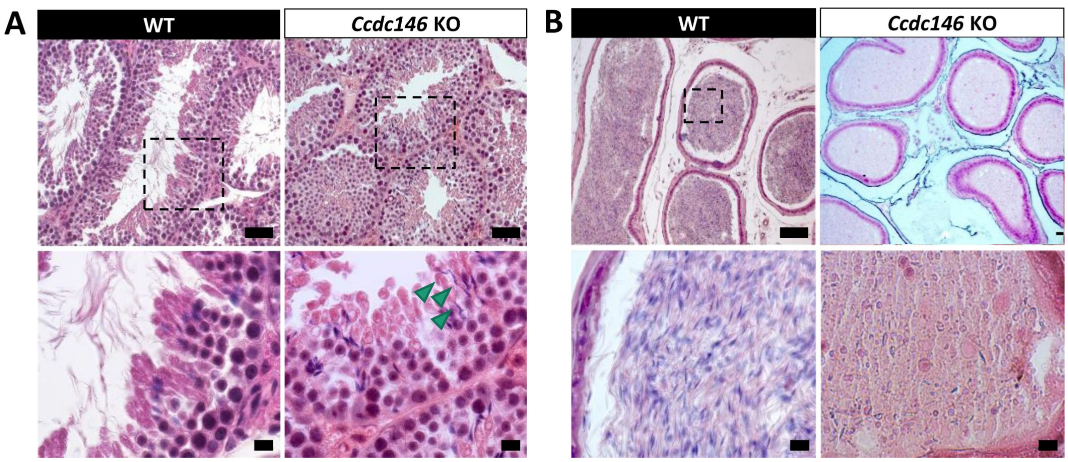

Figure 3

Histological evidence that spermiogenesis is disrupted in *Ccdc146* KO males and leads to a strong decrease in sperm concentration in the epididymis.

(A) Transversal sections of WT and KO testes stained with hematoxylin and eosin. The upper images show the sections at low magnification (Scale bars 50 µm) and the lower images are an enlargement of the dotted square (scale bars 10 µm). In the KO, spermatid nuclei were very elongated and remarkably thin (green arrow heads) and no flagella were visible within the seminiferous tubule lumen. (B) Transversal sections of WT and KO epididymides stained with hematoxylin and eosin. Despite similar epididymis section diameters in WT and KO testes, KO lumen were filled with round cells and contained few spermatozoa with abnormally-shaped heads and flagella. The upper images show the sections at low magnification (scale bars 50 µm) and the lower images are an enlargement of the dotted square (scale bars 10 µm).

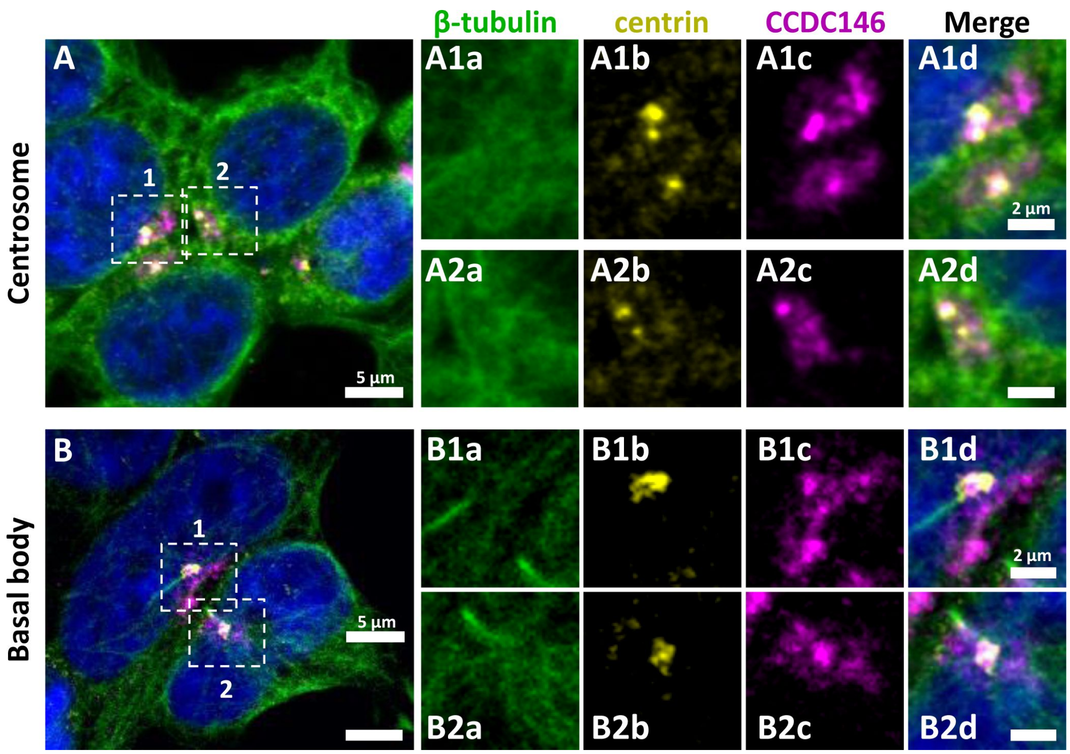

Figure 4

CCDC146 has a centriolar and pericentriolar localization in interphase somatic HEK-293T cells.

HEK-293T cells were immunolabeled for β-tubulin (green), centrin (yellow) and CCDC146 (magenta). DNA was stained with Hoechst (blue). (A) CCDC146 localized to centrioles and/ or to the pericentriolar material in interphase cells. The centrosome area of two cells are shown enlarged in A1 and A2. (B) In serum starved cells with primary cilia, CCDC146 localized to the basal body of primary cilia. The basal body of two cells are shown enlarged in B1 and B2. CCDC146 was also present as dotted signal resembling the pattern for centriolar satellite proteins. Scale bars µm and 2 µm on zoomed images.

at its ends (**Figure 5BC**). The co-labeling intensified in the midzone during chromatid separation (**Figure 5D**). Finally, the separation structure between the two cells, the midbody, was also strongly stained (**Figure 5E**). As HEK-293T cells are an immortalized cell line, we therefore verified that this labeling pattern was not due to an aberrant expression profile and that it also reflected the situation in primary cell lines. Identical labeling profiles were observed in freshly-prepared Human foreskin fibroblasts (HFF cells) (**Figure 5—Figure supplement 1**).

5/ CCDC146 is present in epididymal sperm and its expression peaks at spermatid elongation stage during spermatogenesis

The presence of CCDC146 in the mature epididymal spermatozoa was assessed by WB (**Figure 6A**). To overcome a lack of specific antibodies recognizing mouse CCDC146 (the commercial Ab works for human CCDC146 only), a mouse strain expressing a HA-CCDC146 knock-in (KI) was created. Because the functional domains of the protein are unknown, and to limit as such as possible the risk of interfering with the structure of the protein, we choose to insert the hemagglutinin (HA) tag, which contains only 9 amino acids (YPYDVPDYA). HA sequence was inserted by the CRISPR/Cas9 system into the coding sequence of the *Ccdc146* gene between the two first codons to produce a tagged protein at the N-terminus domain (**Figure 6—Figure supplement 1**). This insertion induced no phenotypic changes, and both female and male mice were viable with normal fertility. Interestingly, the protein was retained in epididymal sperm, where one band was observed around 120 kDa (**Figure 6A**). The theoretical MW of the tagged protein is around 116.2 kDa (115.1 + 1.1); the observed MW was slightly higher, suggesting that the protein migrates at a different weight or some post-translational modifications. The band is specific because it was observed in HA-CCDC146 sperm only and not in WT epdididymal sperm. The protein is therefore present during spermatid differentiation and conserved in mature sperm. To better characterize the presence of CCDC146 in sperm, sperm flagella and heads were purified after mild sonication (**Figure 6B**), and protein extracts were analyzed by WB (**Figure 6A**). In the flagella fraction, the HA Ab revealed a band with an intensity similar to that of the whole sperm at around 120 kDa whereas no staining was observed in head fraction.

From our immunofluorescence analysis of somatic cell lines, it was clear that CCDC146 expression is associated with the cell cycle. In the testis, a wide variety of cell types co-exist, including both somatic and germline cells. The germline cells can be further subcategorized into a wide variety of cells, some engaged in proliferation (spermatogonia), others in meiosis (spermatocytes), or in differentiation (spermatids). To better understand the role of CCDC146 in spermatogenesis, and thus how its absence leads to sperm death and malformation, we initially studied its expression during the first wave of spermatogenesis [45]. Results from this study should shed light on when CCDC146 is required for sperm formation. Expression of *Ccdc146* was normalized with two different housekeeping genes, *Actb* and *Hprt* (**Figure 6C**). Surprisingly, RT-PCR failed to detect *Ccdc146* transcripts on day 9 after birth in proliferating spermatogonia from HA-CCDC146 males. Transcription of *Ccdc146* started on day 18, concomitantly with the initiation of meiosis 2. Expression peaked on day 26, during the differentiation of spermatids.

6/ In sperm, CCDC146 is present in the flagellum, not in the centriole

To attempt to elucidate the function of CCDC146 in sperm cells, we next studied its localization by IF in mouse and human sperm. We focused successively on the anterior segment (head, neck and beginning of the intermediate piece) of the sperm and then on the flagellum.

The IF experiments were first performed on murine epididymal spermatozoa. For this study, we used the mouse model expressing HA-tagged CCDC146 protein. In conventional IF, using an HA Tag Alexa Fluor® 488-conjugated Antibody (anti-HA-AF488-C Ab), a punctiform labeling was observed

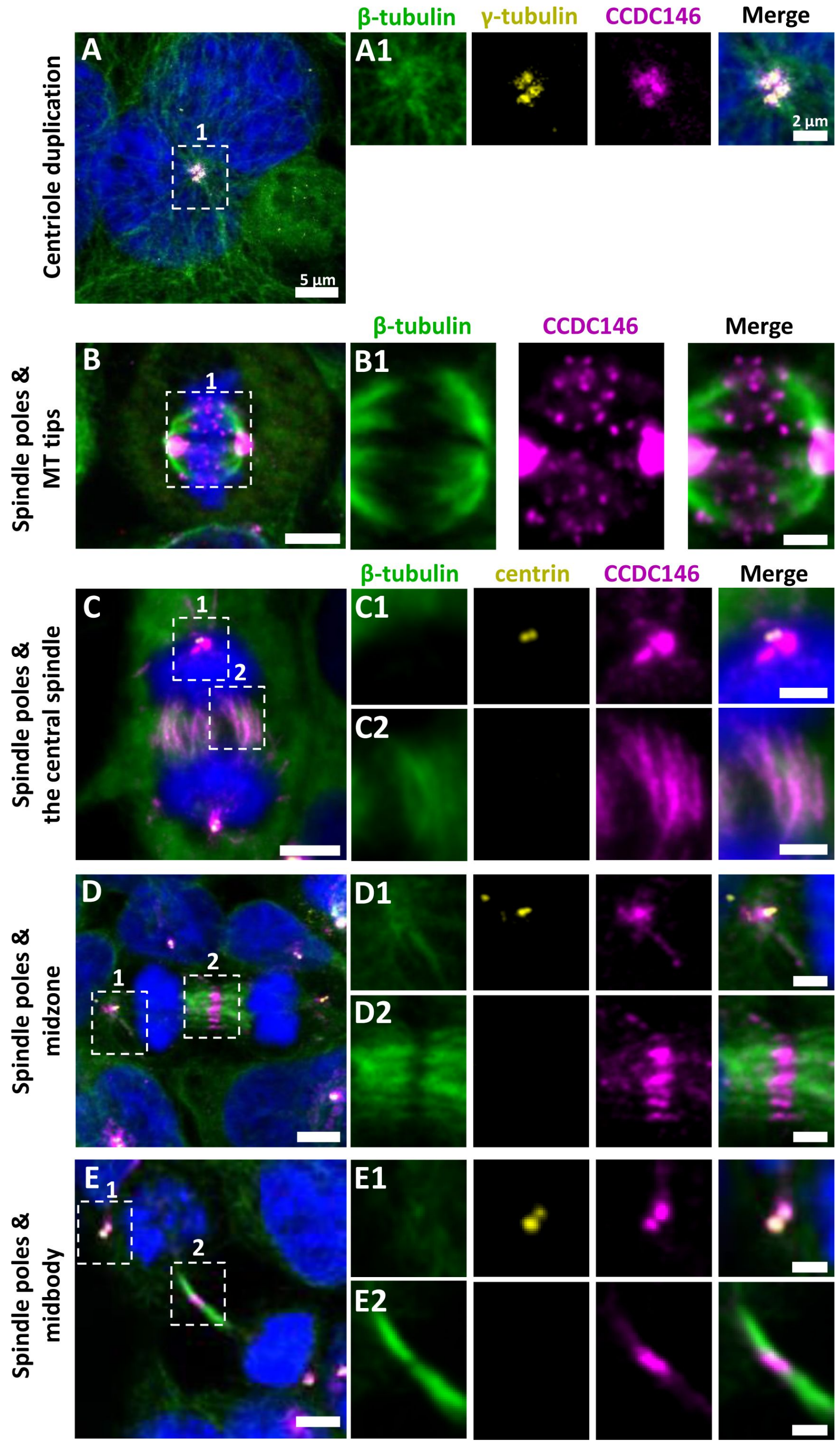

Figure 5

CCDC146 is a microtubule-associated protein (MAP) associating with microtubule-based structures throughout the cell cycle.

HEK-293T cells were immunolabeled with anti-β-tubulin (panels A-E, green), anti-γ tubulin (yellow, panel A), anti-centrin (yellow panels C-E), and anti-CCDC146 (panels A-E, magenta) Abs. DNA was stained with Hoechst (blue). (A) In synchronized HEK-293T cells, CCDC146 is observed associated with mother centrioles and their corresponding procentrioles during centriole duplication. (B-E) In non-synchronized cells, CCDC146 is observed associated with spindle poles (B-E) and with microtubule (MT) tips during metaphase (B), with the central spindle during anaphase (C), with the midzone during telophase (D), and with the midbody during cytokinesis (E). Images on the right show the enlargement of the dotted square in the left image. Scale bar of zoomed images 2 µm.

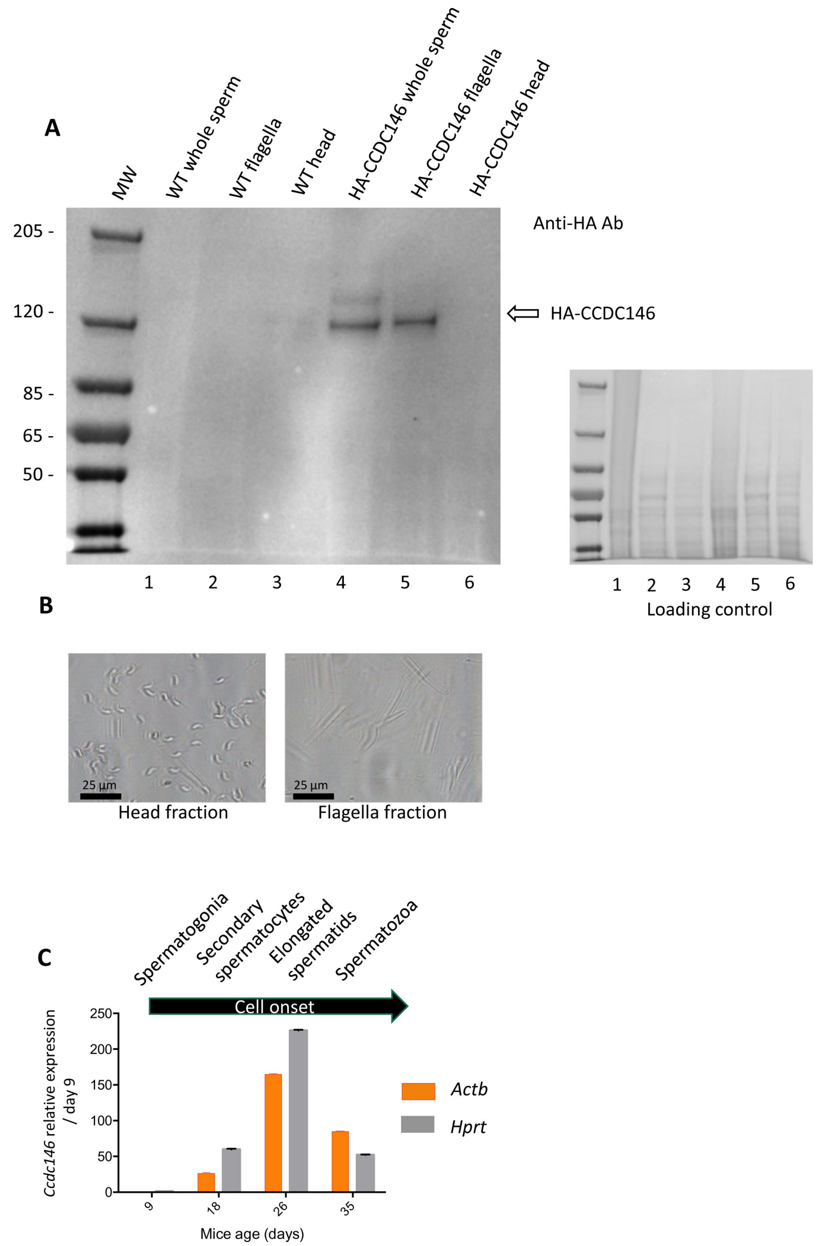

Figure 6

CCDC146 protein is present in epididymal spermatozoa in mouse and *Ccdc146* mRNA is expressed in late spermatocyte and in spermatids;

(A) CCDC146 is retained in epididymal sperm. Western blot of HA-tagged epididymal whole sperm shows a band (arrow) corresponding to HA-tagged CCDC146 (HA-CCDC146), whereas no band was observed with WT epididymal sperm, demonstrating the specificity of the band observed in HA- CCDC146 sperm lane. Sperm extracts obtained from flagella and head fractions of WT and HA-CCDC146 were also analysed and the presence of HA-CCDC146 was revealed by an anti-HA Ab in flagella fraction from HA-CCDC146 sperm only. Loading control (ponceau staining) of the gel is shown on the right. (B) Image of heads and flagella fractions. (C) mRNA expression levels of *Ccdc146* relative to *Actb* and *Hprt* in HA-CCDC146 mouse pups' testes during the first wave of spermatogenesis (n=3). Extremely low *Ccdc146* expression was detected at day 9, corresponding to testes containing spermatogonia and Sertoli cells only. *Ccdc146* expression was observed from postnatal day 18 (formation of secondary spermatocytes), peaked at day 26 (formation of elongated spermatids), and subsequently decreased from day 35 (formation of spermatozoa), suggesting that *Ccdc146* is particularly expressed in elongated spermatids during spermatogenesis.

along the whole flagellum (**Figure 7A**) on HA-CCDC146 only. We validated the specificity of the punctiform staining by performing a statistical comparison of the density of dots in the principal piece of WT and HA-CCDC146 sperm (**Figure 7B**). This study was carried out by analyzing 58 WT spermatozoa and 65 HA-CCDC146 spermatozoa coming from 3 WT and 3 KI males. We found a highly significant difference, with a p-value <0.0001, showing that the signal obtained on spermatozoa expressing the tagged protein is highly specific. To enhance resolution, ultrastructure expansion microscopy (U-ExM), an efficient method to study in detail the ultrastructure of organelles [46], was used. We observed the same punctiform staining all along the axoneme, including the midpiece (**Figure 7C**), using another anti-HA Ab (#2). Unexpectedly, the mouse flagellum presented breaks that most likely resulted from the expansion procedure (**Figure 7D**). Interestingly, strong HA-labeling was observed at the level of these breaks suggesting that CCDC146 epitopes are buried inside the axonemal structure and become accessible mostly on blunt or broken microtubule doublets. The same pattern was observed with a third different anti-HA Ab (#3) (**Figure 7—Figure supplement 1**).

In humans, the spermatozoon retains its two centrioles [32], and they are observable in the neck, as shown by anti-centrin and anti-tubulin labeling (**Figure 8A1ac, A2ac and A3ac**). No colocalization of the CCDC146 label with centrin was observed on human sperm centrioles (**Figure 8A1d, A2d and A3d**), suggesting that the protein is not present in or around this structure. However, two unexpected labeling events were observed: sub-acrosomal labeling and labeling of the midpiece (**Figure 8A1b, A2b and A3b**). At the flagellum level, faint staining was observed along the whole length (**Figure 8A4b**). To enhance resolution, U-ExM was also used (**Figure 8B—Figure supplement 1A**). Once again, no CCDC146 labeling was observed on sperm centrioles (**Figure 8B6** merge), confirming the conventional IF results. Moreover, U-ExM unveiled that the observed CCDC146 midpiece staining (**Figure 8A**) seems in fact associated with isolated structures, which we hypothesize could be mitochondria, now visible by expansion (**Figure 8B1**). This suggests that the labeling on the midpiece observed in IF corresponds to non-specific mitochondrial labeling. This conclusion is supported by the fact that the same isolated structures were also labeled with anti-POC5 Ab (**Figure 8—Figure supplement 1B**). In contrast, at the flagellum level, clear punctiform labeling was observed along the whole length of the principal piece (**Figure 8B1 and B5**), confirming that the protein is present in the sperm flagellum.

The study of CCDC146 localization was made in two species and with two different Abs. We observed slight differences in the CCDC146 staining between human and mouse. However, the antibodies do not target the same part of the CCDC146 protein (the tag is placed at the N-terminus of the protein, and the HPA020082 Ab targets the last 130 amino acids of the Cter), a difference that may change their accessibility to the antigenic site and explain the difference. Nevertheless, overall, the results are fairly consistent and both antibodies target the flagellum. It should be noted that unlike in humans, in mice, centrioles are no longer present in epididymal spermatozoa [33]. Therefore, the WB result, showing the presence of the protein in sperm flagella fraction in mouse sperm, demonstrate unequivocally that the protein is located in the flagellum.

7/ CCDC146 labeling associates with microtubule doublets

We next wanted to determine whether CCDC146 staining was associated with the axoneme or accessory structures of the flagellum (outer dense fibers or fibrous sheath), and if yes, whether it was associated with microtubule doublets or the central pair. To do so, we used U-ExM on human sperm and quantified the relative position of each CCDC146 dot observed (outside the axoneme, outer left and right; microtubule doublets left and right and central pair – **Figure 9A-C**). The different parts of the flagellum were identified from the tubulin staining (**Figure 9B**). The distribution of localizations was summarized in a bar graph, generated for two distinct sperm cells. Labeling was preferentially located on the right and left doublets (**Figure 9D**), indicating that CCDC146 associates more with microtubule doublets. Analysis of protein distribution in mice was not easy, because signal tended to concentrate at breaks. Nevertheless, using U-ExM, on some spermatozoa with frayed microtubule doublets, i.e. with the flagellum taking on the shape of a

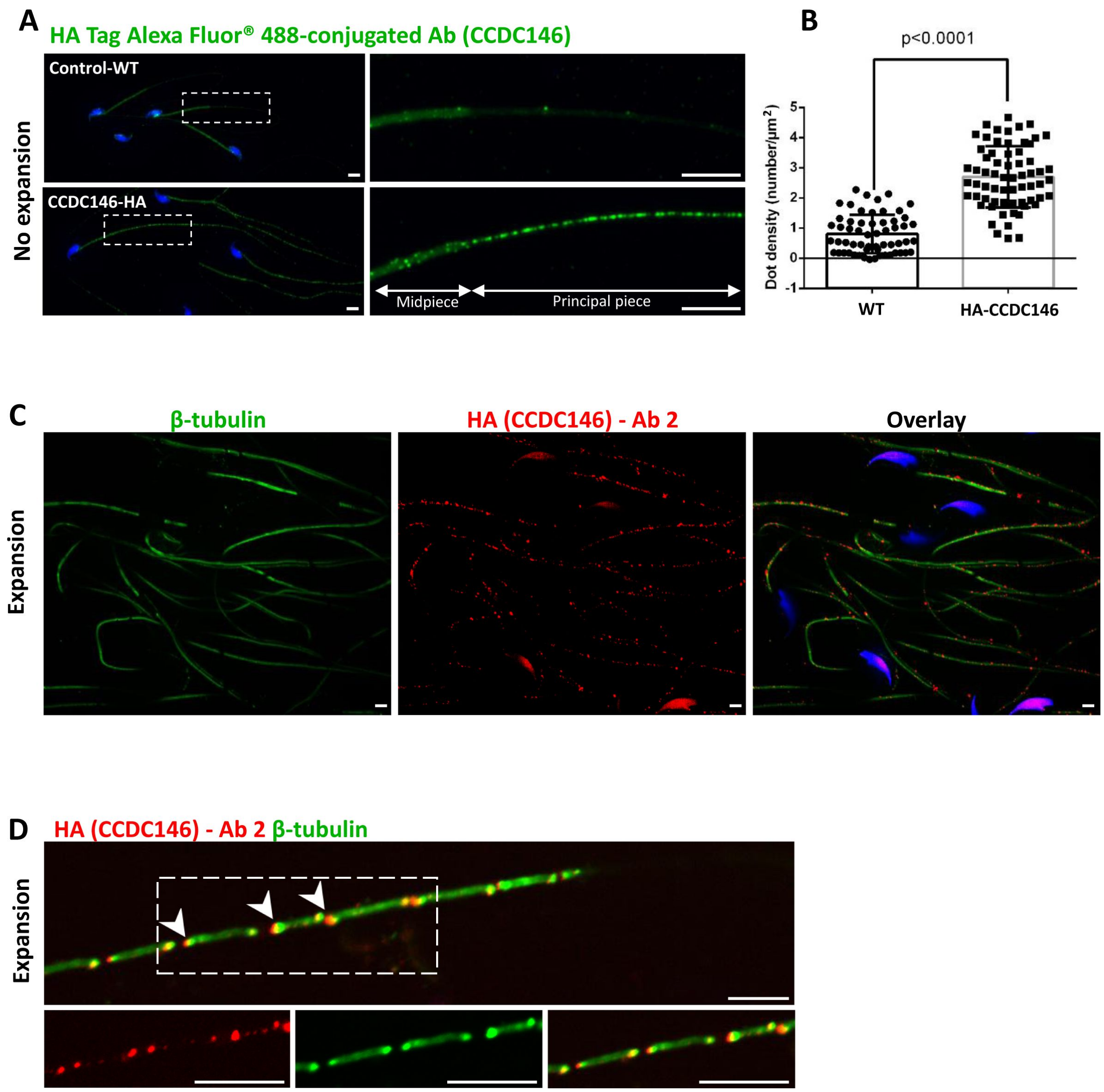

Figure 7

CCDC146 localizes to the flagellum of mouse epididymal spermatozoa.

(A) Mouse epididymal spermatozoa observed with conventional IF. WT and HA-CCDC146 sperm were labeled with anti-HA Tag Alexa Fluor® 488-conjugated 3 (green) Abs (anti-HA #1). DNA was stained with Hoechst (blue). The upper image shows the staining of a WT sperm and the lower images, the staining of a HA-CCDC146 sperm. A punctuate signal is observed in HA-CCDC146 sperm (Scale bars 5 µm). (B) Quantification of the density of dots per square µm (µm^2) and statistical significance between WT and HA-CCDC146 sperm was assessed by Mann-Whitney test, p value as indicated. (C) Mouse epididymal spermatozoa observed with expansion microscopy. HA-CCDC146 sperm were immunolabeled with anti-β-tubulin (red), anti-HA #2 (green) Abs and DNA was stained with Hoechst (blue). The right image shows the sperm with merged immunostaining. (Scale bars 5 µm) (D) Mouse epididymal spermatozoa observed with expansion microscopy. The lower images show the staining, HA-CCDC146 (red), tubulin (green) and merge observed in the principal piece of the flagellum. Strong red punctiform signals were observed at the level of axonemal breakages induced by the expansion process. White arrows indicate the zones of the micro breaks. Scale bars 10 µm.

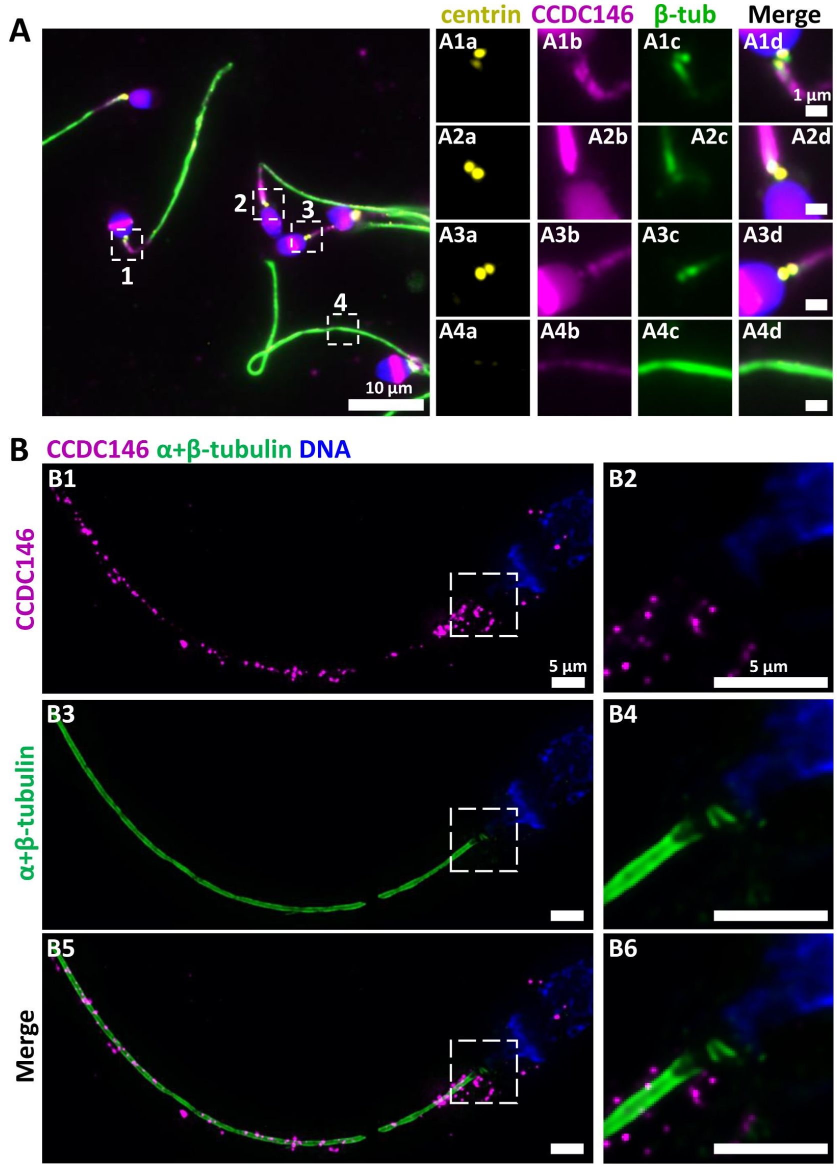

Figure 8

CCDC146 localizes to the flagellum but not to the centrioles of ejaculated human spermatozoa.

(A) Human ejaculated sperm were immunolabeled with Abs recognizing centrin (yellow), CCDC146 (magenta), and β-tubulin (green). DNA was stained with Hoechst (blue). (A1-A3) enlargement of dotted square focused on sperm neck: no colocalization between CCDC146 and centrin. (A4) A faint signal for CCDC146 is present along the length of the sperm flagellum. Scale bar of zoomed images: 1 µm. (B) Human ejaculated sperm observed by expansion microscopy. Sperm were immunolabeled with anti-CCDC146 (magenta) and anti-β-tubulin (green) abs, and DNA was stained with Hoechst (blue). B1 and B5 show strong staining for CCDC146 in the axoneme, B3 shows the localization of the axoneme through tubulin staining. B2, B4 and B6 show enlargements of the dotted squares focused on the sperm neck. CCDC146 did not colocalize with the centrioles at the base of the axoneme. The CCDC146 staining observed probably corresponds to non-specific labeling of mitochondria, as suggested by **Figure 8—figure supplement 1**. Scale bars 5 µm.

hair, we could find that isolated doublets carried the punctiform labeling confirming the results of analyses on human spermatozoa (**Figure 9E**). Taken together, these results from mouse and human sperm demonstrate that CCDC146 is an axonemal protein, probably associated with microtubule doublets.

8/ CCDC146 is solubilized by the sarkosyl detergent

The presence of extensive labeling at axoneme breaks suggests that the antigenic site is difficult to access in an intact flagellum. We therefore hypothesized that CCDC146 could be a MIP. MIPs are generally resistant to solubilization by detergents. However, N-lauroylsarcosine (sarkosyl) can solubilize microtubule doublets, with increasing concentrations destabilizing first the A-tubule, then the B-tubule [47, 48]. Microtubule solubilization allows release of the MIPs contained within the tubules, and the method is recognized [49, 50]. To test whether our hypothesis that CCDC146 may be a MIP, we treated spermatozoa from HA-tagged CCDC146 mice with sarkosyl and performed a WB on the supernatant (**Figure 10A**). The protein was effectively solubilized, leading to the appearance of bands at around 120 kDa on an SDS-PAGE gel following migration of extracts from sperm treated with 0.2 and 0.4% sarkosyl concentrations. Interestingly, a second band around 90 kDa was also immunodecorated by anti-HA antibody in the sarkosyl treated sample, a band not present in the protein extracts from WT males. This band may correspond to proteolytic fragment of CCDC146, the solubilization of microtubules by sarkosyl may have made CCDC146 more accessible to endogenous proteases. The other detergents and buffers tested – RIPA, CHAPS or Tris-HCl – barely solubilized CCDC146 or led to no solubilization (**Figure 10B**). This result confirms the unique action of sarkosyl on CCDC146 and strengthens the hypothesis that the protein is associated with the doublets of microtubules. To confirm the action of sarkosyl on the accessibility of the antigenic site, murine spermatozoa were labeled with an anti-HA antibody after treatment with sarkosyl or no treatment. CCDC146 labeling in the principal piece was significantly increased in the presence of sarkosyl (**Figure 10C**). The full image panel can be found in **Figure 10—Figure supplement 1A**. The presence of the protein in the midpiece was difficult to assess because non-specific staining was observed in this area, the non-specific staining being due to secondary antibodies as showed in **Figure 10—Figure supplement 1B**, where they were used alone. Despite all these results, the CCDC146 localization in the lumen of microtubules requires at this stage more proofs.

9/ KO models show defects in tubulin-made organelles

To better characterize the function of CCDC146 in mouse sperm, we went on to perform a detailed morphological analysis of tubulin-made organelles by IF and scanning microscopy, and examined the morphological defects induced by the absence of CCDC146 at the subcellular level by transmission electron microscopy. This work was performed on immature testicular sperm and on seminiferous tubule sections from adult WT and *Ccdc146* KO males. Mouse testicular sperm were used because they still contain centrioles that become disassembled as sperm transit through the epididymis [51]. We mainly focused our analyses on the centrioles, the manchette, and the axoneme.

The connecting piece between the head and the flagellum, known as the sperm head-tail coupling apparatus (HTCA), is a complex structure containing several substructures including both centrioles, the capitulum and the segmented columns [31]. The distal centriole is embedded in the segmented column and the axoneme emerges from the distal centriole. This structure has a specific shape when observed by scanning electron microscopy (**Figure 11A**). In *Ccdc146* KO sperm, there was a great variability in the morphological defects of the connecting piece, some were almost intact (**Figure 11A2**), whereas others were severely damaged (**Figure 11A3 and A4**). In WT testicular sperm, IF experiments show that the centrioles, identified by anti-tubulin Ab, are very close to each other and adjacent to the nucleus (**Figure 11B1**). For *Ccdc146* KO males, although some sperm cells presented centrioles with almost normal localization (**Figure**

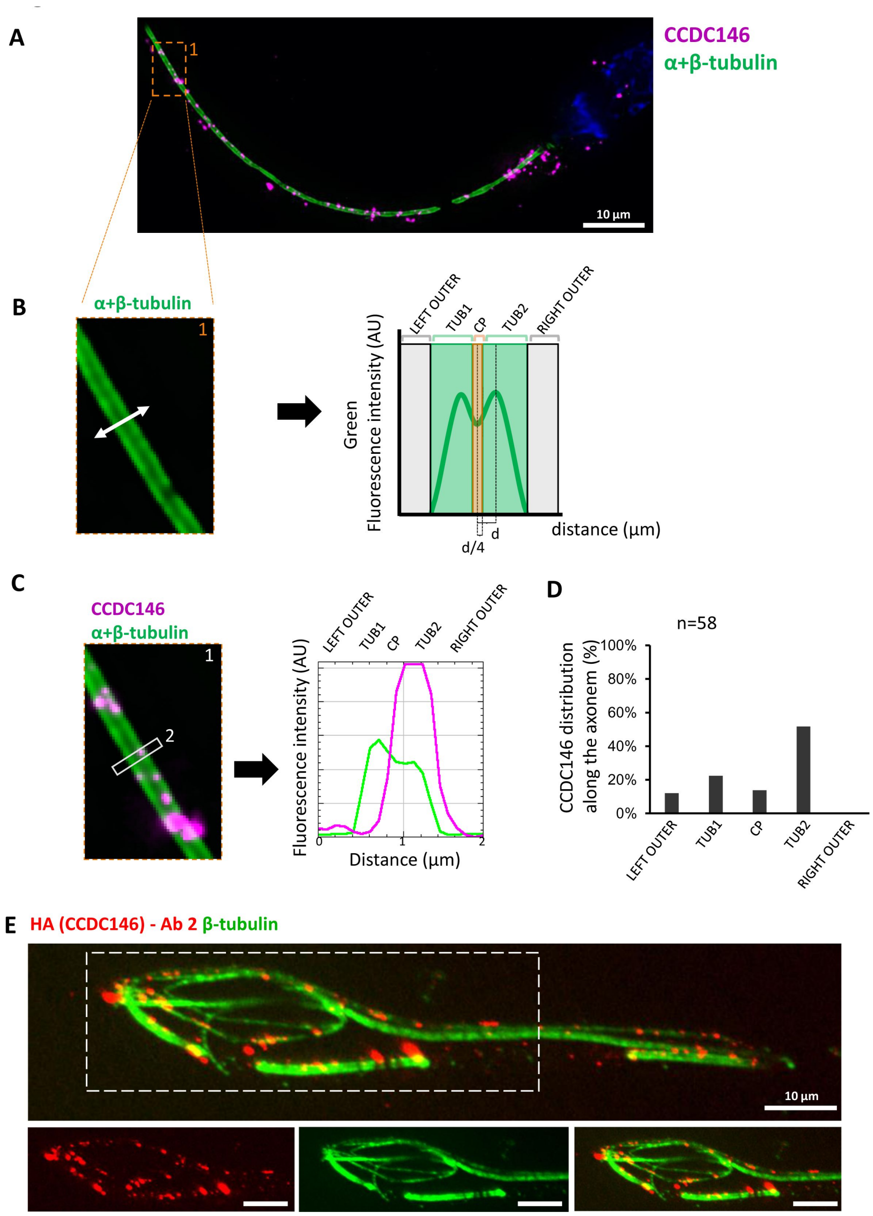

Figure 9

CCDC146 localizes to the microtubule doublets of the axoneme in human and mouse

(A) Sperm was double-stained with anti-tubulin (green) and anti-CCDC146 (magenta) Abs and observed by expansion microscopy. Scale bar 10 µm. (B) Measurement of the green-tubulin signal intensity perpendicularly to the axoneme is quite characteristic with two peaks corresponding to the left and right microtubule doublet. It identifies five axonemal compartments (left outer, left doublet (Tub1), central pair (CP), right doublet (Tub2), and right outer). To determine the CP area, the distance (d) between the center of the flagella and the peak of Tub2 was measured, and all fluorescent dots located in between −d/4 to +d/4 from the center were counted as CP dots. (C) Example of the measurement of the tubulin (green) and the CCDC146 signal intensities measured at the white rectangle (2). The image corresponds to the orange rectangle in A. In this example, the CCDC146 signal is localized in the right doublet of microtubules (Tub2). (D) The position of the CCDC146 signal with respect to the tubulin signal was measured along the entire flagellum. Each CCDC146 signal was assigned to a different compartment of the axoneme, allowing to obtain an histogram showing the distribution of CCDC146 labeling in ejaculated human sperm (n=2, 38 dots analysed). (E) Flagellum of a mouse epididymal spermatozoa observed with expansion microscopy. HA-CCDC146 sperm were immunolabeled with anti-HA #2 (red) and anti-β-tubulin (green). The upper image shows the sperm with merged immunostaining, and the lower images, the staining (red, green, and merge) observed in the principal piece of the flagellum. Scale bars 10 µm.

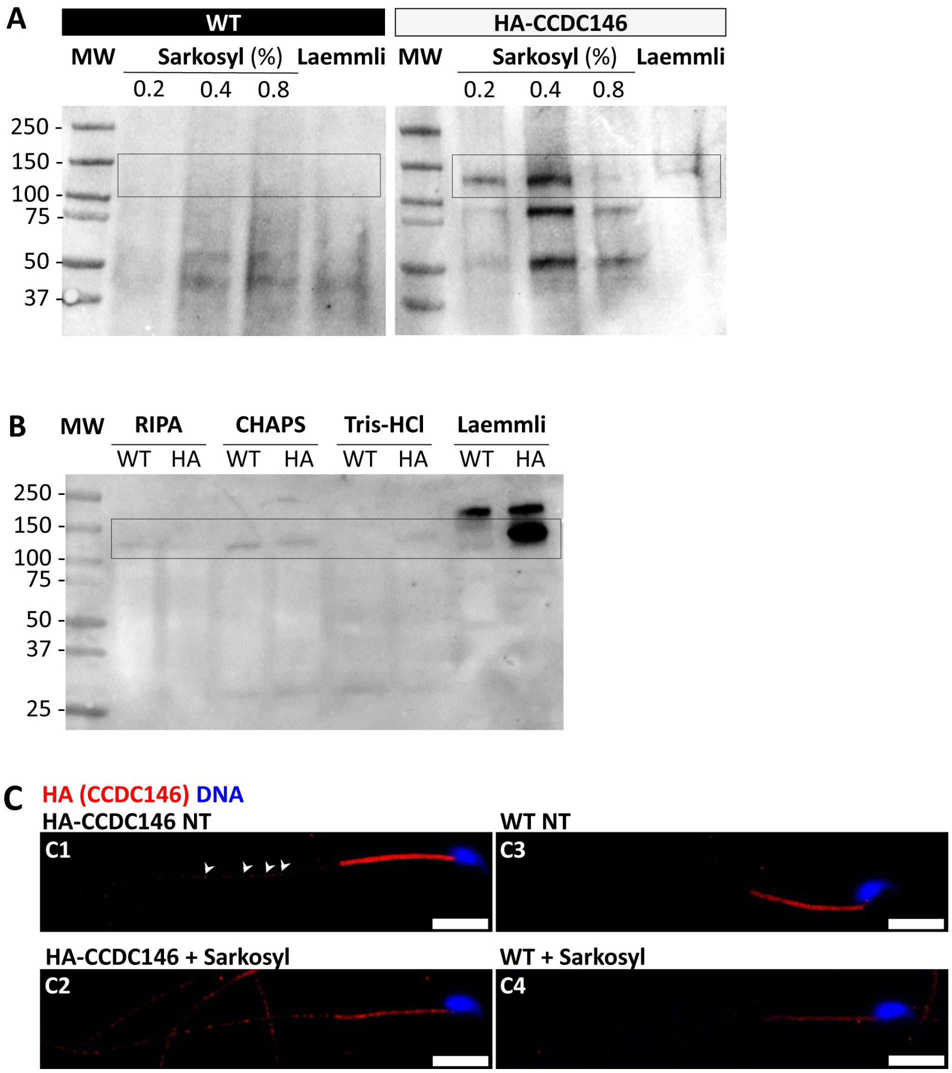

Figure 10

CCDC146 is solubilized by sarkosyl and sarkosyl treatment enhances IF signal

(A) Western blot of WT and HA-CCDC146 sperm extract solubilized with N-lauroylsarcosine (sarkosyl), an anionic detergent. Sarkosyl was used at increasing concentrations (0.2 and 0.4%). The presence of HA-CCDC146 was detected by an anti-HA Ab. (B) Western blot of WT and HA-CCDC146 sperm extracts solubilized with alternative detergents (RIPA, CHAPS, Tris-HCl) and whole sperm extract solubilized in Laemmli. The presence of HA-CCDC146 was revealed by an anti-HA Ab. (C) Epididymal HA-CCDC146 sperm (C1-C2) and WT sperm (C3-C4), treated with sarkosyl (5 min, 0.2% sarkosyl) or not (NT), were immunostained to reveal the HA-tag (red) and counterstained with Hoechst (blue). (C1) Without treatment, a faint CCDC146 signal (white arrow heads) is observed along the flagellum from HA- CCDC146 sperm. (C2) Treatment with sarkosyl enhanced the HA-CCDC146 signal along the sperm flagellum. (C3) The HA signal present the midpiece is likely non-specific since it is present in WT non-treated (NT) sperm. See also **Figure 10—Figure supplement 1B** suggesting that this signal is due to secondary Abs. (C4) The HA signal in WT sperm is not enhanced by sarkosyl treatment. Scale bars 10 µm.

11B2), centriole separation was visible in numerous spermatozoa (**Figure 11B3, B4**), with the structures located far away from the connecting piece (**Figure 11B5**) or duplicated (**Figure 11B6**). These defects were not observed in WT sperm, demonstrating that these defects were caused by the lack of CCDC146. Interestingly, the overall structure of the HTCA under construction in spermatids, observed by TEM, was conserved in *Ccdc146* KO spermatids, with the presence of both centrioles, containing nine triplets of microtubules, as well as accessory cytoskeletal structures, including the capitulum and the segmented columns (**Figure 12A1** for WT and A2 for *Ccdc146* KO). The adjunct of the proximal centriole was also normal in *Ccdc146* KO spermatids (**Figure 12B**) and no difference were observed with adjunct of WT spermatids (**Figure 12—Figure supplement 1**). Remarkably, in a very large proportion of sections, no singlet or doublet of microtubules emerged from the distal KO centriole, suggesting that the process of tubulin polymerization is somehow hampered in these cells (**Figure 12A2**). The absence of microtubules at the end of the distal centriole was confirmed by analysis of serial sections of the sperm centrioles (**Figure 12C**). Moreover, the defects observed in IF experiments, such as duplication, or defective attachment to the nuclear membrane, were frequently confirmed in TEM images (**Figure 12—Figure supplement 2**). Such defects were not observed in WT spermatids.

Manchette formation – which occurs from the round spermatid to fully-elongated spermatid stages – was next studied by IF using β-tubulin Ab. Simultaneously, we monitored formation of the acrosome using an antibody binding to DPY19L2 (**Figure 13**). Initial acrosome binding at step 3 spermatid was not disrupted (not shown), and we observed no differences in covering of the anterior portion of spermatid nucleus at steps 7-8 spermatids (**Figure 13A1/A2**). The development of the manchette in between steps 9-12 spermatids was not hampered, but noticeable defects appeared indicating defective manchette organization– such as a random orientation of the spermatids (**Figure 13B1/B2**) and abnormal acrosome shapes (**Figure 13C1/C2**). Finally, at steps 13-15 spermatids, the manchette was clearly longer and wider than in WT cells (**Figure 13D1/D2** and **Figure 13—Figure supplement 1A**), suggesting that the control of the manchette dynamic is defective: in KO spermatids, the reduction and disappearance of the manchette is hampered and contrary to WT continue to expand (**Figure 13— Figure supplement 1B**). In the mean time, the acrosome is strongly remodeled, as shown in **Figure 16H**, and is detached from the nucleus. This latter morphological defect is associated with a loss of the DPY19L2 staining (**Figure 13D2**). In TEM, microtubules of the manchette were clearly visible surrounding the compacting nucleus of elongating spermatids in both WT and KO. The manchette normally anchors on the perinuclear ring, which is itself localized just below the marginal edge of the acrosome and separated by the groove belt (**Figure 14 A-B** blue arrows) [52]. In KO spermatids, many defects such as marked asymmetry and enlargement were visible (**Figure 14C-F**). The perinuclear ring was no longer localized in the vicinity of the acrosome (**Figure 14 C-F** red arrows) and was often spread into the cytoplasm, providing a large nucleation structure for the manchette (**Figure 14 D-F**, black double arrows), which explains its width and irregularity.

Numerous defects were also observed in the axoneme. These axoneme defects were generally similar to the defects observed in other MMAF mouse models, with disorganization of the axoneme structure (**Figure 15A**) accompanied by fragmentation of the dense fibers and their random arrangement in cell masses anchored to the sperm head (**Figure 15B**). The absence of emerging singlet or doublet microtubules at the base of the distal centriole leads to a complete disorganization of the flagellum and the presence of notably dense fiber rings devoid of internal tubulin elements (**Figure 15BC**, red squares and enlargements). Typical cross sections of the midpiece and principal piece of WT sperm are shown in **Figure 15—Figure supplement 1**

Finally, the shape of the nucleus presented numerous abnormalities during elongation (**Figure 16**). The emergence of head defects was concomitant to the implantation of the manchette (**Figure 16E**, red arrow), with no defects observed on round spermatids at stages III-VIII. Unexpectedly, a distortion was created in the center of the nucleus at the anterior pole, causing the formation of a bilobed compacted nucleus (**Figure 16E-F**, red arrows) and likeky responsible of

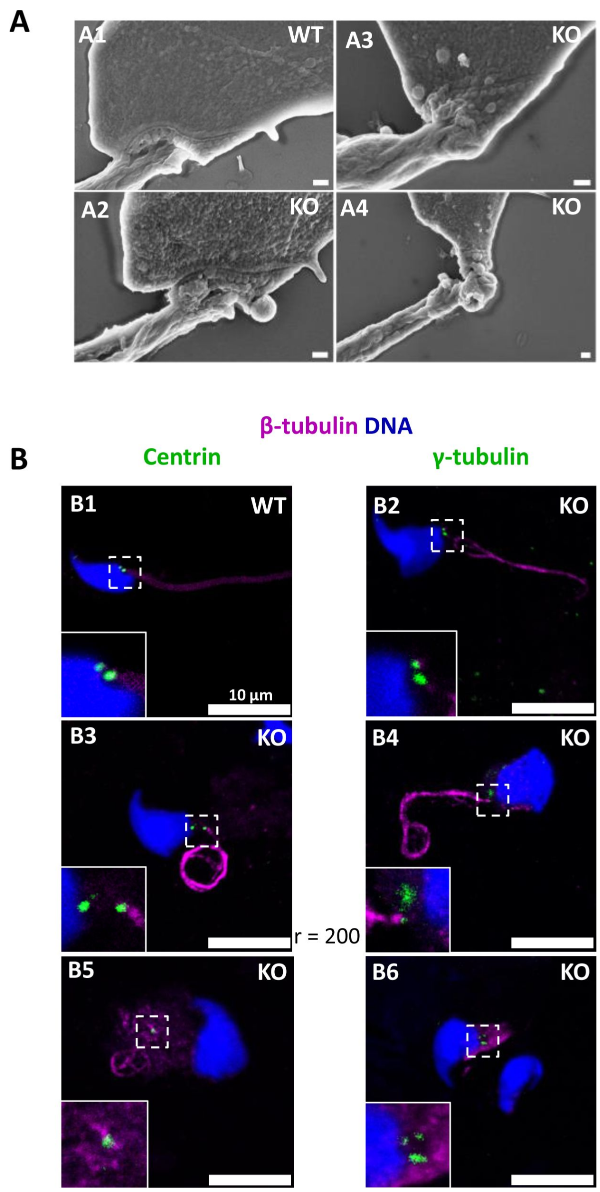

Figure 11

The absence of CCDC146 causes defects of the head-tail coupling apparatus in epididymal spermatozoa and duplication and mislocalization of centriole in testicular sperm.

(A) Scanning electron microscopy of WT and *Ccdc146* KO epididymal spermatozoa showed aberrant head morphologies and irregular head-tail coupling apparatus (HTCA) linking the sperm head with the flagellum. There is a great variability in the morphological damage, with sperm presenting almost intact HTCA (A2) whereas other were strongly impacted by the absence of the protein (A3-4). Scales bars 200 nm. (B) Testicular spermatozoa from WT (B1) and *Ccdc146* KO (B2-B6) mice immunolabeled with anti-β-tubulin (magenta) and anti-centrin (B1-B3) or anti-γ-tubulin (B4-B6) (green) Abs. Centrioles appeared to be normal (B2) in some spermatozoa, separated but partially attached to the head (B3, B4), completely detached from the sperm head (B5) or duplicated (B6). Scale bar 10 µm.

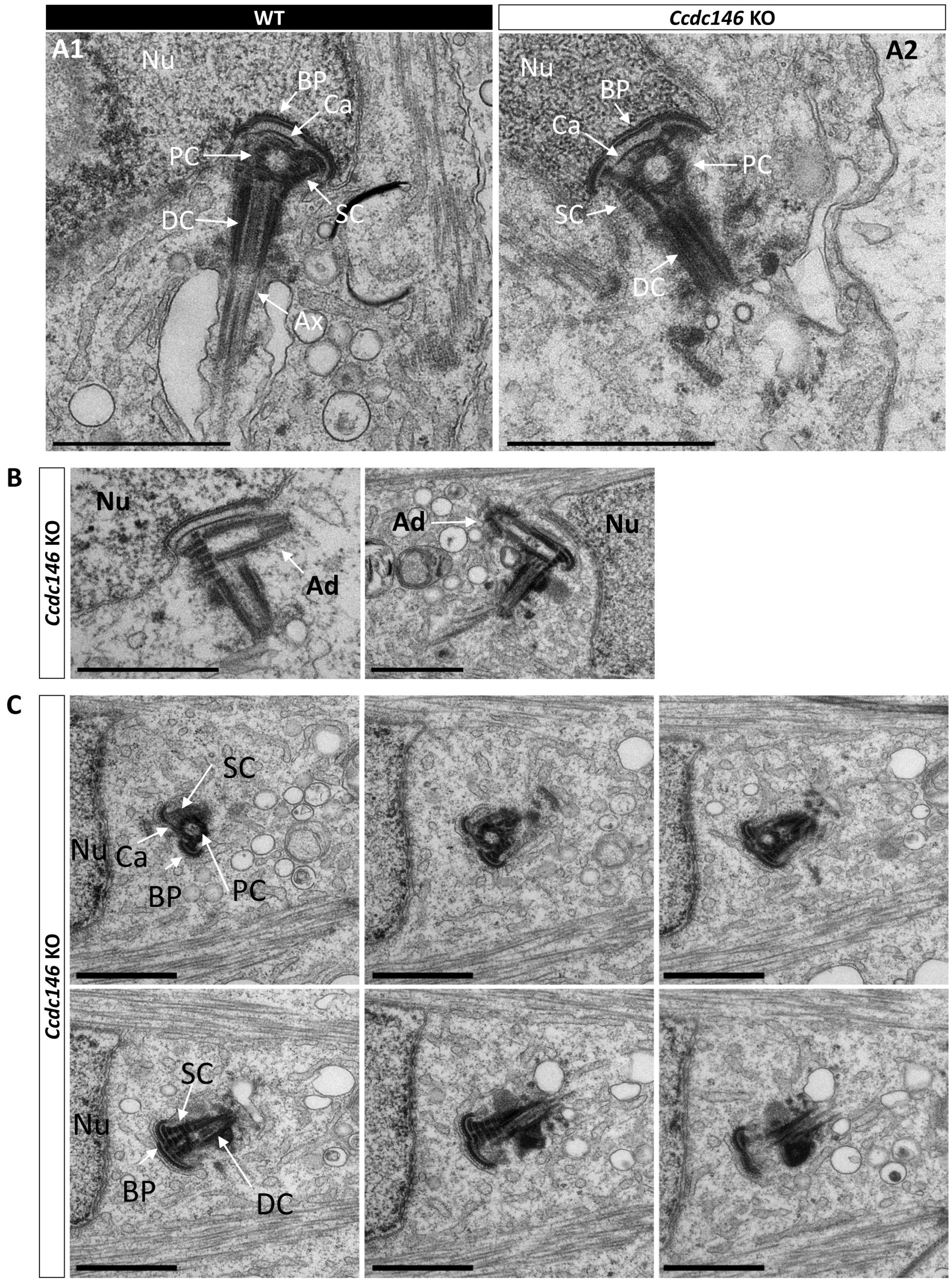

Figure 12

Absence of elongation of axonemal microtubules at the base of the distal centriole.

(A) In WT spermatids (A1), the proximal centriole (PC) is linked to the base of the compacting nucleus (Nu) through the basal plate (BP) and the capitulum (Ca), and the distal centriole (DC) is embedded in the segmented column (SC). All these sperm-specific cytoskeletal structures make up the HTCA. At the base of the distal centriole, axonemal microtubules (Ax) grow. In *Ccdc146* KO elongating spermatids (A2), the overall structure of the HTCA is conserved, with the presence of the centrioles and the accessory cytoskeletal structures. However, no axonemal microtubules are visible, emerging from the DC. (B) The adjunct (Ad) of the proximal centriole is also preserved in *Ccdc146* KO spermatids. (C) Serial sections of the HTCA of a *Ccdc146* KO spermatid confirm the absence of axonemal microtubules at the base of the DC during spermatid elongation. Scale bars 1 μm.

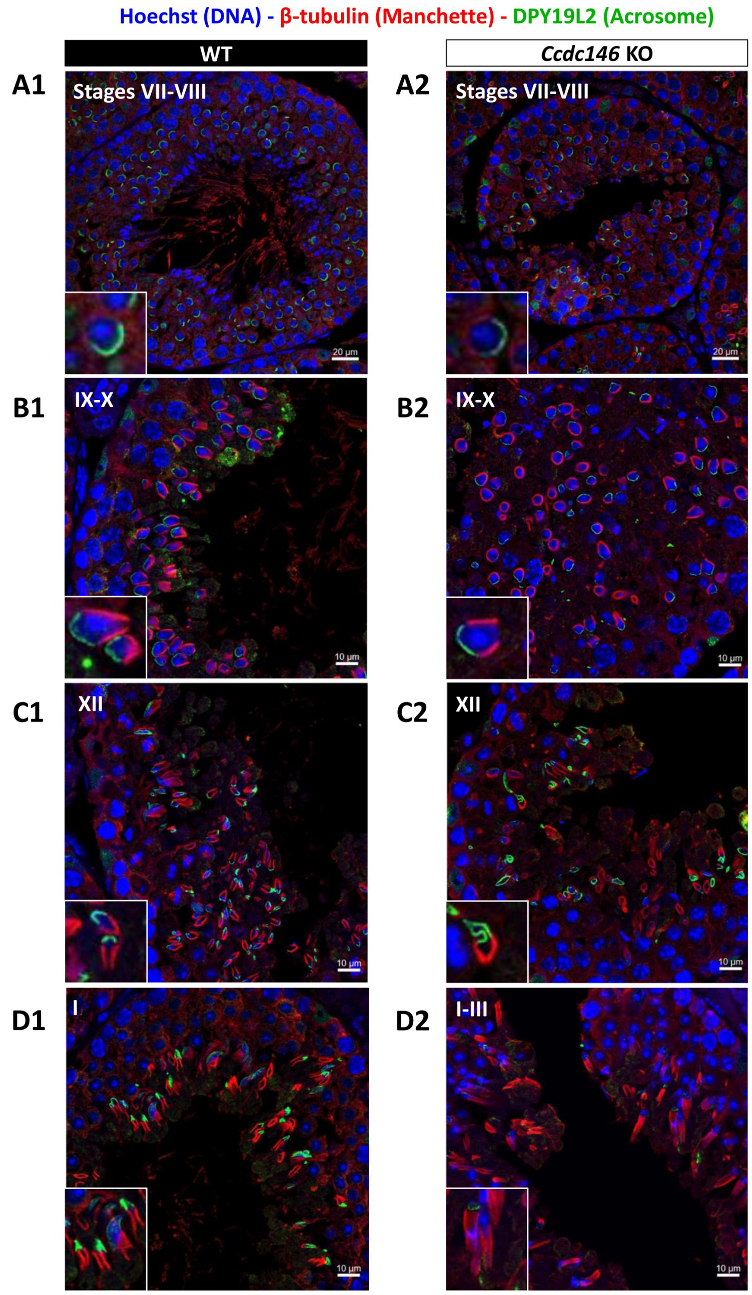

Figure 13

Analysis of stages of spermatogenesis by IF reveals acrosome formation and manchette elongation defects in *Ccdc146* KO spermatids.

Cross sections of WT (A1-D1) and *Ccdc146* KO (A2-D2) testes showing different stages of mouse spermatogenesis (I, VII-XII). Stages were determined by double immunostaining for β-tubulin (red; manchette elongation) and DPY19L2 (green; acrosome localization), and DNA was stained with Hoechst (blue). (A1/A2) At stage VII-VIII, acrosome spreading on round spermatids appeared similar between WT and KO. However, very few mature spermatozoa lined the lumen in the KO. (B1/B2) Cell orientations appeared random from stage IX-X in the KO and the tubules contained more advanced spermatid stages. Random orientations of cells are evidenced by manchette cross-sections in various planes. (C2) Abnormal acrosomes of elongating KO spermatids are observed by stage XII. (D1/D2) The manchette of elongated spermatids at stage I-III was longer in the KO compared with the WT. Insets in A1-D1 and A2-D2 show typical spermatids from stages I-XII from WT and KO males, respectively, showing details of acrosome formation (green) and manchette elongation (red). Scale bars stages VII-VIII 20 µm, scale bars IX-X, XII and I-III 10 µm.

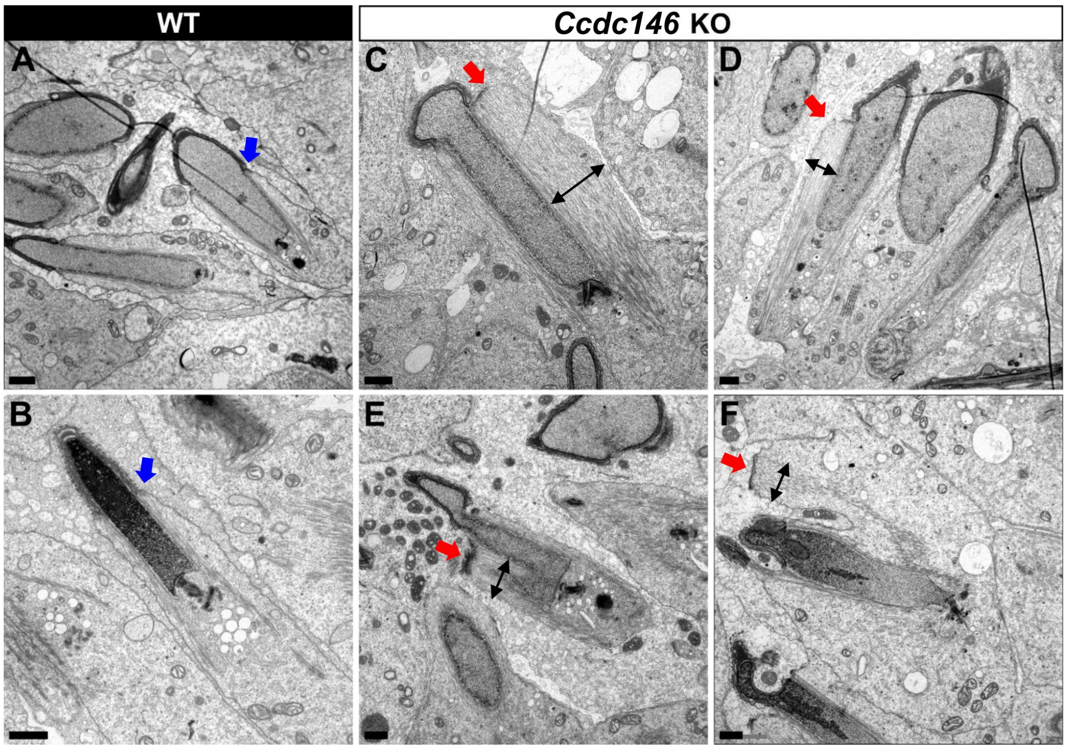

Figure 14

TEM of elongating spermatids from *Ccdc146* KO male shows ultrastructural defects of the manchette.

(A, B) Ultrastructural analysis of the manchette in WT elongating spermatids shows the normal thin perinuclear ring, anchored below the acrosome (blue arrows) and allowing a narrow array of microtubules to anchor. (C-F) In elongating spermatids from *Ccdc146* KO animals, the perinuclear ring was abnormally broad, usually located on one side of the spermatid (red arrows), creating an asymmetric and wide bundle of microtubules. The resulting manchette was wider and often longer than in WT animals (black double arrows). (F) The tubulin nucleation location was sometimes ectopic in the KO (red arrow) and coincided with irregularly shaped sperm heads. Scale bars 1µm.

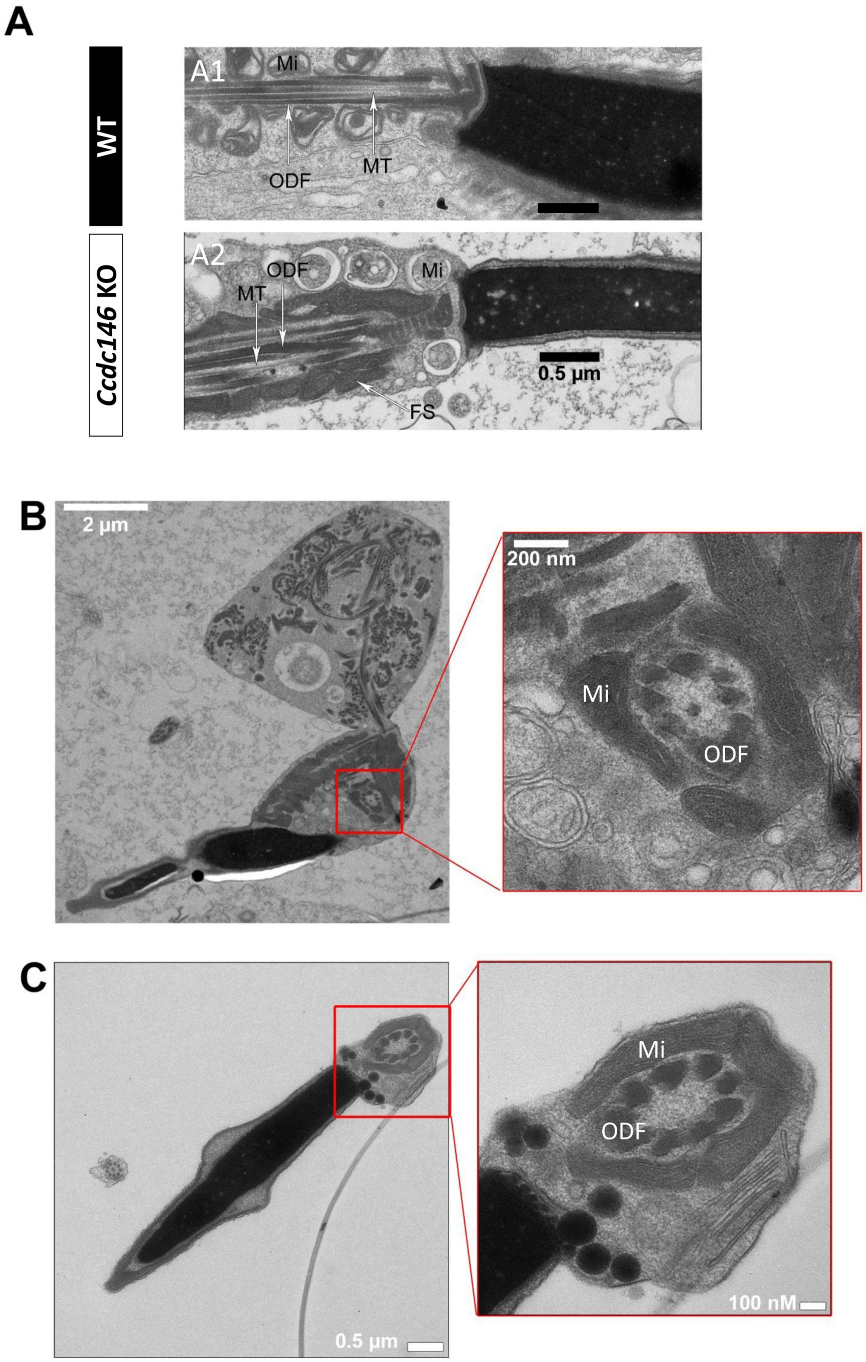

Figure 15

The axonemes of *Ccdc146* KO spermatids present multiple defects visible under TEM.

(A1) A longitudinal section of a WT flagellum shows a typical structure of the principal piece, with outer dense fibers (ODF) at the periphery, microtubules (MT) in the center, and mitochondria (Mi) aligned along the flagellum. (A2) A longitudinal section of a *Ccdc146* KO flagellum shows a disorganized midpiece, with altered mitochondria, the presence of an amorphous fibrous sheath (FS) and altered microtubules. (B) Longitudinal section of a *Ccdc146* KO sperm showing dispersed and non-assembled flagellar material in a cytoplasmic mass. The right-hand image is the enlargement of the red square, showing the presence of an external ring of mitochondria surrounding an outer dense fiber ring devoid of microtubular material. (C) Longitudinal section of another *Ccdc146* KO sperm showing a similar abnormal midpiece structure. The right-hand image is the enlargement of the red square, showing the presence of an external ring of mitochondria surrounding an outer dense fiber ring devoid of microtubular material. Scale bars as indicated.

the presence of visible vacuoles inside the nuclei (**Figure 16G-H**, white arrows). Moreover, acrosomes were severely impacted at steps13-15 spermatid, with strong deformations and detachment (**Figure 16H**, blue arrow heads).

Discussion

This study allowed us to identify and validate the involvement of a new candidate gene in male infertility, *CCDC146*. This gene was first described in mice by our team earlier this year, in the context of a research project on the different types of genetic causes of sperm abnormalities in mouse [53]. However, no data other than the sperm phenotype (MMAF) was presented in this initial report. In this paper, we present for the first time, the evidence of the presence of mutations in humans and a description of the localization of the protein in somatic and germ cells. We also examined the lesion spectrum at the subcellular level, to obtain a detailed view of the molecular pathogenesis associated with defects in this gene.

1/ Genetic complexity of MMAF

The question of how to interpret and annotate the mutations identified by high-throughput sequencing is an important issue in clinical genetics. The most difficult step is certainly the identification of genes associated with a given disease. Beyond pure genetic data and the pathogenicity of mutations, a fundamental element is the reproduction of the disease when the gene is absent in an animal model. When these two elements are combined – KO photocopying disease and pathogenicity of the identified mutations - the probability of mutation causality in the studied disease is very high. Both mutations are predicted to produce a premature stop codons: p.Arg362Ter and p.Arg704serfsTer7, leading either to the complete absence of the protein in case of non-sense mediated mRNA decay or to the production of a truncated protein missing almost two third or one fourth of the protein respectively. *CCDC146* is very well conserved throughout evolution (Figure supplementary 1), including the 3' end of the protein which contains a large coil-coil domain (**Figure 1**). In view of the very high degree of conservation, it is most likely that the 3' end of the protein, absent in both subjects, is critical for the CCDC146 function and hence that both mutations are deleterious. These results combined with the fact that the absence of CCDC146 is deleterious in mice and induces an infertility similar to that observed in the human, enable us to associate *CCDC146* with MMAF syndrome in humans.

It is essential to characterize all genes involved in male infertility. Therefore, the characterization of a new gene involved in MMAF syndrome is not just “one more gene”. First, it further confirms that MMAF syndrome is a heterogeneous recessive genetic disease, currently associated with defects in more than 50 genes. Second, it helps direct the diagnostic strategy to be directed during genome-wide searches. The type of profiling required for MMAF is very different from that performed for instance for cystic fibrosis, where one mutation is responsible for more than 50% of cases. Third, the discovery of all genes involved in MMAF is important in the context of oligogenic heterozygous inheritance of sperm abnormalities in human [53]. Indeed, a homozygous deleterious mutation is found in only half of all MMAF patients, suggesting that other modes of inheritance must be involved. Only by establishing an extended list of genes linked to MMAF will it be possible to determine whether oligogenic heterozygous inheritance is a relevant cause of this syndrome in humans. Finally, some of the genes involved in MMAF syndrome are also involved in ciliary diseases [7]. By further exploring MMAF-type infertility, we can hope to enhance our understanding of another underlying pathology. Although the vast majority of patients with PCD are diagnosed before 5 years of age [54], one study showed that in a cohort of 240 adults with a mean age of 36 years (36 ± 13) presenting with chronic productive cough and recurrent chest infections, PCD was identified for the first time in 10% of patients [55]. This result suggests that infertility diagnosis can occur before PCD diagnosis, and consequently that infertility management might improve PCD diagnosis and care.

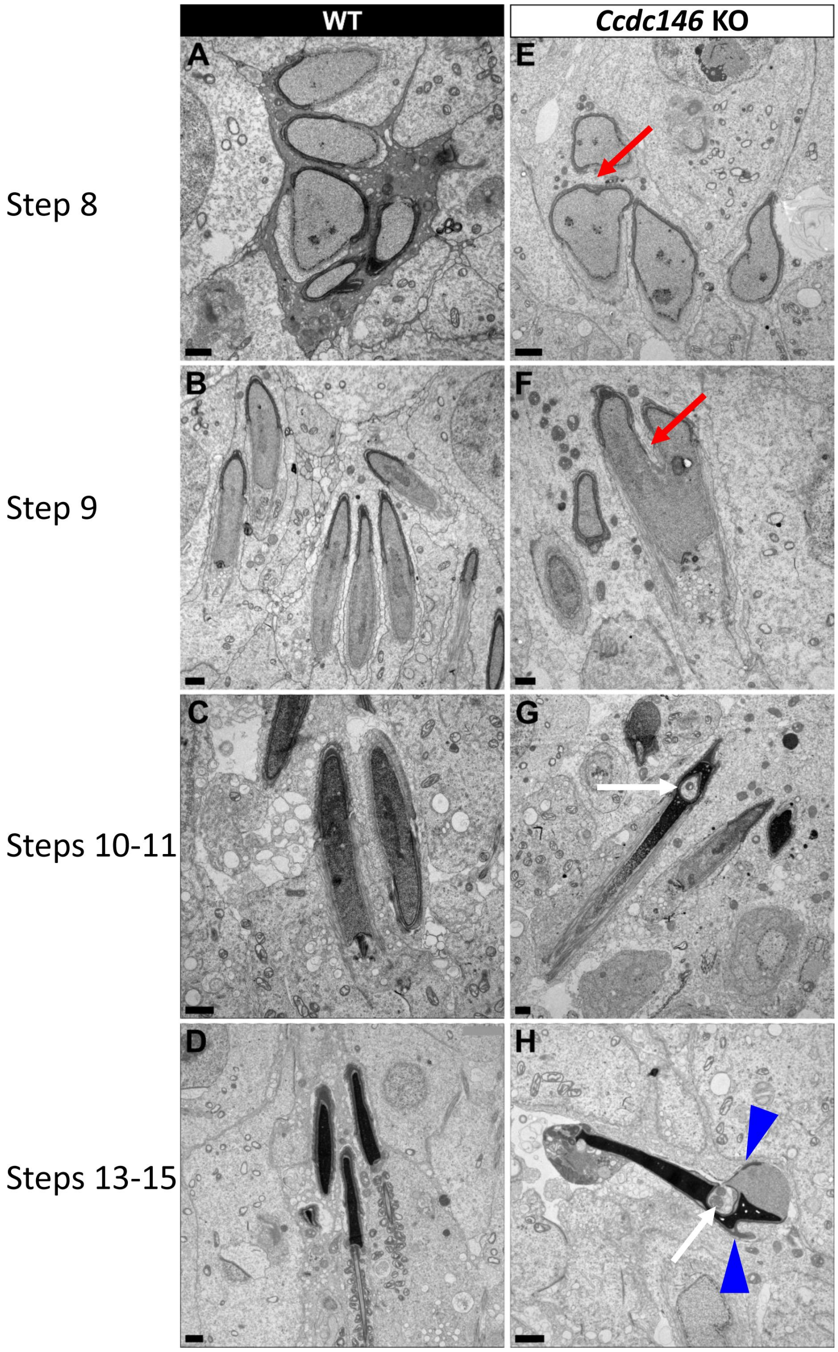

Figure 16.

Spermatid head shape is aberrant in the absence of CCDC146.

Comparative ultrastructural analysis of the spermatid head in WT (A-D) and *Ccdc146* KO (E-H) testis sections. (A, E) Spermatid nuclei at the beginning of elongation. KO spermatid nuclei showing nuclear membrane invaginations and irregular shape that were not present in the WT (red arrow). The acrosome of KO spermatids appeared intact. (B, F) Morphology of nuclei in elongating spermatids. Whereas nucleus elongation is symmetric in the WT, in the *Ccdc146* KO, more pronounced head invaginations are observed (red arrow). (C, G) Elongated spermatids. Although nuclear condensation appeared normal in both WT and KO nuclei, vacuolization is observed in KO nucleus (white arrow). (D, H) Elongated KO spermatids showed malformed elongated nuclear shapes with frequent invaginations (white arrow) and absence of flagella (H) compared to the WT (D). The acrosome of KO spermatids showed numerous defects such as detachment induced by swelling/bubbling of the plasma membrane (blue arrow heads). Scale bars 1 µm.

2/ CCDC146 is present in flagellum and absent of in sperm centriole

Several proteomics studies of the sperm centriole identified CCDC146 as a centrosome-associated protein in bovine sperm [28, 56], a species where centrioles are present in ejaculated sperm. Using a different methodological approach, based on IF with conventional and expansion microscopy, we observed no CCDC146 signal in centrioles from testicular mouse and ejaculated human sperm. The same result was obtained with two different types of antibodies: with human sperm, we used a commercial anti-CCDC146 Ab, whereas with mouse sperm we used several anti-HA antibodies. The same IF approach allowed us to clearly identify the sperm centrosome using a number of antibodies such as anti-POC5 and anti-beta tubulin, ruling out a possible failure of our IF protocol. It is worth noting that centrosomal proteins were isolated from whole flagella using several strategies including sequential use of multiple detergents, and that the centrosomal proteins were purified in the last fractions. These results indicate that CCDC146 is in fact barely soluble in conventional buffers, which could explain why it is co-purified with the centrosomal fraction in proteomics studies.

In epididymal mouse sperm, the biochemical analyses showed the presence of the protein in the flagellum only, and not in the head fractions: the detection of the band appeared very quickly at visualization and became very strong after few minutes, demonstrating that the protein is abundant in the flagella. It is important to note that epididymal sperm do not have centrioles and therefore this signal is not a centriolar signal. We also performed a statistical analysis showing that the immuno-staining observed in the principal piece of mouse sperm is very specific (**Figure 7B**). Altogether, these results demonstrate unequivocally the intracellular localization of CCDC146 in the flagellum of mouse sperm. In human, we obtained very similar results, with a flagellar localization. Using expansion microscopy, we show that the protein is associated with the microtubule doublets in both mouse sperm (**Figure 9E**) and human sperm (**Figure 9 B-D**).

Moreover, CCDC146 was solubilized with the sarkosyl detergent only, known to destabilize microtubules and generally used to identify microtubule inner proteins (MIP) and the CCDC146 fluorescent signal was enhanced when sperm were treated with this peculiar detergent. Expansion microscopy also revealed a strong signal in mouse sperm at the point of axoneme rupture. Taken together, these elements strongly suggest that CCDC146 is a microtubule associate protein (MAP) and may be a MIP. However more experiments need to be performed to validate the inner localization of CCDC146 because there are other proteins, located in the peri-axonemal structure of the axoneme, that are difficult to solubilize as well.

Finally, in human sperm, staining was observed on acrosome and in the midpiece. Staining on acrosome should always be taken with caution in sperm. Indeed, numerous glycosylated proteins are present at the surface of the plasma membrane regarding the outer acrosomal membrane for sperm attachment, and are responsible to numerous non-specific staining. Moreover this acrosomal staining was not observed in mouse sperm, strongly suggesting that it is not specific.

Concerning the staining in the midpiece observed in both conventional and Expansion microscopy, it also seems to be nonspecific and associated with secondary Abs. In **Figure 8—Figure supplement 1B** showing the localization of POC5, a very similar staining of the midpiece was also observed, although POC5 was never described to be present in the midpiece, therefore questioning the specificity of the signal observed with the anti-CCDC146 antibody in the midpiece. POC5 and CCDC146 staining experiments shared the same secondary Ab and the midpiece signal was likely due to it. For midpiece staining observed in **Figure 10C** and **Figure 10—Figure supplement 1A** by IF, it is worth to note that a similar staining was observed when secondary Ab were used alone (see **Figure 10—Figure supplement 1B**). The strong fluorescent signal is clearly due to secondary Abs, likely masking the punctuate staining of the axoneme.

3/ Functional complexity of CCDC146

The results presented in this report show striking differences in localization of CCDC146 between somatic and germ cells. Our results show that the protein displays several subcellular localizations, which vary according to the cell cycle, and the cell nature. In somatic cells, the protein is mostly associated with the centrosome and other microtubular structures – in particular the mitotic spindle – at the end of microtubules and at the level of the kinetochore and the midbody. These observations also identify CCDC146 as a MAP. In contrast, in spermatozoa, our results indicate that CCDC146 is an axonemal protein that seems to be associated with the microtubule doublets. Despite its broad cellular distribution, the association of CCDC146 with tubulin-dependent structures is remarkable. However, centrosomal and axonemal localizations in somatic and germ cells, respectively, have also been reported for CFAP58 [37, 57], thus the re-use of centrosomal proteins in the sperm flagellar axoneme is not unheard of. In addition, 80% of all proteins identified as centrosomal are found in multiple localizations (*https://www.proteinatlas.org/humanproteome/subcellular/centrosome*). The ability of a protein to home to several locations depending on its cellular environment has been widely described, in particular for MAP. The different localizations are linked to the presence of distinct binding sites on the protein. For example, MAP6 binds and stabilizes microtubules, through Mc modules, and associates with membranes and neuroreceptors through palmitoylated cysteines. MAP6 can also localize in the microtubule lumen, in its role as MIP, thanks to its Mn modules. Finally, in addition to its associations with subcellular compartments and receptors, the presence of proline-rich domains (PRD) in the MAP6 sequence, allows it to bind to SH3-domain-containing proteins, and thus triggering activation of signaling pathways [58]. Another example of a protein with multiple localizations is CFAP21, which is a MIP but also a cytosolic calcium sensor. In the latter capacity, CFAP21 modulates the interaction of STIM1 and ORAI1 upon depletion of calcium stores [59, 60]. These examples illustrate the complexity of function of some multiple-domain proteins.

The fact that CCDC146 can localize to multiple subcellular compartments suggests that it also contains several domains. Interestingly, a PF05557 motif (Pfam mitotic checkpoint protein, *https://www.ebi.ac.uk/interpro/protein/UniProt/E9Q9F7/*) has been identified in the mouse CCDC146 sequence between amino acids 130 and 162. Proteins belonging to the "mitotic spindle checkpoint" monitor correct attachment of the bipolar spindle to the kinetochores. The presence of this motif likely explains the ability of CCDC146 to localize to cell cycle-dependent subcellular compartments containing tubulin. However, the most important structural motifs identified in CCDC146 are the coiled-coil domains. Although coiled-coil domains play a structural role in a variety of protein interactions, their presence in CCDC146 remains mysterious, and how they contribute to its function remains to be elucidated. Nevertheless, this motif is compatible with a MIP function for this protein, since several MIP proteins including CCDC11 (FAP53), CCDC19 (FAP45), and CCDC173 (FAP210) are coiled-coil proteins [59]. It is worth noting that the ortholog of *CCDC146* in *Chlamydomonas*, *MBO2*, codes for a protein required for the beak-like projections of doublets 5 and 6, located inside the lumen of the tubule B [61], in the proximal part of the *Chlamydomonas* flagellum. Although no beak-like projections are present in the mammalian axoneme, the location inside the tubule seems to be evolutionarily conserved. A more recent study of MBO2 showed that the protein is also present all along the flagellum of *Chlamydomonas* and is tightly associated with microtubule doublets [62]. These observations support our results showing association of CCDC146 with this axonemal structure.

The results presented here also show a striking difference in the phenotype induced by the lack of the protein in somatic and male germ cells. This protein is essential for spermatogenesis, and its absence leads to immotile non-functional sperm and to complete infertility in both humans and mice. Conversely, both patients and *Ccdc146* KO mice seem to be healthy and present no other conditions such as primary ciliary dyskinesia (PCD). CCDC146, despite its wide expression profile in many tissues therefore seems to be dispensable except during spermatogenesis. Nevertheless, because this protein is localized on the tips of the spindle and may be involved in mitotic checkpoints, its absence may lead to late proliferative disorders. This hypothesis is supported by the fact that *CCDC146* is reported to be down-regulated in thyroid cancer [63]. The link between infertility and risk of cancer was recently underlined, with mutations found in genes like *FANCM* [64]. Therefore, it would be interesting to monitor aging in *Ccdc146* KO mice and to study their life expectancy and cancer rates compared to WT mice.

4/ TEM reveals that lack of CCDC146 severely impacts microtubule-based organelles

No morphological defects were observed before the elongating spermatid stage, and in round spermatids, the acrosome started to spread on the nucleus in a normal way (**Figure 13**). Morphological defects appeared clearly from the onset of spermatid elongation. This result indicates that the protein is only necessary for late spermiogenesis, from the phase corresponding to flagellum biogenesis. All the organelles composed mainly of tubulin were strongly affected by the absence of the protein. The manchette structure in elongating *Ccdc146* KO spermatids was asymmetric, abnormally broad, and ectopic, leading to the formation of aberrantly-shaped sperm heads. So far, manchette defects have been associated with defects in intra-flagellar transport (IFT), and intra-manchette transport (IMT) [65]. CCDC146 was only localized to the sperm axoneme by IF, and no signal was observed in the manchette, suggesting that CCDC146 is probably not involved in the transport machinery. Moreover, our results indicated that the manchette was remarkably long in elongated spermatids. A similar phenotype was observed in Katanin80-deficient animals [66]. Katanin80 is a microtubule-severing enzyme that is important for manchette reduction. Interestingly, the absence of WDR62, a scaffold protein involved in centriole duplication, leads to defective katanin80 expression, and the presence of elongated manchettes in mice [67]. In combination with our results, this detail suggests that the manchette's structure and its reduction are influenced by centrosomal proteins, possibly through katanin80 defects. The precise molecular link between CCDC146 and manchette assembly and reduction remains to be identified.

The HTCA was also aberrant in *Ccdc146* KO spermatids. Centrioles in elongating spermatids were frequently displaced from their implantation fossa at the nuclear envelope. We observed some correctly lodged centrioles in round spermatids; however, we are unable to determine with certainty whether the majority of centrioles failed to correctly attach or whether they detached from the nuclear envelope during spermatid elongation. Defects in cohesion of the HTCA have been associated with the acephalic spermatozoa syndrome, and were shown to involve a number of proteins such as SUN5, SPATA6 and ODF1 [4, 68-70]. Here, we did not observe any sperm decapitation, suggesting that CCDC146 is involved in a different pathway controlling the HTCA. Moreover, elongating *Ccdc146* KO spermatids displayed supernumerary centrioles. Abnormal centriole numbers have also been reported in the absence of a few other centrosome-associated proteins including DZIP1 [36] and CCDC42 [41], and of microtubule-regulating proteins such as katanin like-2 [71] and tubulin deglutamylase CCP5 [72].

Overall, these results suggest a relationship between the manchette and the centrioles. In the literature there are limited information about this relationship. Studies of the spermiogenesis defects observed in different models deficient for centrosomal proteins show some common features such as abnormal manchette and duplicated centrosomes. The absence of these proteins does not appear to be directly responsible for these defects, rather it seems to modify the expression of microtubule regulatory proteins such as katanins [66, 67, 71]. Moreover, there is a report showing that both organelles share molecular components [73].These modifications could explain the pleiotropic effect of the absence of CCDC146 on microtubule-based organelles.

In conclusion, by characterizing the genetic causes of human infertility, we not only improve the diagnosis and prognosis of these pathologies but also pave the way for the discovery of new players in spermatogenesis. We are constantly adding to the number of proteins present in the flagella of the mammalian spermatozoa that are necessary for its construction and functioning. This study showed that CCDC146, a protein previously described as a centrosomal protein in somatic and germ cells, localizes in spermatozoa's axonemal microtubule doublets. The presence of CCDC146 in somatic cells' centrosomes gives weight to the idea of a centrosome with a dynamic composition, allowing it to fulfill its multitude of functions throughout all the phases of cellular life.

Material and methods

Human subjects and controls

We analyzed WES data from a cohort of 167 MMAF individuals previously established by our team [9]. All individuals presented with a typical MMAF phenotype characterized by severe asthenozoospermia (total sperm motility below 10%) with at least three of the following flagellar abnormalities present in >5% of the spermatozoa: short, absent, coiled, bent or irregular flagella. All individuals had a normal somatic karyotype (46,XY) with normal bilateral testicular size, normal hormone levels and secondary sexual characteristics. Sperm analyses were carried out in the source laboratories during routine biological examination of the individuals according to World Health Organization (WHO) guidelines [74]. Informed and written consents were obtained from all the individuals participating in the study and institutional approval was given by the local medical ethical committee (CHU Grenoble Alpes institutional review board). Samples were stored in the Fertithèque collection declared to the French Ministry of health (DC-2015-2580) and the French Data Protection Authority (DR-2016-392).

Sanger sequencing

CCDC146 single nucleotide variants identified by exome sequencing were validated by Sanger sequencing as previously described [9]. PCR primers used for each individual are listed in Table supplementary 1.

Cell culture

HEK-293T (Human Embryonic Kidney) and HFF (Human Foreskin Fibroblasts) cells were grown in D10 medium consisting in DMEM with GlutaMAX (Dulbecco's Modified Eagle's Medium, Sigma Aldrich) supplemented with 10% heat-inactivated fetal bovine serum (FBS, Life Technologies) and 10% of penicillin-streptomycin (Sigma Aldrich) in a 5% CO2 humidified atmosphere at 37 °C. HEK-293T cells were divided twice weekly by 1/10 dilution. HFFs cells were divided 1/5 one time a week.

Ethics statement

Breeding and experimental procedures were carried out in accordance with national and international laws relating to laboratory animal welfare and experimentation (EEC Council Directive 2010/63/EU, September 2010). Experiments were performed under the supervision of C.L. (agreement 38 10 38) in the Plateforme de Haute Technologie Animale (PHTA) animal care facility (agreement C3851610006 delivered by the Direction Départementale de la Protection des Populations) and were approved by the ethics committee of the PHTA and by the French government (APAFlS#7128- 2016100609382341.v2).

Generation of *Ccdc146* KO and HA-tagged CCDC146 mice

Ccdc146 KO mice were generated using the CRISPR/Cas9 technology as previously described [53]. Briefly, to maximize the chances of generating deleterious mutations, two gRNAs located in two distinct coding exons located at the beginning of the targeted gene were used. For each gene, the two gRNAs (5'-CCT ACA GTT AAC ATT CGG G-3' and 5'-GGG AGT ACA ATA TTC AGT AC-3'), respectively targeting exons 2 and 4, were inserted into two distinct plasmids, each plasmid also contained the Cas9 sequence. The Cas9 gene was controlled by a CMV promoter and the gRNA and its RNA scaffold by a U6 promoter. Full plasmids (pSpCas9 BB-2A-GFP (PX458)) containing the specific sgRNA were ordered from Genescript (*https://www.genscript.com/crispr-synthetic-sgrna.html*). Both plasmids were co-injected into the zygotes' pronuclei at a concentration of 2.5 ng/mL. Plasmids were directly injected as delivered by the supplier, without in vitro production and purification of Cas9 proteins and sgRNA.

HA-Ccdc146 knock-in mice were also generated by CRISPR/Cas9. Twenty-seven nucleotides encoding the HA (hemagglutinin) tag (5'-TAC CCA TAC GAT GTT CCA GAT TAC GCT TAG-3') were inserted immediately after the start codon of *Ccdc146*. One plasmid containing one sgRNA (5'-TAC TTT AGA ACT GTG AAA AA-3') and Cas9 was injected (5 ng/µL) with a single-stranded DNA (150 nucleotides, 50 ng/µl) as a template for the homology directed repair (HDR) **Figure 6—Figure supplement 1**. PCR primers used for genotyping are listed in Table supplementary 1. Genetically modified *Ccdc146* strains were bred in the Grenoble university animal platform (HTAG) and housed under specific-pathogen-free conditions. Animals were euthanized by cervical dislocation at the indicated ages.

Phenotypic analysis of *Ccdc146* KO mice

Fertility test – Three adult males of each genotype were housed individually with two fertile WT B6D2 females for 12 weeks. The date of birth and the number of pups were recorded.

Sperm analysis – Epididymal sperm were obtained by making small incisions in the mouse caudae epididymides placed in 1 mL of warm M2 medium (Sigma Aldrich), and the sperm were allowed to swim up for 10 min at 37 °C. Sperm samples (10 µL) were used for Computer-assisted semen analysis (CASA, Hamilton Thorn Research, Beverley, MA, USA) using a 100-µm-deep analysis chamber (Leja Products B.V., Nieuw-Vennep, the Netherlands). A minimum of 100 motile sperm was recorded in each assay. The remaining sperm samples were washed in 1X phosphate buffered saline (PBS, Life technologies), 10 µL were spread onto slides pre-coated with 0.1% poly-L-lysine (Epredia), fixed in 70% ethanol (Sigma Aldrich) for 1 h at room temperature (RT) and submitted to a papanicolaou staining (WHO laboratory manual) to assess sperm morphology. Images were obtained using a Zeiss AxioImager M2 fitted with a 40X objective (color camera AxioCam MRc) and analyzed using ZEN (Carl Zeiss, version 3.4).

Testis and epididymides histology

Testis and epididymides samples from 8-16-week-old mice were fixed for 24 h in PBS/4% PFA (Electron Microscopy Sciences), dehydrated in a graded ethanol series, embedded in paraffin wax, sectioned at 5 µm and placed onto Superfrost slides (Fischer scientific). For both, slides were deparaffinated and rehydrated prior to use. Tissue morphology and structure were observed after coloration by Mayer's hematoxylin and eosin phloxine B (WHO protocols) using a Zeiss AxioImager M2 (color camera AxioCam MRc).

Terminal deoxynucleotidyl transferase dUTP nick-end labeling (TUNEL) assay on testes

Testes samples from three adult individuals for each genotype were analyzed. Apoptotic cells in testis sections were identified using the Click-iT™ Plus TUNEL Assay kit (Invitrogen) in line with the manufacturer's instructions. DNA strand breaks for the positive control were induced by DNase I treatment. Each slide contained up to eight testis sections. DNA was stained with Hoechst (2 µg/mL). Images were acquired and reconstituted using a Zeiss Axioscan Z1 slide scanner and analyzed by Fiji [75]. The total number of seminiferous tubules and the number of tubules containing at least one TUNEL-positive cell were counted in each testis section.

Conventional Immunofluorescence

(IF) Somatic Cells – HEK-293T or HFFs (10 000 cells) were grown on 10-mm coverslips previously coated with poly-D-lysine (0.1 mg/mL, 1 h, 37 °C, Gibco) placed in a well on a 24-well plate. For cell synchronization experiments, cells underwent S-phase blockade with thymidine (5 mM, Sigma-Aldrich) for 17 h followed by incubation in a control culture medium for 5 h, then a second blockade at the G2- M transition with nocodazole (200 nM, Sigma-Aldrich) for 12 h. Cells were then fixed with cold methanol (Sigma Aldrich) for 10 min at different times for IF labelling. After washing twice in PBS, non-specific sites were blocked with PBS/5% FBS/5% NGS (normal goat serum, Life technologies) for 1 h at RT. After washing three times in PBS/1% FBS, primary antibody was added in PBS/1% FBS and incubated overnight at 4°C. Coverslips were washed three times in PBS/1% FBS before adding secondary antibody in PBS/1% FBS and incubating for 2 h at RT. After washing three times in PBS, nuclei were stained with 2 µg/mL Hoechst 33342 in PBS (Sigma Aldrich). Coverslips were once again washed three times in PBS, then carefully placed on Superfrost slides (cells facing the slide) and sealed with nail polish.

Spermatogenic cells – Seminiferous tubules were isolated from mouse testes (8–16 weeks old). After removing of the tunica albuginea, the testes were incubated at 37 °C for 1 h in 3 mL of a solution containing (mM): NaCl (150), KCl (5), CaCl2 (2), MgCl2 (1), NaH2PO4 (1), NaHCO3 (12), D-glucose (11), Na-lactate (6), HEPES (10) pH 7.4, and collagenase type IA (1 mg/mL – Sigma Aldrich). Tubules were rinsed twice in collagenase-free medium and cut into 2-mm sections. Spermatogenic cells were obtained by manual trituration and filtered through a 100-µm filter. The isolated cells were centrifuged (10 min, 500 g), resuspended in 500 µL PBS and 50 µL was spread onto slides pre-coated with 0.1% poly-L-lysine (Epredia) and allowed to dry. Dried samples were fixed for 5 min in PBS/4% PFA. After washing twice in PBS, slides were placed in PBS/0.1% Triton/5% BSA (Euromedex) for 90 min at RT. Following two washes in PBS, the primary antibody was added in PBS/1% BSA overnight at 4 °C. After washing three times in PBS/1% BSA, secondary antibody was added in PBS/1% BSA for 2 h at RT. After washing three times in PBS, nuclei were stained with 2 µg/mL Hoechst 33342 in PBS (Sigma Aldrich) and slides were mounted with DAKO mounting media (Agilent).

Testis sections – After deparaffination and rehydration, testis sections were subjected to heat antigen retrieval for 20 min at 95 °C in a citrate-based solution at pH 6.0 (VectorLabs). Tissues were then permeabilized in PBS/0.1% Triton X-100 for 20 min at RT. After washing three times in PBS (5 min each), slides were incubated with blocking solution (PBS/10% BSA) for 30 min at RT. Following three washes in PBS, slides were incubated with primary antibodies in PBS/0.1% Tween/5% BSA overnight at 4 °C. Slides were washed three times in PBS before applying secondary antibodies in blocking solution and incubating for 2 h at RT. After washing in PBS, nuclei were stained with 2 µg/mL Hoechst 33342 in PBS (Sigma Aldrich) for 5 min at RT. Slides were washed once again in PBS before mounting with DAKO mounting media (Agilent).

Spermatozoa - Mouse spermatozoa were recovered from the caudae epididymides. After their incision, sperm were allowed to swim in 1 mL of PBS for 10 min at 37 °C. They were washed twice with 1 mL of PBS 1X at 500 g for 5 min, and 10 µL of each sample was smeared onto slides pre-coated with 0.1% poly-L-lysine (Epredia). Sperm were fixed in PBS/4% PFA for 45 s, washed twice in PBS and permeabilized in PBS/0.1% Triton for 15 min at RT. After incubation in PBS/0.1% Triton/2% NGS for 2 h at RT, primary antibody was added in PBS/0.1% Triton/2% NGS overnight at 4 °C. After washing three times in PBS/0.1% Triton, the secondary antibody was applied in PBS/0.1% Triton/2% NGS for 90 min at RT. Slides were washed three times in PBS before staining nuclei with 2 µg/mL Hoechst 33342 in PBS (Sigma Aldrich). Slides were washed once in PBS before mounting with DAKO mounting media (Agilent).

For human spermatozoa, the protocol was based on that of Fishman *et al.* [32]. Straws were thawed at RT for 10 min and resuspended in PBS. Sperm were washed twice in 1 mL of PBS (10 min, 400 g), 50 µL was spread onto slides pre-coated with 0.1% poly-L-lysine and left to dry. Dry slides were fixed in 100% ice-cold methanol for 2 min and washed twice in PBS. Cells were permeabilized in PBS/3% Triton X-100 (PBS-Tx) for 1 h at RT. Slides were then placed in PBS-Tx/1% BSA (PBS-Tx-B) for 30 min at RT. Sperm were then incubated with primary antibodies in PBS-Tx-B overnight at 4 °C. After washing three times in PBS-Tx-B for 5 min each, slides were incubated with secondary antibodies for 1 h at RT in PBS-Tx-B. After washing three times in PBS, nuclei were stained with 2 µg/mL Hoechst 33342 in PBS (Sigma Aldrich) and slides were mounted with DAKO mounting media (Agilent).

Sarkosyl protocol for mouse sperm - Mouse sperm from caudae epididymides were collected in 1 mL PBS 1X, washed by centrifugation for 5 min at 500 g and then resuspended in 1 mL PBS 1X. Sperm cells were spread onto slides pre-coated with 0.1% poly-L-lysine (Epredia), treated or not for 5 min with 0.2% sarkosyl (Sigma) in Tris-HCl 1 mM, pH7.5 at RT and then fixed in PBS/4% PFA for 45 s at RT. After washing twice for 5 min with PBS, sperm were permeabilized with PBS/0.1% Triton X-100 (Sigma Aldrich) for 15 min at RT and unspecific sites were blocked with PBS/0.1% Triton X-100/2% NGS for 30 min at RT. Then, sperm were incubated overnight at 4 °C with primary antibodies diluted in PBS/0.1% Triton X-100/2% NGS. Slides were washed three times with PBS/0.1% Triton X-100 before incubating with secondary antibody diluted in PBS/0.1% Triton X-100/2% NGS for 90 min at RT. Finally, sperm were washed three times in PBS/0.1% Triton X-100, adding 2 µg/mL Hoechst 33342 (Sigma Aldrich) during the last wash to counterstain nuclei. Slides were mounted with DAKO mounting media (Agilent).

Image acquisition – For all immunofluorescence experiments, images were acquired using 63X oil objectives on a multimodal confocal Zeiss LSM 710 or Zeiss AxioObserver Z1 equipped with ApoTome and AxioCam MRm or NIKON eclipse A1R/Ti2. Images were processed using Fiji [75] and Zeiss ZEN (Carl Zeiss, version 3.4). Figures for cultured cells were arranged using QuickFigures [76].

Image analysis – To quantify the number of spots in the flagella the brightfield and fluorescent images were preprocessed with a home-made macro in ImageJ (*https://imagej.net/ij/*) to improve the contrast and decrease noise. Masks of whole cells, midpieces and head were then realised using Ilastik [77] on brightfield and fluorescent images. The masks of midpieces and head were then subtracted from the mask of the whole cell to obtain a mask of the flagella. This mask was used as an ROI to quantify intensity maxima in the flagella of control and CCDC146 expressing cells. To compare the number of spots in images with a random distribution, we quantified the total number of intensity maxima in the fluorescent image and a new image was generated with a random distribution of the same number of spots. Subsequently, the same quantification of intensity maxima in the flagella was carried out on the random images. A second home-made macro in ImageJ allows to automatize these two steps. Both macros are available on demand.

Expansion microscopy

(U-ExM) Coverslips used for either sample loading (12 mm) or image acquisition (24 mm) were first washed with absolute ethanol and dried. They were then coated with poly-D-lysine (0.1 mg/mL) for 1 h at 37 °C and washed three times with ddH2O before use. Sperm from cauda epididymides or human sperm from straws were washed twice in PBS. $1{\times}10^6$ sperm cells were spun onto 12-mm coverslips for 3 min at 300 g. Crosslinking was performed in 1 mL of PBS/1.4% formaldehyde/2% acrylamide (ThermoFisher) for 5 h at 37 °C in a wet incubator. Cells were embedded in a gel by placing a 35-µL drop of a monomer solution consisting of PBS/19% sodium acrylate/10% acrylamide/0.1% N,N'- methylenbisacrylamide/0.2% TEMED/0.2% APS (ThermoFisher) on parafilm and carefully placing coverslips on the drop, with sperm facing the gelling solution. Gelation proceeded in two steps: 5 min on ice followed by 1 h at 37 °C. Coverslips with attached gels were transferred into a 6-well plate for incubation in 5 mL of denaturation buffer (200 mM SDS, 200 mM NaCl, 50 mM Tris in ddH2O, pH 9) for 20 min at RT until gels detached. Then, gels were transferred to a 1.5-mL microtube filled with fresh denaturation buffer and incubated for 90 min at 95 °C. Gels were carefully removed with tweezers and placed in beakers filled with 10 mL ddH2O to cause expansion. The water was exchanged at least twice every 30 min. Finally, gels were incubated in 10 mL of ddH2O overnight at RT. The following day, a 5- mm piece of gel was cut out with a punch. To remove excess water, gels were placed in 10 mL PBS for 15 min, the buffer was changed, and incubation repeated once. Subsequently, gels were transferred to a 24-well plate and incubated with 300 µL of primary antibody diluted in PBS/2% BSA at 37 °C for 3 h with vigorous shaking. After three washes for 10 min in PBS/0.1% Tween 20 (PBS-T) under agitation, gels were incubated with 300 µL of secondary antibody in PBS/2% BSA at 37 °C for 3 h with vigorous shaking. Finally, gels were washed three times in PBS-T for 10 min with agitation. Hoechst 33342 (2 µg/mL) was added during the last wash. The expansion resolution was between 4X and 4.2X depending on sodium acrylate purity. For image acquisition, gels were placed in beakers filled with 10 mL ddH2O. Water was exchanged at least twice every 30 min, and then expanded gels were mounted on 24-mm round poly-D-lysine-coated coverslips, placed in a 36-mm metallic chamber for imaging. Confocal microscopy was performed using either a Zeiss LSM 710 using a 63x oil objective or widefield was performed using a Leica THUNDER widefield fluorescence microscope, using a 63x oil objective and small volume computational clearing.

Ccdc146 expression

Testis from HA-CCDC146 pups were collected at days 9, 18, 26 and 35 after birth and directly cryopreserved at −80 °C before RNA extraction (n=3 for each day). RNA was extracted as follow. Frozen testes were placed in RLT buffer (Qiagen)/1% β-mercaptoethanol (Sigma), cut in small pieces and lysis performed for 30 min at RT. After addition of 10 volumes of TRIzol (5 min, RT, ThermoFisher) and 1 volume of chloroform (2 min, RT, Sigma Aldrich), the aqueous phase was recovered after centrifugation at 12 000 g, 15 min, 4 °C. RNA was precipitated by the addition of one volume of isopropanol (Sigma) and of glycogen (20 mg/mL, ThermoFisher) as a carrier, tube was placed overnight at −20 °C. The day after, after centrifugation (15 min, 12 000 g, 4 °C), RNA pellet was washed with ethanol 80%, air-dried and resuspended in 30 µl of ultrapure RNAse free water (Gibco). RNA concentrations were determined by using the Qubit RNA assay kit (ThermoFisher). 800 ng of total RNA were used to perform the RT step using the iScript cDNA synthesis kit (Bio-Rad) in a total volume of 20 µL. Gene expression was assessed by qPCR (1 µL of undiluted cDNA in a final volume of 20 µL with the appropriate amount of primers, see table 1) using the SsoAdvanced Universal SYBR Green Supermix (Bio-Rad). The qPCR program used was 94 °C 15 min, (94 °C 30 s, 58 °C 30 s, 72 °C 30 s) x40 followed by a melt curve analysis (58 °C 0.05 s, 58–95 °C 0.5-°C increment 2–5 s/step). Gene expression was calculated using the $2^{-\Delta CT}$ method. Results are expressed relative to *Ccdc146* expression on day 9.

	Forward primer		Reverse primer	
Genes	Sequence	Concentration (nM)	Sequence	Concentration (nM)
Ccdc146	5’-TGCTGCATGACGCCGTGATG-3’	750	5’-GGAGACCTCCGTGGAGAATGCTTC-3’	500
Hprt	5’-CCTAATCATTATGCCGAGGATTTGG-3’	500	5’-TCCCATCTCCTTCATGACATCTCG-3’	250
Actb	5’-CTTCTTTGCAGCTCCTTCGTTGC-3’	250	5’-AGCCGTTGTCGACGACCAGC-3’	250

CCDC146 detection by WB

Protein extraction from whole sperm head and sperm flagella – Spermatozoa from HA- CCDC146 males (9 weeks old) were isolated from both epididymis in 1 mL PBS and washed twice with PBS by centrifugation at RT (5 min, 500 g). Half of the sperm were incubated with 1X protease inhibitor cocktail (mini Complete EDTA-free tablet, Roche Diagnostic), incubated for 15 min on ice and sonicated. Separated flagella were isolated by centrifugation (600 g, 20 min, 4 °C) in a Percoll gradient. Percoll concentrations used were 100%, 80%, 60%, 34%, 26%, 23%, and sperm flagella were isolated from the 60% fraction and heads form the 100% fraction. Samples were washed with PBS by centrifugation (500 g, 10 min, 4 °C). Whole sperm or sperm flagella were incubated in 2X Laemmli buffer (Bio-Rad), heated at 95 °C for 10 min and centrifuged (15 000 g, 10 min, 4 °C). The supernatants were incubated with 5% β-mercaptoethanol, boiled (95 °C, 10 min), cooled down and placed at −20 °C until use.

Proteins solubilization from total sperm – HA-CCDC146 sperm were recovered from caudae epididymides in 1 mL PBS and washed twice in 1 mL PBS by centrifugation 500 g, 5 min. Lysis was then performed for 2 h at 4 °C on wheel in either Chaps buffer (10 mM Chaps / 10 mM HEPES / 137 mM NaCl / 10% Glycerol), in RIPA Buffer (Pierce IP Lysis buffer, ThermoFisher), in Tris 10 mM / HCl 1 M buffer or in Tris 10 mM / HCl 1 M / 0.2 to 0.8% sarkosyl buffer. After centrifugation 15 000 g, 4 °C, 15 min, the supernatants were recovered and 5% of β-mercaptoethanol added. After boiling (95 °C, 10 min), the samples were cooled down and placed at −20 °C until use.

Western blot – The different protein lysates were fractionated on 5-12% SDS-PAGE precast gels (Bio-Rad) and transferred onto Trans-Blot Turbo Mini 0.2 µm PVDF membranes using the Trans-Blot Turbo Transfer System (Bio-Rad) and the appropriate program. Membranes were then blocked in PBS/5% milk/0.1% Tween 20 (PBS-T) for 2 h at RT before incubating with the primary antibody in PBS- T overnight at 4 °C with agitation. Membranes were then washed three times for 5 min in PBS-T and incubated with the secondary HRP-antibody in PBS-T for 1 h at RT. After three washes (PBS-T, 10 min), the membrane was revealed by chemiluminescence using the Clarity ECL substrate (Bio-Rad) and images were acquired on a Chemidoc apparatus (Bio-Rad).

Transmission electron microscopy

Testis from adult mice were fixed in PBS/4% paraformaldehyde (PFA). They were then decapsulated from the tunica albuginea and the seminiferous tubules were divided into three to four pieces using a razor blade (Gillette Super Sliver). The seminiferous tubules were incubated for 60 min at RT in fixation buffer (100 mM HEPES pH7.4, 4 mM CaCl2, 2.5% glutaraldehyde, 2% PFA, all from Sigma Aldrich) and then the buffer was exchanged with fresh fixation buffer and the samples left overnight fixed at 4 °C. After washing three times for 10 min in 100 mM HEPES pH 7.4, 4 mM CaCl2, samples were post-fixed in 1% osmium tetroxide (Carl Roth, Karlsruhe, DE) in distilled water for 120 min at 4 °C. After three additional 10-min washes in distilled water, the tissue pieces were embedded in 1.5% DifcoTM Agar noble (Becton, Dickinson and Company, Sparks, MD, US) and dehydrated using increasing concentrations of ethanol. The samples were then embedded in glycidyl ether 100 (formerly Epon 812; Serva, Heidelberg, Germany) using propylene oxide as an intermediate solvent according to the standard procedure. Ultrathin sections (60-80 nm) were cut with a diamond knife (type ultra 35°; Diatome, Biel, CH) on the EM UC6 ultramicrotome (Leica Microsystems, Wetzlar, Germany) and mounted on single-slot pioloform-coated copper grids (Plano, Wetzlar, Germany). Finally, sections were stained with uranyl acetate and lead citrate [78]. The sectioned and contrasted samples were analyzed under a JEM-2100 transmission electron microscope (JEOL, Tokyo, JP) at an acceleration voltage of 80 kV. Images were acquired using a 4080 x 4080 charge-coupled device camera (UltraScan 4000 - Gatan, Pleasanton, CA, US) and Gatan Digital Micrograph software. The brightness and contrast of images were adjusted using the ImageJ program.

Scanning electron microscopy

The two epididymides of mature males were recovered in 1 mL of 0.1 M sodium cacodylate buffer (pH 7.4, Electron Microscopy Sciences) and the sperm were allowed to swim for 15 min at 37 °C. After centrifugation for 10 min, 400 g, RT, the supernatant was discarded and the pellet resuspended in primary fixating buffer (2% glutaraldehyde/0.1 M sodium cacodylate buffer, pH 7.4, Electron Microscopy Sciences) for 30 min at 4 °C. After washing three times in 0.1 M sodium cacodylate buffer (400 g, 10 min), the pellet was submitted to post fixation using 1% Osmium tetroxide 2% (OsO_4, Electron Microscopy Sciences) in 0.1 M sodium cacodylate buffer for 30 min at 4 °C. Fixed cells were washed three times in 0.1 M sodium cacodylate buffer (400 g, 5 min), the sample was then placed on a coverslip and treated with Alcian blue 1% (Electron Microscopy Sciences) to improve attachment. The sample was then dehydrated in graded ethanol series: 50%, 70%, 80%, 90%, 96% and 100% (10 min, once each). Final dehydration was performed for 10 min in a v/v solution of 100% ethanol/100% Hexamethyldisilazane (HMDS) followed by 10 min in 100% HMDS. Samples were left to dry overnight before performing metallization. Samples were analyzed using a Zeiss Ultra 55 microscope at the C.M.T.C. – Consortium des Moyens Technologiques Communs (Material characterization platform), Grenoble INP.

Statistical analysis

Statistical differences were assessed by applying unpaired t tests, Mann-Whitney tests or one-way ANOVA using GraphPad Prism 8 and 9. Histograms show mean ± standard deviation, and p-values were considered significant when inferior to 0.05.

Antibodies used

Primary antibodies				
Target	**Host species**	**Reference**		**Dilution**
CCDC146	Rabbit	Atlas Antibodies	HPA020082	IF: 1/200 U-ExM: 1/200
Centrin, 20H5	Mouse	Merck	04-1624	IF: 1/200
γ-tubulin	Mouse	Santa Cruz Biotechnology	sc-17787	IF: 1/500
β-tubulin	Guinea pig	Geneva Antibody Facility	AA344-GP	IF: 1/500
β-tubulin	Rabbit	Cell Signaling Technology	2128	IF: 1/100
PCM1 (G-6)	Mouse	Santa Cruz Biotechnology	sc-398365	IF: 1/200
α-tubulin β-tubulin *	Mouse	Geneva Antibody Facility	AA-345 AA-344	U-ExM: 1/250 IF: 1/500
POC5	Rabbit	Bethyl	A303-341A	IF: 1/250 U-ExM: 1/200
High Affinity (HA)	Rat	Roche	11867423001	IF: 1/400 U-ExM: 1/400 WB: 1/2500
High Affinity (HA)	Rabbit	Cell Signaling Technology	3724	U-ExM: 1/100
High Affinity (HA)	Rabbit	Sigma Aldrich	H6908	U-ExM: 1/200
HA Tag Alexa Fluor®-488-conjugated Antibody		Cell Signalling	28427	IF:1/400
Dpy19L2	Rabbit	**		IF: 1/100

Target	Fluorophore	Reference		Dilution
Goat anti-Rabbit	Alexa Fluor 488	Jackson ImmunoResearch	111-545-144	IF: 1/800 U-ExM: 1/250
Goat anti-Mouse	DyLight 549	Jackson ImmunoResearch	115-505-062	IF: 1/400
Goat anti-Guinea Pig	Alexa Fluor 647	Invitrogen	A-21450	IF: 1/800
Goat anti-Rabbit	Alexa Fluor 568	Life Technologies	A11036	U-ExM: 1/250
Goat anti-Mouse	Alexa Fluor 488	Life Technologies	A11029	U-ExM: 1/250
Goat anti-Rat	Alexa Fluor 549	Jackson ImmunoResearch	112-505-175	IF: 1/800
Goat Anti-Rat	HRP conjugate	Merck	AP136P	1/10 000

*α-tubulin and β-tubulin were used together and noted as α+β-tubulin

**Dpy19l2 antibodies are polyclonal antibodies produced in rabbit that were raised against RSKLREGSSDRPQSSC and CTGQARRRWSAATMEP peptides corresponding to amino acids 6-21 and 21-36 of the N-terminus of mouse Dpy19l2, as described in [79].

Acknowledgements

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Figure Supplementary 1. Amino-acid conservation of CCDC146 orthologs from man to *X. tropicalis*. The amino acids altered and missing from the presence of the variant p.Arg704serfsTer7 are highlighted in yellow.

	Patient ID	Age	Semen volume (mL)	pH	Viscosity (normal:1, abnormal:2)	Sperm concentration (million/mL)	Total sperm number (million/ejaculate)	Non sperm cells/Round cells (million/ml)	Polynuclear Neutrophils	Total Motility (a+b+c) after 1h (%)	Progressive Motility (PR; a+b)	Vitality (%)	Sperm morphology : normal forms (%)	Multiple Anomalies Index (MAI)	Tapered head	Thin head	Microcephalic	Macrocephalic	Multiple heads	abnormal base (Abnormal post-acrosomal region)	Abnormal acrosomal region	Excess residual cytoplasm	Thin midpiece	Bent or misaligned tail	No tail	Short tail	Irregulary shaped tail	Coiled tail	Multiple tails
Lower Reference Limits (WHO)			1.5	≥7.2		15	39			40	32	58	4	–	–	–	–	–	–	–	–	–	–		–	–	–	–	–
CCDC146	08IF039	48	5.5	NA	2	35	193	3.2	NA	15	10	42	1	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	>10%	>10%	>10%	>10%	>10%
CCDC146	09IF026	52	2	7.7	1	42	84	1	0	10	7	62	0	2.4	5	2	6	2	0	6	34	11	3	NA	35	37	67	18	3

Figure 1—Figure supplement 1.

Detailed semen parameters of both MMAF individuals harboring a *CCDC146* variant.

Values are percentages unless specified otherwise. NA: not available, Reference limits (5th centiles and their 95% confidence intervals) according to World Health Organization (WHO) standards (Cooper et al., 2010). Patient 08IF39 variant c.1084C>T and patient O9IF26 variant c.2112Del.

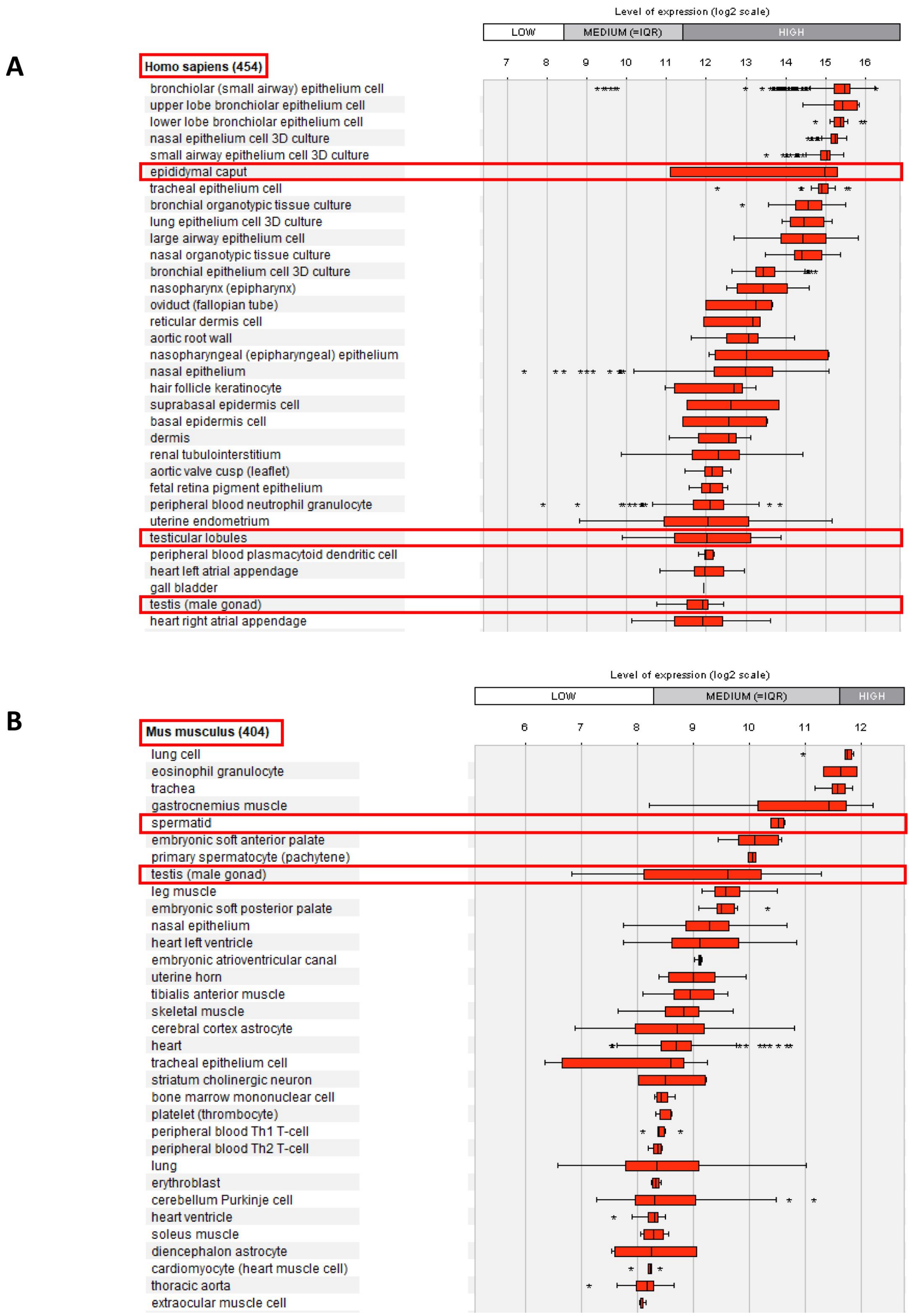
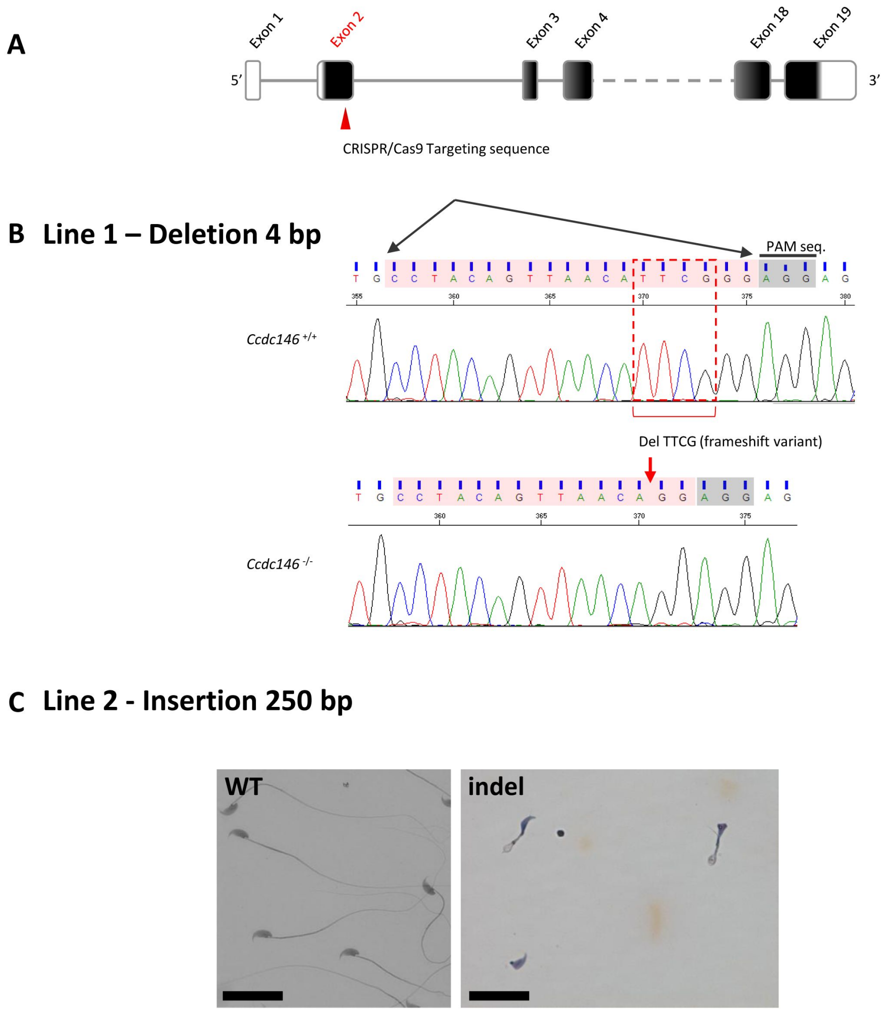


Figure 1—Figure supplement 2.

Relative mRNA expression levels for human and mouse *Ccdc146* transcripts.

(A) *Ccdc146* mRNA levels measured in different tissues/cells in humans using Affymetrix microarrays (data available from the Genevestigator database, *https://genevestigator.com*). Red rectangles highlight the high expression level in male reproductive organs. (B) Similar data for mice. The level of expression is medium in mice. Data were generated with Genevestigator (Hruz T, Laule O, Szabo G, Wessendorp F, Bleuler S, Oertle L, Widmayer P, Gruissem W and P Zimmermann (2008) Genevestigator V3: a reference expression database for the meta-analysis of transcriptomes. Advances in Bioinformatics 2008, 420747)



D

	Genotypes – line 2							
	WT	WT	WT	Htz	Htz	Htz	KO	KO
Sperm count (x 10^6/ml)	35	15.5	18.8	24.8	26.8	14.2	2	9.1
Vitality (%)	81.2	85.7	86.6	79.3	84.4	81.5	78.6	78.9
Progressive Motility	ND	ND	ND	ND	ND	ND	0	0
Typical forms	ND	ND	ND	ND	ND	ND	0	0
Flagellar defects	ND	ND	ND	ND	ND	ND	0	0

Figure 2—Figure supplement 1.

Molecular strategy used to generate *Ccdc146* KO mice by CRISPR/Cas9.

(A) The exonic structure of mouse *Ccdc146* is shown and the coding sequence indicated in black. (B) Exon 2, the first coding sequence, was targeted by an RNA guide (5′-CCT ACA GTT AAC ATT CGG G-3) and the Cas9 induced in line 1 a deletion of four nucleotides upstream the PAM sequence, as indicated by the red box. Electropherogram presenting the WT and the homozygote deletion are shown. (C) In line 2, the Cas9 induced an insertion of 250 nucleotides. Sperm morphology of F2 males from line 2 was severely damaged. Scale bars 30 µm. (D) Comparison of sperm parameters in WT, heterozygote and KO animals from line 2; ND not determined.

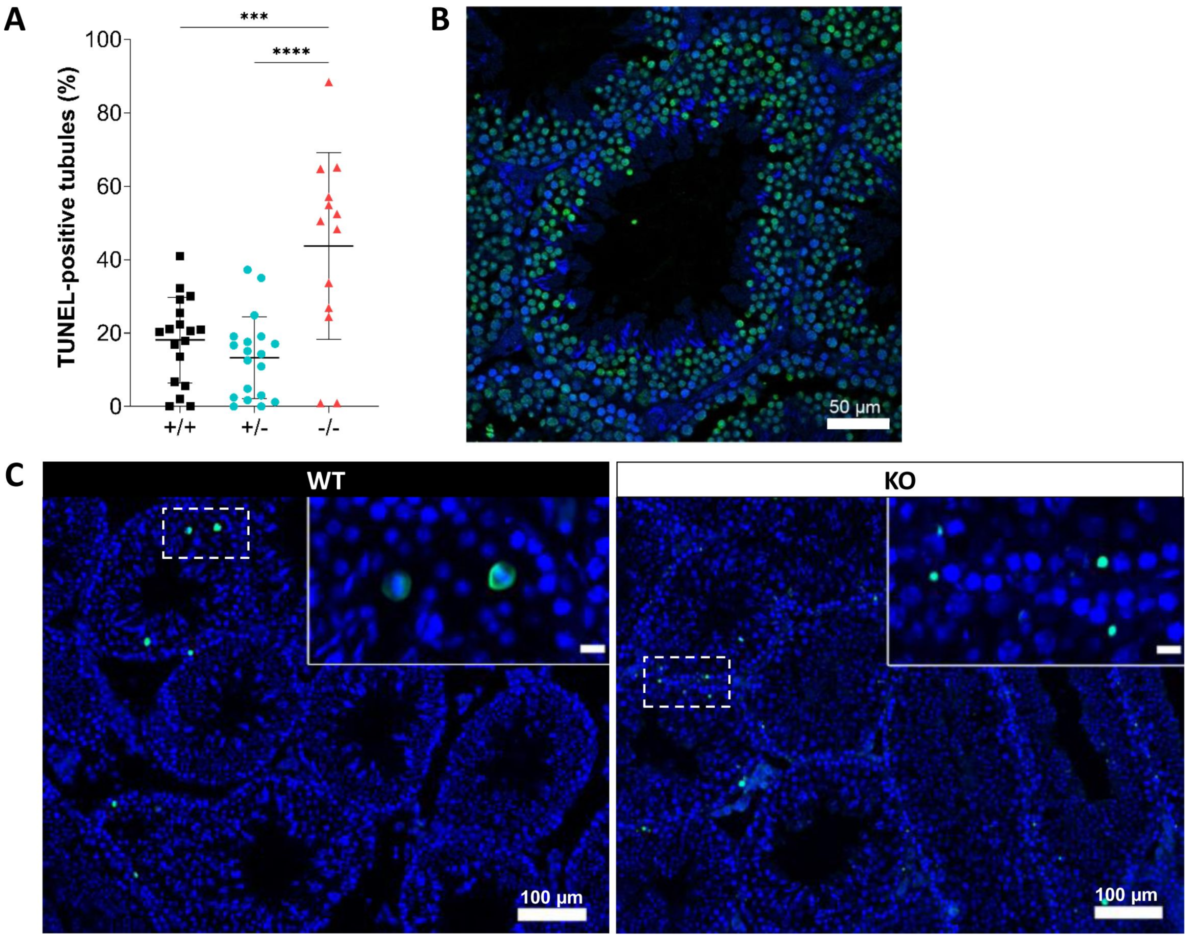


Figure 2—Figure supplement 2.

Increased levels of apoptosis in testes from *Ccdc146* KO mice.

The TUNEL assay was used to visualize double-strand DNA breaks, as an indication of the level of apoptosis during WT and KO spermatogenesis. (A) Comparison of the % of tubules per testis cross-section containing at least one fluorescent (TUNEL-positive) cell in WT, heterozygote, and *Ccdc146* KO animals. Number of sections counted per genotype n=13-20, 3 different mice per genotype. (B) Control testis section treated with H202. (C) The majority of TUNEL-positive cells in the WT corresponded to pachytene cells undergoing meiosis (C, WT zoomed image) whereas the localization of TUNEL-positive cells in KO was more scattered. Statistical comparisons according to ordinary one-way ANOVA test (**** $p<0.0001$; *** $p<0.001$, ** $p<0.01$, * $p<0.05$). Scale bars of zoomed images 50 µm.

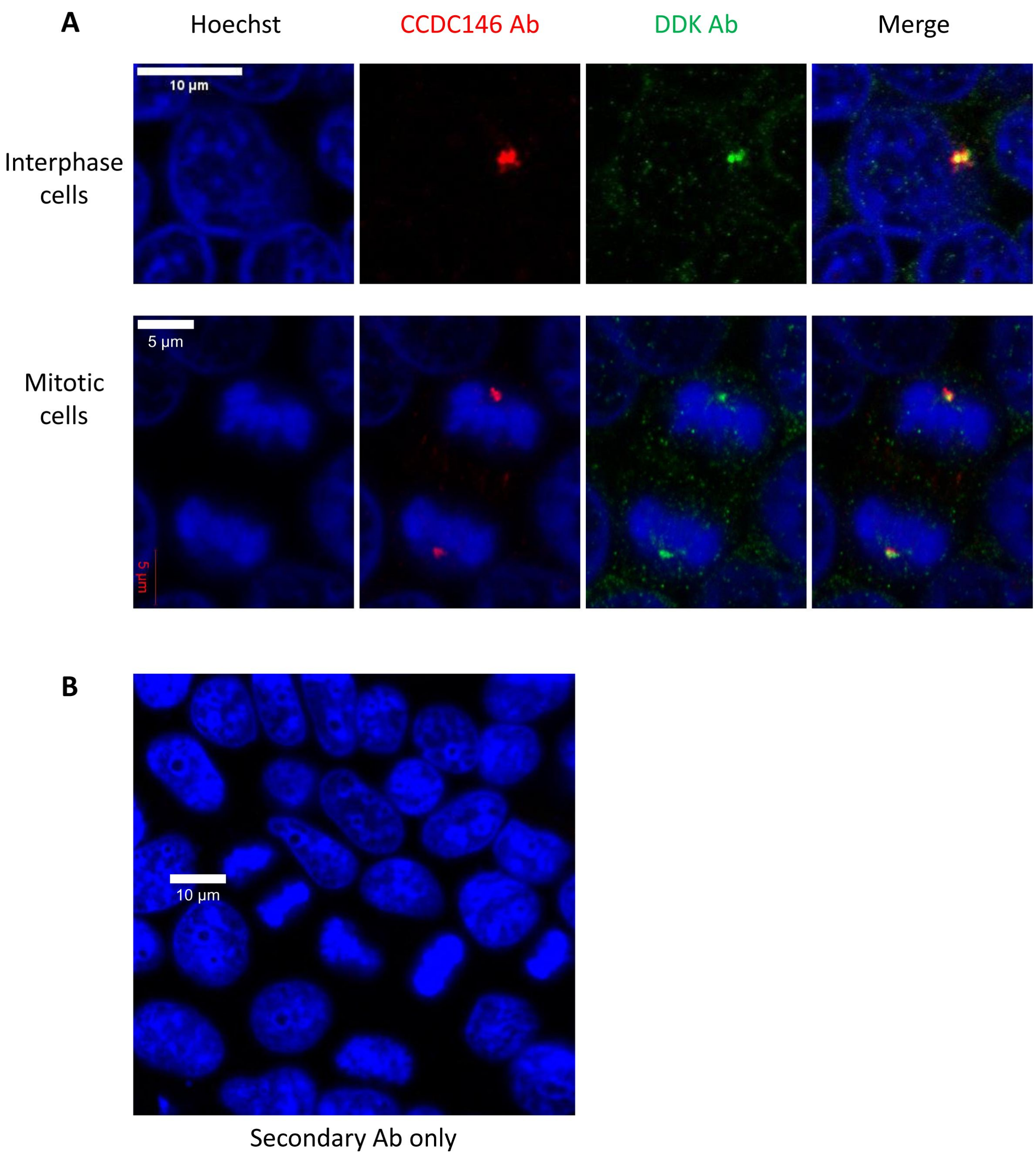


Figure 4—Figure supplement 1.

DDK and CCDC146 Abs immunodecorate the same cellular components.

HEK-293T cells were transfected with a plasmid containing the sequence of DDK-CCDC146 (human). After cell fixation, cells were double immunolabeled for CCDC146 (red) and DDK (green) with the corresponding Abs. (A) Immunostaining of interphase and mitotic cells. Both antibodies stain the centrosome. (B) Same experiment with secondary Ab only. Scale bars as indicated.

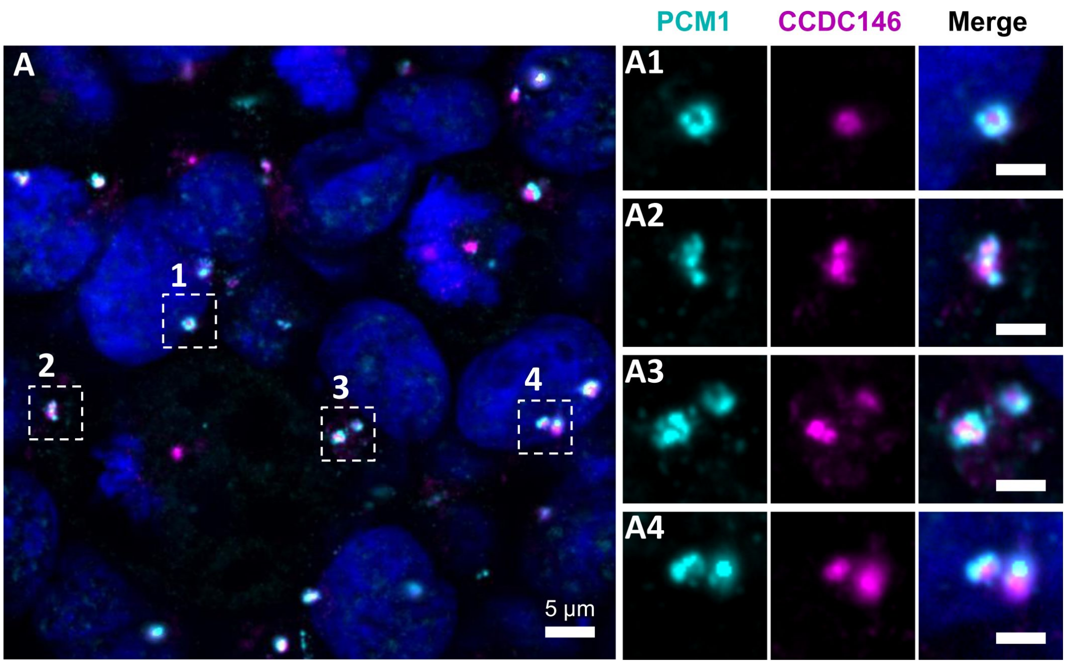


Figure 4—Figure supplement 2.

CCDC146 does not colocalize with the centriolar satellite marker PCM1.

(A) HEK-293T cells were double immunolabeled for PCM1 (cyan) and CCDC146 (magenta). (A1- A4) Images on the right show the enlargement of the dotted squares in the left image. PCM1 surrounds the CCDC146 signal, but no colocalization is observed, suggesting that CCDC146 is not a centriolar satellite protein. DNA was stained with Hoechst (blue). Scale bars on zoomed images represent 2 µm.

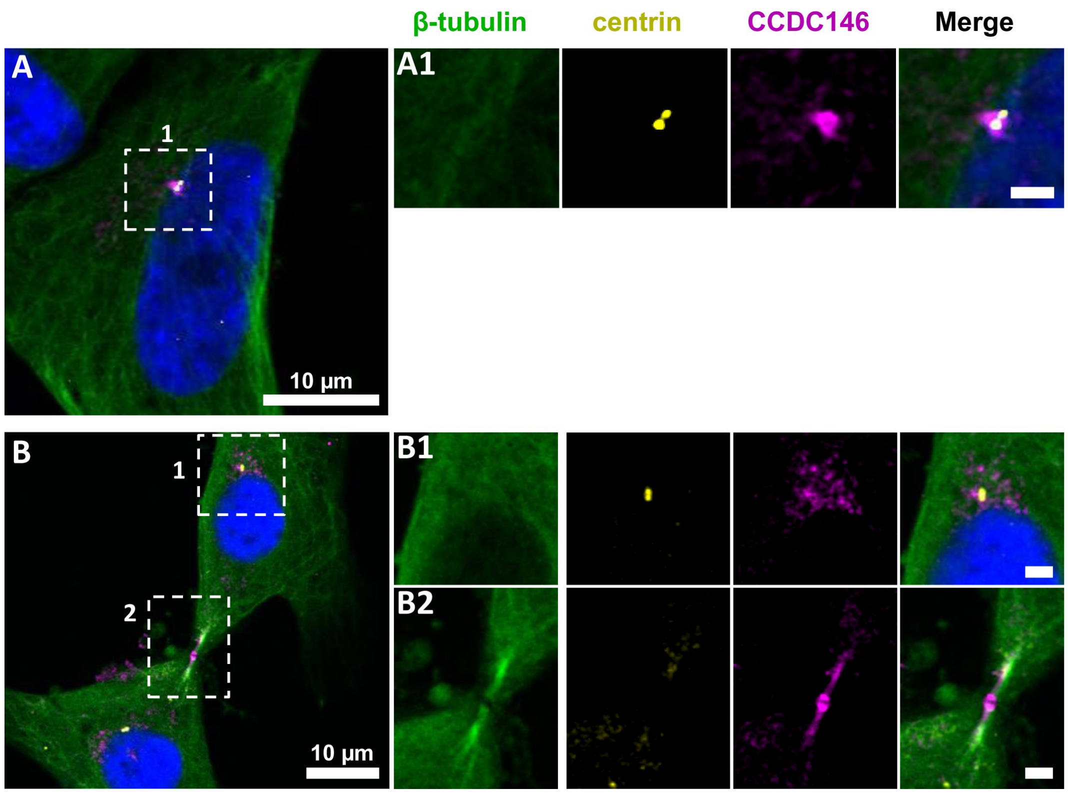


Figure 5—Figure supplement 1.

CCDC146 shows a similar localization to the centrosome and to the midbody in primary HFF cells.

(A) Primary human foreskin fibroblast (HFF) cells were triple immunolabeled with anti-β-tubulin (green), anti-centrin (yellow, showing the centrioles) and anti-CCDC146 (magenta). (A1) The right-hand images show the enlargement of the dotted squares in the left-hand image. CCDC146 localized to and around the centrioles. (B) Staining of the centrosome area (B1) and (B2) shows co-localization of the CCDC146 staining with the midbody during cytokinesis. Scale bars of zoomed images 2 µm.

A

Strand	Sequence	PAM	On-target score	Off-target score
+	TACTTTAGAACTGTGAAAA A	TGG	27.9	29.0

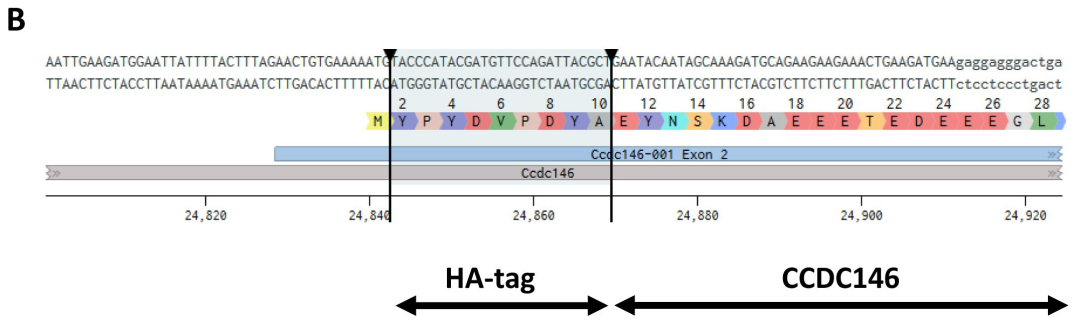


Figure 6—Figure supplement 1.

Molecular strategy used to generate HA-tagged CCDC146 mice by CRISPR/Cas9

(A) Exon 2, the first coding sequence, was targeted by an RNA guide (5′- TAC TTT AGA ACT GTG AAA AAT GG −3′). (B) Using a single-stranded DNA (ssDNA) template, the HA sequence (5′-TAC CCA TAC GAT GTT CCA GAT TAC GCT-3′) was inserted upstream of the PAM sequence.

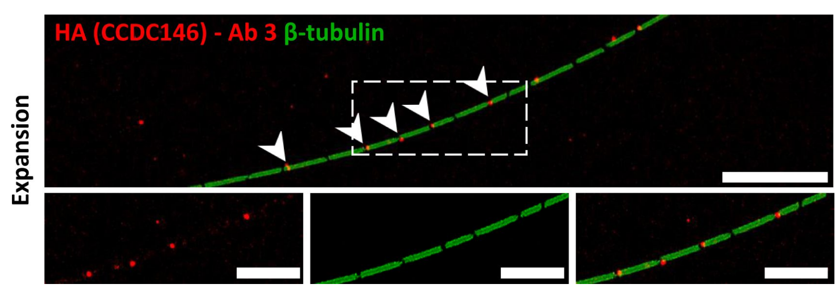


Figure 7—Figure supplement 1.

Staining by Anti-HA Abs of axonemal breaks induced by expansion.

Mouse epididymal spermatozoa observed with expansion microscopy. HA-CCDC146 sperm were immunolabeled with anti-β-tubulin (green) and anti-HA #3 (red) Abs. The right image shows the sperm with merged immunostaining. Expansion induced axonemal breaks associated with strong HA- CCDC146 staining (red). Scale bars correspond to 10 µm.

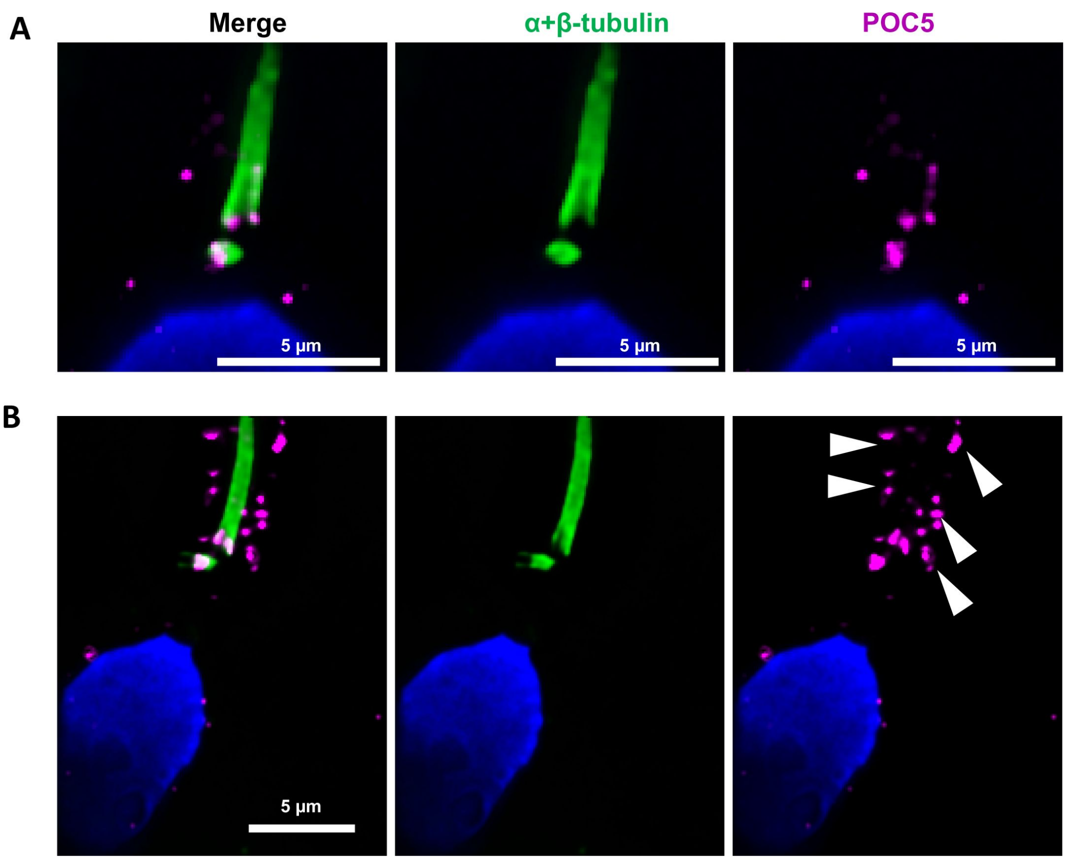
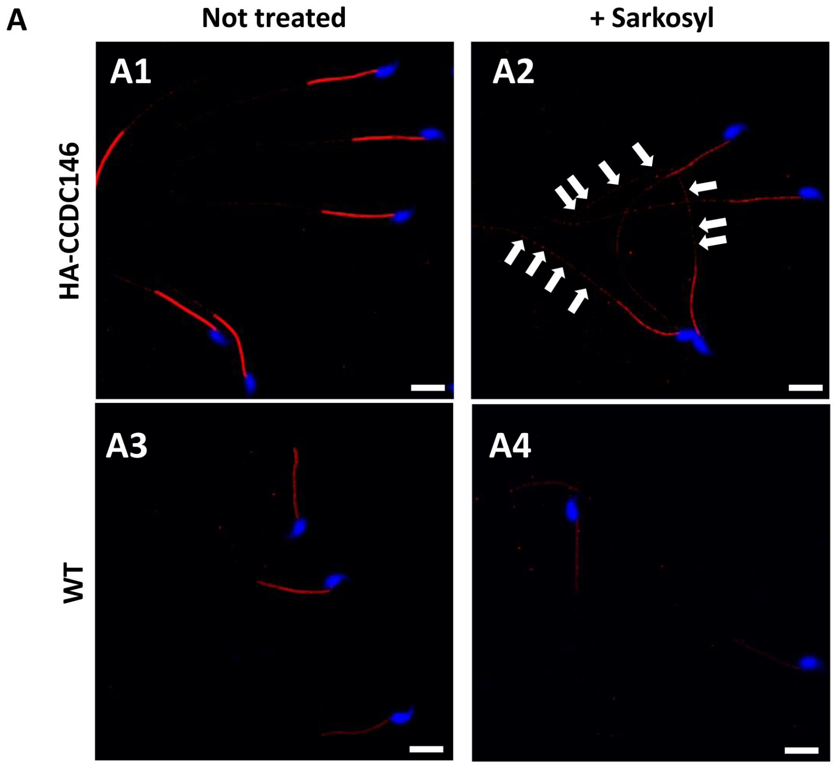


Figure 8—Figure supplement 1.

Centrioles are identified by anti-POC5 Abs in expanded human ejaculated spermatozoa.

Human control sperm were co-stained, after expansion, with anti-α+β- tubulin (green) and anti-POC5 (magenta) Abs. (A) The centrosomal protein POC5 locates to centrioles at the base of the axoneme. (B) Apart the centriole staining, a scattered staining was also observed in the midpiece around the axoneme (white arrows heads). Scale bars 5 µm.



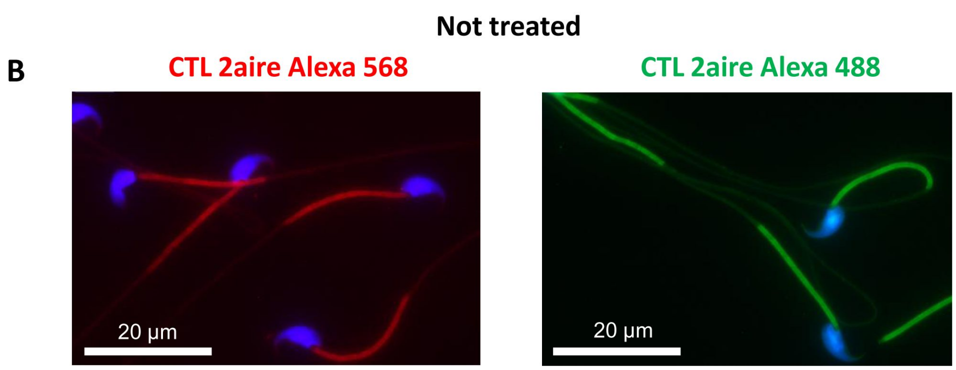


Figure 10—Figure supplement 1.

Sperm sarkosyl treatment corroborates the presence of CCDC146 along the mouse flagellum.

HA-CCDC146 sperm (A1-A2) and epididymal WT (A3-A4) not treated or treated (5 min, 0.2% sarkosyl), were immunostained with anti-HA Ab (red) and counterstained with Hoechst (blue). (A1) Without treatment, a faint HA-CCDC146 signal was observed along the HA- CCDC146 principal piece. The strong staining in the midpiece is not specific (see panel B). (A2) After treatment with sarkosyl, the HA-CCDC146 signal along the sperm principal piece was enhanced (white arrows), whereas the signal in the midpiece decreased. (A3) WT untreated (NT) sperm exhibited almost no HA-CCDC146 signal in the principal piece. The midpiece is stained but it is likely not specific (see panel B). (A4) The HA signal is not enhanced in WT principal piece by sarkosyl treatment, suggesting that the enhanced signal observed with sarkosyl on HA-CCDC146 sperm is specific. Scale bars 10 µm. (B) HA-CCDC146 sperm were only immunolabeled with secondary antibodies used to reveal HA staining. Strong staining is observed on the midpiece, confirming its non-specific nature. Scale bars 20 µm.

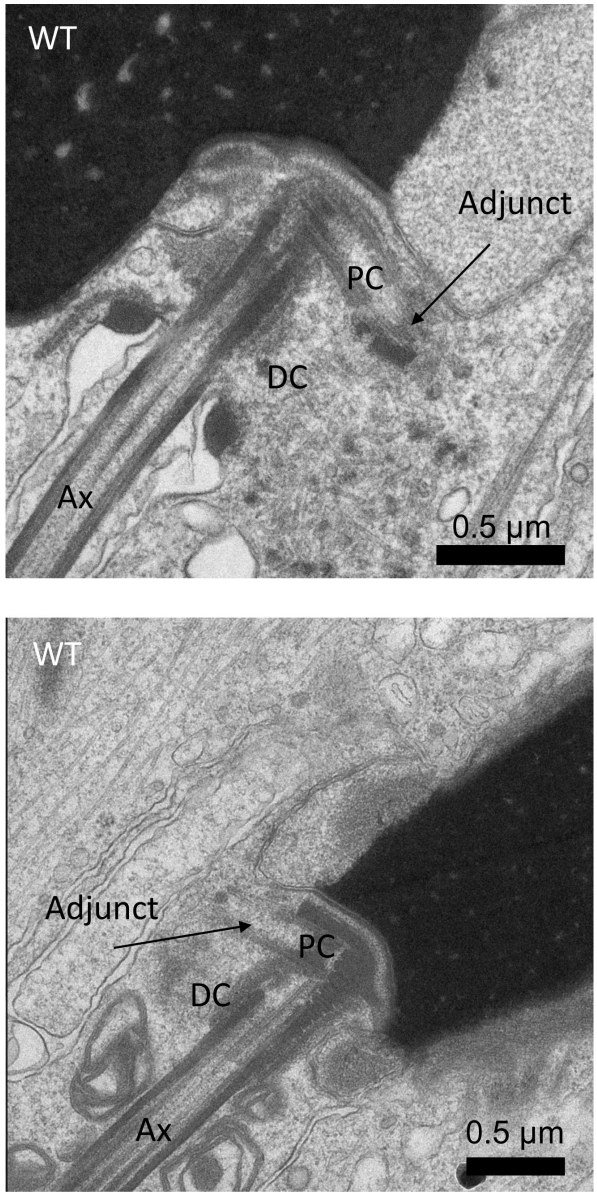


Figure 12—Figure supplement 1.

Ultrastructure of WT sperm showing the head to tail coupling apparatus (HTCA) in elongating spermatids.

The adjunct is connected to the proximal centriole (PC). The axoneme (Ax) is elongated from the distal centriole (DC). Scale bars 0.5 μm.

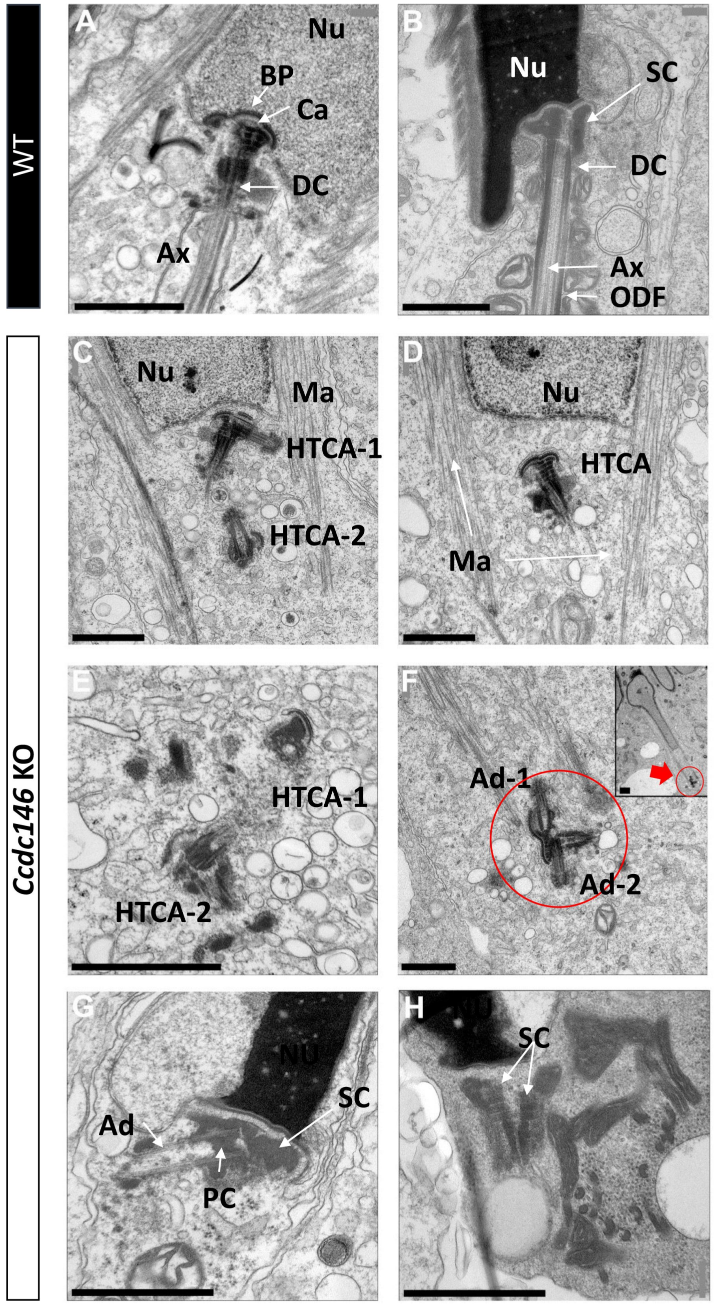


Figure 12—Figure supplement 2.

Lack of CCDC146 causes centriole duplication and mislocalization in *Ccdc146* KO spermatids.

Ultrastructural analysis of centrioles in adult mouse WT (A, B) and *Ccdc146* KO (C-H) testis sections. (A) In WT spermatids, the axoneme is attached to the base of the compacting nucleus (Nu) through the basal plate (BP) and the capitulum (Ca), and the distal centriole (DC) is embedded in the segmented column (SC). These sperm-specific cytoskeletal structures make up the head-to-tail coupling apparatus (HTCA). (B) In WT elongated spermatids, the different components of the axonemal structures (Ax) and outer dense fibers (ODF) were visible downstream the distal centriole. (C) In *Ccdc146* KO elongating spermatids, the overall structure of the HTCA was conserved, with the presence of the centrioles and the accessory cytoskeletal structures. However, the HTCAs were often duplicated (C, E, F) and separated from their usual nuclear attachment site (C-F, H), and sometimes misplaced far away from the nucleus (F), the red arrow in F indicating the misplaced centrioles at the end of the manchette. The axoneme is missing (C-F). In elongated spermatids with condensed nucleus, malformed and detached centrioles with poorly-assembled or missing flagella compared to the WT (B) can be seen. Manchette (Ma), Adjunct (Ad), proximal centriole (PC). Scale bars 1µm.

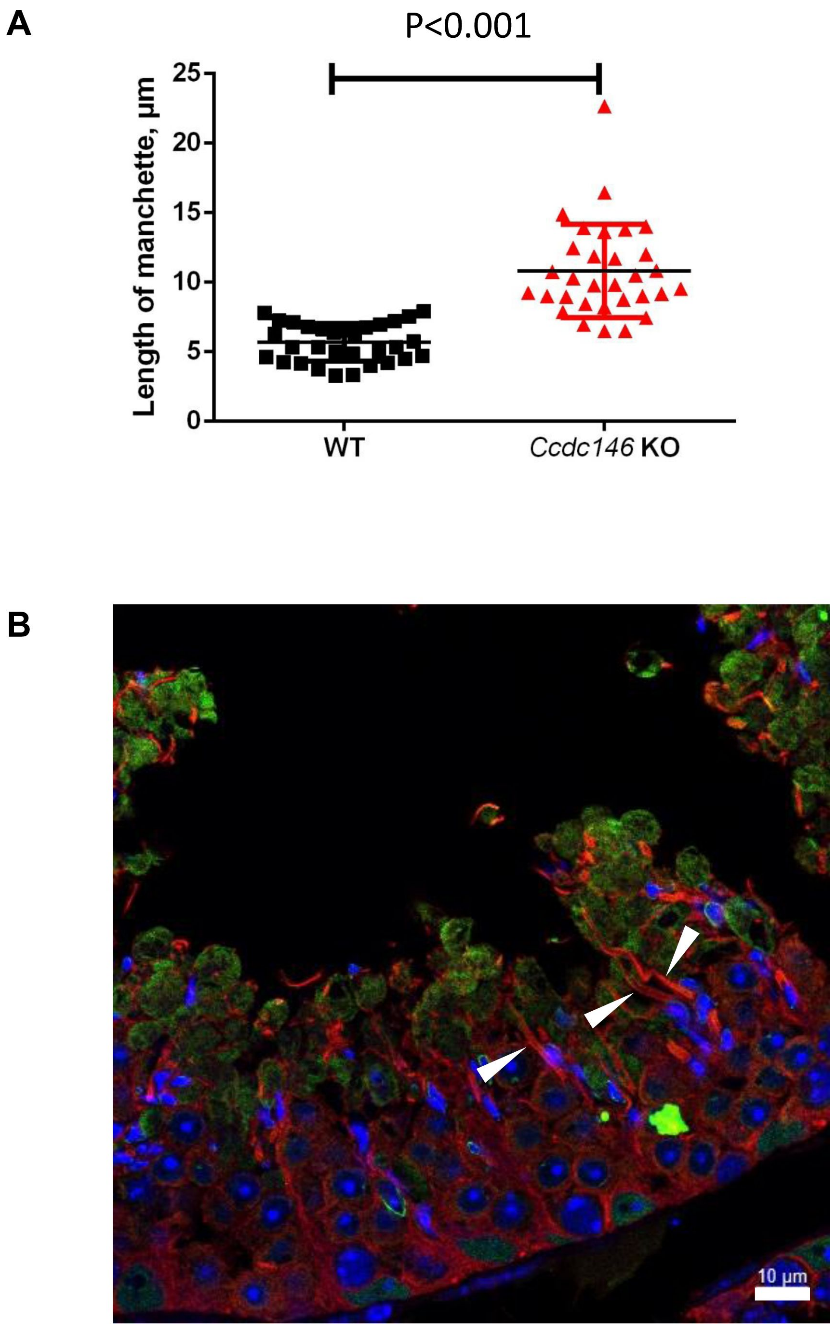


Figure 13—Figure supplement 1.

The manchette of elongating spermatids from *Ccdc146* KO male is longer than those from WT males.

(A) Comparison of the length of the manchette of elongating spermatids measured in cross-sections of seminiferous tubules at stage I for WT and stages I-III for *Ccdc146* KO spermatids. Statistical significance between WT and *Ccdc146*-KO spermatids was assessed by Mann-Whitney test, p value as indicated. (B) Examples of very long manchettes (white arrow heads) in *Ccdc146* KO cross-sections of seminiferous tubules at stage I-III. WT cross-section is presented **Figure 13D1**. Scale bars as indicated.

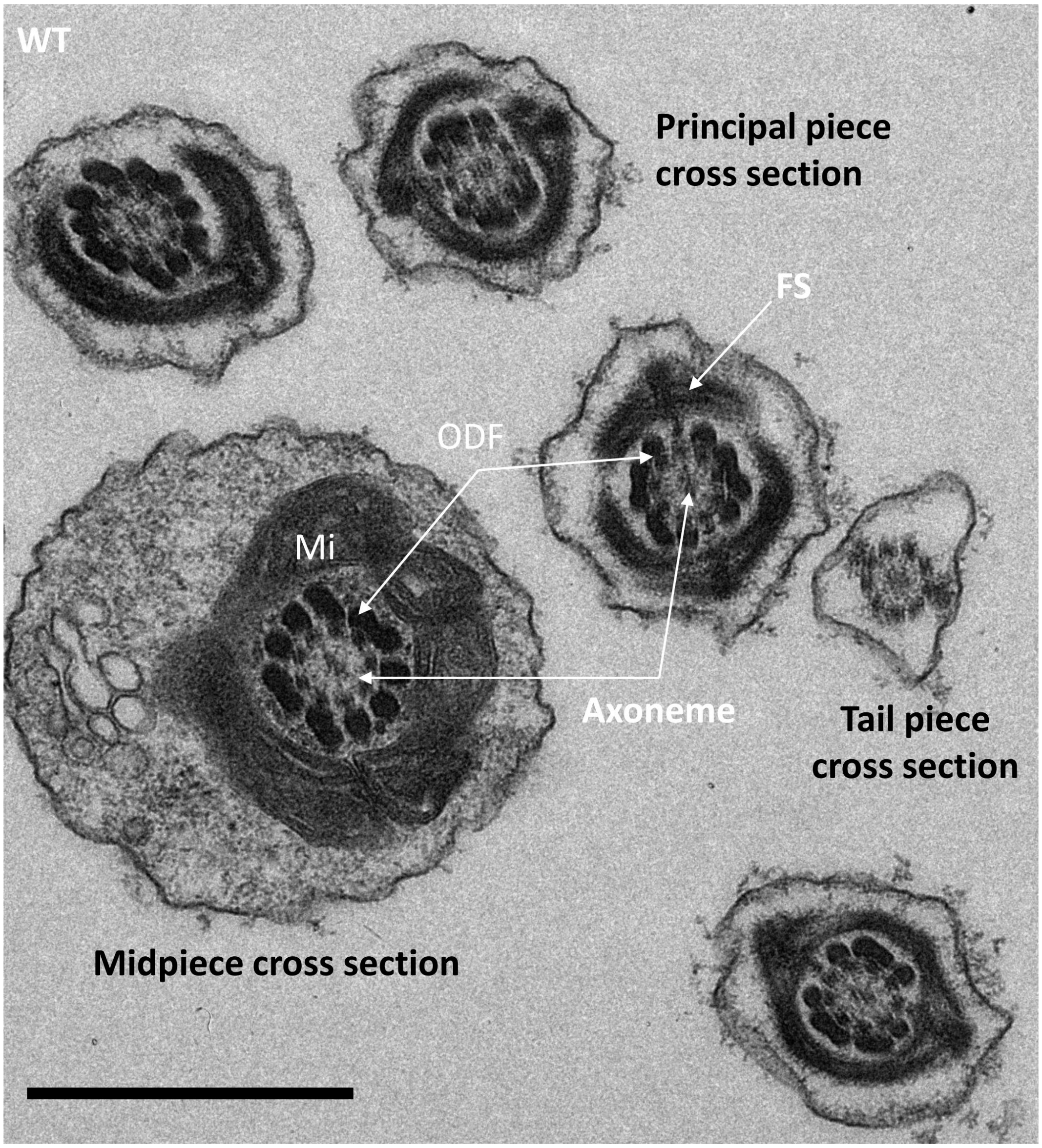


Figure 15—Figure supplement 1.

Cross-sections of the midpiece, principal piece and tail piece of WT elongating spermatids showing the ultrastructure of the different pieces.

In the midpiece cross section, the axoneme is surrounded by 9 outer dense fibers (ODF) and an external layer of mitochondria (Mi). In the principal piece, the ODF are surrounded by the fibrous sheet (FS). Scale bar 1 μm.

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Reviewer #1 (Public Review):

Here, Muronova et al., demonstrate the physiological importance of a centriole and microtubule-associated protein, CCDC146, in sperm flagellar formation and male reproduction. This study identifies novel causal variants to cause male infertility and resolves the pathogenicity by the mutation with characterizing mouse models. Furthermore, the authors' claims are well supported by the biochemical and imaging approaches used in this study.

https://doi.org/10.7554/eLife.86845.2.sa1

Reviewer #3 (Public Review):

Male infertility is an important health problem. Among pathologies with multiple morphological abnormalities of the flagellum (MMAF), only 50% of the patients have no identified genetic causes. It is thus primordial to find novel genes that cause the MMAF syndrome. In the current work, the authors follow up the identification of two patients with MMAF carrying a mutation in the CCDC146 gene. To understand how mutations in CCDC146 lead to male infertility, the authors generated two mouse models: a CCDC146-knockout mouse, and a knockin mouse in which the CCDC146 locus is tagged with an HA tag. Male CCDC146-knockout mice are infertile, which proves the causative role of this gene in the observed MMAF cases. Strikingly, animals develop no other obvious pathologies, thus underpinning the specific role of CCDC146 in male fertility. The authors have carefully characterised the subcellular roles of CCDC146 by using a combination of expansion and electron microscopy. They demonstrate that all microtubule-based organelles, such as the sperm manchette, the centrioles, as well as the sperm axonemes are defective when CCDC146 is absent. Their data show that CCDC146 is a microtubule-associated protein, and indicate, but do not prove beyond any doubt, that it could be a microtubule-inner protein (MIP).

This is a solid work that defines CCDC146 as a novel cause of male infertility. The authors have performed comprehensive phenotypic analysis to define the defects in CCDC146 knockout mice. The manuscript text is well written and easy to follow also for non-specialists. The introduction and discussion chapters contain important background information that allow to put the current work into the greater context of fertility research. Overall, this manuscript provides convincing evidence for CCDC146 being essential for male fertility and illustrates this with a large panel of phenotypic observations. Together, the work provides important first insights into the role of a so-far unexplored proteins, CCDC146, in spermatogenesis, thereby broadening the spectrum of genes involved in male infertility.

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Author Response

The following is the authors’ response to the original reviews.

First of all, we'd like to thank the three reviewers for their meticulous work that enable us to present now an improved manuscript and substantial changes were made to the article following reviewers' and editors' recommendations. We read all their comments and suggestions very carefully. Apart from a few misunderstandings, all comments were very pertinent. We responded positively to almost all the comments and suggestions, and as a result, we have made extensive changes to the document and the figures. This manuscript now contains 16 principal figures and 15 figure supplements.

The number of principal figures is now 16 (1 new figure), and additional panels have been added to certain figures. On the other hand, we have added 7 additional figures (supplement figures) to answer the reviewers' questions and/or comments.

Main figures

▪ Figures 1, 4, 5, 10, 11, 12, 13, 14: unchanged ▪ Figure 7 and 8 were switched.

▪ Figure 2: we added panel F in response to reviewer 3's and request for sperm defect statistics

▪ Figure 3: the contrast in panel B has been taken over to homogenize colors

▪ Figure 6: This figure was recomposed. The WB on testicular extract was suppressed and we present a new WB allowing to compare the presence of CCDC146 in the flagella fraction. Using an anti-HA Ab, we demonstrate that the protein is localized in the flagella in epididymal sperm. Request of the 3 reviewers.

▪ Figure 7 (old 8): to avoid the issue of the non-specificity of secondary antibodies, we performed a new set of IF experiments using an HA Tag Alexa Fluor® 488-conjugated Antibody (anti-HA-AF488-C Ab) on WT and HA-CCDC146 sperm. These results are now presented in figure 7 panel A (new). The specificity of the signal obtained with the anti-HA-AF488-C Ab on mouse spermatozoa was evaluated by performing a statistical study of the density of dots in the principal piece of the flagellum from HA-CCDC146 and WT sperm. These results are now presented in figure 7 panel B (new). This study was carried out by analyzing 58 WT spermatozoa and 65 CCDC146 spermatozoa coming from 3 WT and 3 KI males. We found a highly significant difference, with a p-value <0.0001, showing that the signal obtained on spermatozoa expressing the tagged protein is highly specific. We have added a paragraph in the MM section to describe the process of image analysis. We finally present new images obtained by ExM showing no staining in the midpiece (figure 7C new). Altogether, these results demonstrate unequivocally the presence of the protein in the flagellum. Moreover, the WB was removed and is now presented in figure 6 (improved as requested).

▪ Figure 8. Was old figure 7

▪ Figure 9: figure 9 was recomposed and improved for increased clarity as suggested by reviewer 2 and 3.

▪ Figure 16 was before appendix 11

Figure supplements and supplementary files

▪ Figure 1-Figure supplement 1 New. Sperm parameters of the 2 patients. requested by editor (remark #1) by the reviewer 1 (Note #3)

▪ Figure 2-Figure supplement 1 new. Sperm parameters of the line 2 (KO animals) requested by the reviewer 1 (Note #5)

▪ Figure 4-Figure supplement 1 New. Experiment to evaluate the specificity of the human CCDC146 antibody. Minimal revision request and reviewer 1 note #8

▪ Figure 6-Figure supplement 1 New. Figure recomposed; Asked by reviewer 2 note #4 and reviewer 3

▪ Figure 8-Figure supplement 1 New. We now provide new images to show the non-specific staining of the midpiece of human sperm by secondary Abs in ExM experiments; Asked by reviewer 2

▪ Figure 10-Figure supplement 1 New. We added new images to show the non-specific staining of the midpiece of mouse sperm by secondary Abs in IF (panel B). Rewiever 1 note #9 and reviewer 2 note #5

▪ Figure 12-Figure supplement 1 New. Control requested by reviewer 3 Note #23

▪ Figure 13-Figure supplement 1 New. We provide a graph and a statistical analysis demonstrating the increase of the length of the manchette in the Ccdc146 KO. Requested by editor and reviewer 3 Note 24

▪ Figure 15-Figure supplement 1 New. Control requested by reviewer 2. Minor comments

▪ Figure supplementary 1 New. Answer to question requested by reviewer 2 note #1

All the reviewers' and editors' comments have been answered (see our point to point response) and we resubmit what we believe to be a significantly improved manuscript. We strongly hope that we meet all your expectations and that our manuscript will be suitable for publication in "eLife". We look forward to your feedback,

> *Point by point answer*
>
> *Please note that there has been active discussion of the manuscript and the summarize points below is the minimal revision request that the reviewers think the authors should address even under this new review model system. It was the reviewers' consensus that the manuscript is prepared with a lot of oversights - please see all the minor points to improve your manuscript.*

All minimal revision requests have been addressed

> *Minimal revision request*
>
> 1. *Clinical report/evaluation of the two patients should be given as it was not described even in their previous study as well as full description of CCDC146.*

We provide now a new Figure 1-figure supplement 1 describing the patients sperm parameters

> 1. *Antibody specificity should be provided, especially given two of the reviewers were not convinced that the mid piece signal is non-specific as the authors claim. As both KO and KI model in their hands, this should be straightforward.*

To validate the specificity of the Antibody, we transfected HEK cells with a human DDK-tagged CCDC146 plasmid and performed a double immunostaining with a DDK antibody and the CCDC146 antibody. We show that both staining are superimposable, strongly suggesting that the CCDC146 Ab specifically target CCDC146. This experiment is now presented in Figure 4-Figure supplement 1. Next, to avoid the issue of the non-specificity of secondary antibodies, we performed a new set of IF experiments using an HA Tag Alexa Fluor® 488-conjugated Antibody (anti-HA-AF488-C Ab) on WT and HA-CCDC146 sperm. These results are now presented in figure 7 panel A (new). The specificity of the signal obtained with the anti-HA-AF488-C Ab on mouse spermatozoa was evaluated by performing a statistical study of the density of dots in the principal piece of the flagellum from HA-CCDC146 and WT sperm. These results are now presented in figure 7 panel B (new). This study was carried out by analyzing 58 WT spermatozoa and 65 CCDC146 spermatozoa coming from 3 WT and 3 KI males. We found a highly significant difference, with a p-value <0.0001, showing that the signal obtained on spermatozoa expressing the tagged protein is highly specific. We have added a paragraph in the MM section to describe the process of image analysis. We finally present new images obtained by ExM showing no staining in the midpiece (figure 7C new). Altogether, these results demonstrate unequivocally the presence of the protein in the flagellum.

> 1. *The authors should improve statistical analysis to support their experimental results for the reader can make fair assessment. Combined with clear demonstration of ab specificity, this lack of statistical analysis with very few sample number is a major driver of dampening enthusiasm towards the current study.*

Several statistical analyses were carried out and are now included:

1. distribution of the HA signal in mouse sperm cells (see point 2 Figure 7 panel B)
2. quantification and statistical analyses of the defect observed in Ccdc146 KO sperm (figure 2 panel E)
3. Quantification and statistical analyses of the length of the manchette in spermatids 13-15 steps (Figure 13-Figure supplement 1 new)

> 1. *The authors need to clarify (peri-centriolar vs. centriole)*

In figure 4A, we have clearly shown that the protein colocalizes with centrin, a centriolar core protein in somatic cells. This colocalization strongly suggests that CCDC146 is therefore a centriolar protein, and this is now clearly indicated lines 211-212. However, its localization is not restricted to the centrioles and a clear staining was also observed in the pericentriolar material (PCM). The presence of a protein in PCM and centriole was already described, and the best example is maybe gamma-tubulin (PMID: 8749391).

> *or tone down (CCDC146 to be a MIP) of their claim/description.*

Concerning its localization in sperm, we agree with the reviewer that our demonstration that CCDC146 is MIP would deserve more results. Because of that, we have toned down the MIP hypothesis throughout the manuscript. See lines 491495

> *Testis-specific expression of CCDC146 as it is not consistent with their data.*

We have also modified our claim concerning the testis-expression of CCDC146. Line 176

> ***Reviewer #1 (Recommendations For The Authors):***
>
> *Major comments*
>
> 1. *As described in general comments, this study limits how the CCDC146 deficiency impairs abnormal centriole and manchette formation. The authors should explain their relationship in developing germ cells.*

In fact, there are limited information about the relationship between the manchette and the centriole. However, few articles have highlighted that both organelles share molecular components. For instance, WDR62 is required for centriole duplication in spermatogenesis and manchette removal in spermiogenesis (Commun Biol. 2021; 4: 645. doi: 10.1038/s42003-021-02171-5). Another study demonstrates that CCDC42 localizes to the manchette, the connecting piece and the tail (Front. Cell Dev. Biol. 2019 https://doi.org/10.3389/fcell.2019

.00151). These articles underline that centrosomal proteins are involved in manchette formation and removal during spermiogenesis and support our results showing the impact of CCDC146 lack on centriole and manchette biogenesis. This information is now discussed. See lines 596-603

> 1. *The authors generated knock-in mouse model. If then, are the transgene can rescue the MMAF phenotype in CCDC146-null mice? This reviewer strongly suggest to test this part to clearly support the pathogenicity by CCDC146.*

We indeed wrote that we created a "transgenic mice", which was misleading. We actually created a CCDC16 knock-in expressing a tagged-protein. The strain was actually made by CRISPR-Cas9 and a sequence coding for the HA-tag was inserted just before the first amino acid in exon 2, leading to the translation of an endogenous HA-tagged CCDC146 protein. We have removed the word transgenic from the text and made changes accordingly (see lines 250-253). We can therefore not use this strain to rescue the MMAF phenotype as suggested by the reviewer.

> 1. *Although the authors cite the previous study (Coutton et al., 2019), the study does not describe any information for CCDC146 and clinical information for the patients. The authors must show the results for clinical analysis to clarify the attended patients are MMAF patients without other phenotypic defects.*

We have now inserted a table, indicating all sperm parameters for the patients harboring a mutation in the CCDC146 gene (Figure 1-Figure supplement 1) and is now indicated lines 159-160

> 1. *The authors describe CCDC146 expression is dominant in testes, However, the level in testis is only moderate in human (Supp Figure 1). Thus, this description is not suitable.*

In Figure 1-figure supplement 2 (old FigS1), the median of expression in testis is around 12 in human, a value considered as high expression by the analysis software from Genevestigator. However, for mouse, it is true that the level of expression is medium. We assumed that reviewer's comment concerned testis expression in mouse. To take into account this remark, we changed the text accordingly. See line 176.

> 1. *Although the authors mentioned that two mice lines are generated, only one line information is provided. Authors must include information for another line and provide basic characterization results to support the shared phenotype within the lines.*

We now provide a revised Figure 2-figure supplement 1CD, presenting the second line and the corresponding text in the main text is found lines 178-183.

> 1. *In somatic cells, the CCDC146 localizes at both peri-centriole and microtubule but its intracellular localization in sperm is distinguished. The authors should explain this discrepancy.*

The multi-localization of a centriolar protein is already discussed in detail in discussion lines 520-526. We have written:

"Despite its broad cellular distribution, the association of CCDC146 with tubulin-dependent structures is remarkable. However, centrosomal and axonemal localizations in somatic and germ cells, respectively, have also been reported for CFAP58 [37, 55], thus the re-use of centrosomal proteins in the sperm flagellar axoneme is not unheard of. In addition, 80% of all proteins identified as centrosomal are found in multiple localizations (https://www.proteinatlas.org/humanproteome/subcellular/centrosome). The ability of a protein to home to several locations depending on its cellular environment has been widely described, in particular for MAP. The different localizations are linked to the presence of distinct binding sites on the protein.... "

> 1. *Authors mention CCDC146 is a centriolar protein in the title and results subtitle. However, the description in results part depicts CCDC146 is a peri-centriolar protein, which makes confusion. Do the authors claim CCDC146 is centrosomal protein?*

In figure 4A, we have clearly shown that the protein colocalizes with centrin, a centriolar core protein. This colocalization strongly suggests that CCDC146 is therefore a centriolar protein in somatic cells, and is now clearly indicated lines 211-212. However, its localization is not restricted to the centrioles and a clear staining was also observed in the pericentriolar material (PCM). The presence of a protein in PCM and centriole was already described and the best example is maybe gamma-tubulin (PMID: 8749391).

> 1. *Verification of the antibody against CCDC146 must be performed and shown to support the observed signal are correct. 2nd antibody only signal is not proper negative control.*

It is a very important remark. The commercial antibody raised against human CCDC146 was validated in HEK293-cells expressing a DDK-tagged CCDC146 protein. Cells were co-marked with anti-DDK and anti-CCDC146 antibodies. We have a perfect colocalization of the staining. This experiment is now presented in Figure 4-figure supplement 1 and presented in the text (lines 206-208).

> 1. *In human sperm, conventional immunostaining reveals CCDC146 is detected from acrosome head and midpiece. However, in ExM, the signal at acrosome is not detected. How is this discrepancy explained? The major concern for the ExM could be physical (dimension) and biochemical (properties) distortion of the sample. Without clear positive and negative control, current conclusion is not clearly understood. Furthermore, it is unclear why the authors conclude the midpiece signal is non-specific. The authors must provide experimental evidence.*

Staining on acrosome should always be taken with caution in sperm. Indeed, numerous glycosylated proteins are present at the surface of the plasma membrane regarding the outer acrosomal membrane for sperm attachment and are responsible for numerous nonspecific staining. Moreover, this acrosomal staining was not observed in mouse sperm, strongly suggesting that it is not specific.

Concerning the staining in the midpiece observed in both conventional and Expansion microscopy, it also seems to be nonspecific and associated with secondary Abs.

For IF, we now provide new images showing clearly the nonspecific staining of the midpiece when secondary Ab were used alone (see Figure 10-figure supplement 1B).

For ExM, we provide new images in Figure 8-figure supplement 1B (POC5 staining) showing a staining of the midpiece (likely mitochondria), although POC5 was never described to be present in the midpiece. Both experiments (CCDC146 and POC5 staining by ExM) shared the same secondary Ab and the midpiece signal was likely due to it.

Moreover, we now provide new images (figure 7C) in ExM on mouse sperm showing no staining in the midpiece and demonstrating that the punctuated signal is present all along the flagellum. Finally, we would like to underline that we now provide new IF results, using an anti-HA conjugated with alexafluor 488 and confirming the ExM results.

These points are now discussed lines 498-502 for acrosome and lines 503-511 for midpiece staining.

> 1. *For intracellular localization of the CCDC146 in mouse sperm, the authors should provide clear negative control using WT sperm which do not carry the transgene.*

This experiment was performed.

To avoid the issue of the non-specificity of secondary antibodies, we performed a new set of IF experiments using an HA Tag Alexa Fluor® 488-conjugated Antibody (anti-HA-AF488-C Ab) on WT and HA-CCDC146 sperm. These results are now presented in figure 7 panel A (new). The specificity of the signal obtained with the anti-HA-AF488-C Ab on mouse spermatozoa was evaluated by performing a statistical study of the density of dots in the principal piece of the flagellum from HA-CCDC146 and WT sperm. These results are now presented in figure 7 panel B (new). This study was carried out by analyzing 58 WT spermatozoa and 65 CCDC146 spermatozoa coming from 3 WT and 3 KI males. We found a highly significant difference, with a p-value <0.0001, showing that the signal obtained on spermatozoa expressing the tagged protein is highly specific. We have added a paragraph in the MM section to describe the process of image analysis. We finally present new images obtained by ExM showing no staining in the midpiece (figure 7C new). Altogether, these results demonstrate unequivocally the presence of the protein in the flagellum.

> 1. *Current imaging data do not clearly support the intracellular localization of the CCDC146. Although western blot imaging reveal that CCDC146 is detected from sperm flagella, this is crude approach. Thus, this reviewer highly recommends the authors provide more clear experimental evidence, such as immuno EM.*

We provide now a WB comparing the presence of the protein in the flagellum and in the head fractions; see new figure 6. We show that CCDC146 is only present in the flagellum fraction; The detection of the band appeared very quickly at visualization and became very strong after few minutes, demonstrating that the protein is abundant in the flagella. It is important to note that epididymal sperm do not have centrioles and therefore this signal is not a centriolar signal. We also now provide new statistical analyses showing that the immuno-staining observed in the principal piece is very specific (Figure 7B). Altogether, these results demonstrate unequivocally the intracellular localization of CCDC146 in the flagellum. This point is now discussed lines 480-489

> 1. *Although sarkosyl is known to dissociate tubulin, it is not well understood and accepted that the enhanced detection of CCDC146 by the detergent indicates its microtubule inner space. Sperm axoneme to carry microtubule is also wrapped peri-axonemal components with structural proteins, which are even not well solubilized by high concentration of the ionic detergent like SDS.*

We agree with the reviewer that the solubilization of the protein by sarkozyl is not a proof of the presence of the protein inside microtubule. Taking into account this point, the MIP hypothesis was toned down and we now discuss alternative hypothesis concerning these results; See discussion lines 490-497

> 1. *SEM image is not suitable to explain internal structure (line 317-323).*

We agree with the reviewers and changes were made accordingly. See lines 354-357

> *Minor comments*
>
> 1. *In main text, supplementary figures are cited "Supp Figure". And the corresponding legends are written in "Appendix - Figure". Please unify them.*

Done Labelled now “Figure X-figure supplement Y”

> 1. *Line 159, "exon 9/19" is not clear.*

We have written now exons 9 and indicated earlier that the gene contains 19 exons

> 1. *Line 188, "positive cells" are vague.*

Positive was changed by “fluorescent”

> 1. *Representative TUNEL assay image for knockout testes were not shown in Supp Figure 3B.*

It was a mistake now Figure 2-figure supplement 2C

> 1. *Please provide full description for "IF" and "AB" when described first.*

Done

> 1. *Line 262, It is unclear what is "main piece".*

Changed to principal piece

> 1. *Line 340, Although the "stage" information might be applicable, this is information for "seminiferous tubule" rather than "spermatid". This reviewer suggests to provide step information rather than stage information.*

We agree with the reviewer that there was a confusion between “stage” and “step”. We change to step spermatids

> 1. *Line 342, Step 1 is not correct in here.*

OK corrected. now steps 13-15 spermatids

> 1. *Line 803, "C." is duplicated.*

Removed

> 1. *Figure 3A, it will be good to mark the defective nuclei which are described in figure legends.*

These cells are now indicated by white arrow heads

> 1. *Figure 5, Please provide what MT stands for.*

Now explained in the legend of figure 5

> 1. *Figure 6. Author requires clear blot images for C. In addition, Panel B information is not correct. If the blot was performed using HA antibody, then how "WT" lane shows bands rather than "HA" bands?*

The reviewer is correct. It was a mistake; The figure was recomposed and improved.

> ***Reviewer #2 (Recommendations For The Authors):***
>
> *Overall, editing oversights are present throughout the manuscript, which has made the review process quite difficult. Some repetitive figures can be removed to streamline to grasp the overall story easier. Some claims are not fully supported by evidence that need to tone down. Some figures not referenced in the main text need to be mentioned at least once.*

All figures are now referenced in the text

> *Major comments:*
>
> 1. *163-164 - Please clarify the claim that there is going to be an absence of the protein or nonfunctional protein, especially for the patient with a deletion that could generate a truncated protein at two third size of the full-length protein. Similarly, 35% of the protein level is present for the patient with a nonsense mutation. Some in silico structural analysis or analysis of conserved domains would be beneficial to support these claims.*

Both mutations are predicted to produce a premature stop codons: p.Arg362Ter and p.Arg704serfsTer7, leading either to the complete absence of the protein in case of non-sense mediated mRNA decay or to the production of a truncated protein missing almost two third or one fourth of the protein respectively. CCDC146 is very well conserved throughout evolution (Figure supplementary 1), including the 3' end of the protein which contains a large coil-coil domain (Figure 1B). In view of the very high degree of conservation, it is most likely that the 3' end of the protein, absent in both subjects, is critical for the CCDC146 function and hence that both mutations are deleterious. This explanation is now added to the discussion. see lines 439-448

> 1. *173, 423 - Please clearly state a rationale of your mouse model design (i.e., why a mouse model that recapitulate human mutation is not generated) as the truncations identified in human patients are located further towards the C-terminus, and it is not clear whether truncated proteins are present, and if so, they could still be functional. Basically, the current mouse model supports the causality of the human mutations.*

This is an important question, which goes beyond the scope of this article, and raises the question of how to confirm the pathogenicity of mutations identified by high-throughput sequencing. The production of KO or KI animals is an important tool to help confirm one' suspicions but the first element to take into consideration is the nature of the genetic data.

Here we had two patients with homozygous truncating variants. In human, it is well established that the presence of premature stop codons usually induces non-sense mediated mRNA decay (NMD), inducing the complete absence of the protein or a strong reduction in protein production. In the unlikely absence of NMD in our two patients, the identified variants would induce the production of proteins missing 60% and 30% of their C terminal part. Often (and it is particularly true for structural proteins) the production of abnormal proteins is more deleterious than the complete absence of the protein (and it is most likely the purpose of NMD, to limit the production of abnormal "toxic" proteins). For these reasons, to try to recapitulate the most likely consequences of the human variants, without risking obtaining an even more severe effect, we decided to introduce a stop codon in the first exon in order to remove the totality of the protein in the KO mice.

The second element is to interpret the phenotype of the KO animals. Here, the human sperm phenotype is perfectly recapitulated in the KO mice.

Overall, we have strong genetic arguments in human and the reproduction of the phenotype in KO mice confirming the pathogenicity of the variants identified in men.

This point is now discussed see lines 433-438

> 1. *Figure 6A - the labelling is misleading as it seems to suggest that the specific cells were isolated from the testes for RT-PCR.*

We have modified the labelling to avoid any confusion.

> *Figure 6B -Signal of HA-tag is shown in WT, not in transgenic. Please check the order of the labels. Figure 6C - This blot is NOT a publication-quality figure. The bands are very difficult to observe, especially in lane D18. Because it is one of the important data of this study, replacing this figure is a must.*

The figure has been completely remade, including new results. See new figure 6. Figure 6C was suppressed.

> 1. *Supplementary fig 6 is also not a publication-level figure, and the top part seems largely unnecessary (already in the figure legend).*

The figure has been completely remade as well (now Figure 6-Figure Supplement 1).

> 1. *261/267- The conclusion that mitochondrial staining in the flagellum (in both mice and humans) is non-specific is not convincing. Supplementary fig 8 shows that the signal from secondary only IF possibly extends beyond the midpiece - but it is hard to determine as no mitochondrial-specific staining is present. Either need to tone down the conclusion or provide supporting experimental evidence.*

First, to avoid the issue of the non-specificity of secondary antibodies, we performed a new set of IF experiments using an HA Tag Alexa Fluor® 488-conjugated Antibody (anti-HA-AF488-C Ab) on WT and HA-CCDC146 sperm. These results are now presented in figure 7 panel A (new). The specificity of the signal obtained with the anti-HA-AF488-C Ab on mouse spermatozoa was evaluated by performing a statistical study of the density of dots in the principal piece of the flagellum from HA-CCDC146 and WT sperm. These results are now presented in figure 7 panel B (new). This study was carried out by analyzing 58 WT spermatozoa and 65 CCDC146 spermatozoa coming from 3 WT and 3 KI males. We found a highly significant difference, with a p-value <0.0001, showing that the signal obtained on spermatozoa expressing the tagged protein is highly specific. We have added a paragraph in the MM section to describe the process of image analysis. We finally present new images obtained by ExM showing no staining in the midpiece (figure 7C new). Altogether, these results demonstrate unequivocally the presence of the protein in the flagellum. These experiments are now described lines 271-279

Second, we provide new images of the signal obtained with secondary Abs only that shows more clearly that the secondary Ab gave a non-specific staining (Figure 10-Figure supplement 1B). This point is discussed lines 503-511

> 1. *Figure 9 A - Please relate the white line to Fig. 9B label in X-axis. The information from Fig 9A+D and 9E+F are redundant. The main text nor the figure legends indicate why these specific two sperm were chosen for quantification and demonstrating the outcomes. One of them could be moved to supplementary information or removed, or the two could be combined.*

As suggested by the reviewer, we have combined the two sperm to demonstrate that CCDC146 staining is mostly located on microtubule doublets. Moreover, the figure was recomposed to make it clearer.

> *Minor comments:*
>
> *All of the supplementary figures are referred to as Supp Fig X in the text, however, they are actually titled Appendix - Figure X. This needs to be consistent.*

The figures are now referred as figure supplement x in both text and figures

> *Line 125 - edit spacing.*

We think this issue (long internet link) will be curated later and more efficiently by the journal, during the step of formatting necessary for publication.

> *144 - With which to study □ with which we studied?*

We made the change as suggested.

> *151 - Supp Fig 1 - the text says that the gene is highly transcribed in human and mouse testes, but the information in the figure states that the level in mouse tissues is "medium"*

We have corrected this mistake in the text; See line 176

> *165 - The two mutations are most likely deleterious. Please specifically mention what analyses done to predict the deleterious nature to support these claims.*

Both variants, c.1084C>T and c.2112del, are extremely rare in the general population with a reported allele frequency of 6.5x10-5 and 6.5x10-06 respectively in gnomAD v3. Moreover, these variants are annotated with a high impact on the protein structure (MoBiDiC prioritization algorithm (MPA) score = 10, DOI: 10.1016/j.jmoldx.2018.03.009) and predicted to induce each a premature termination codon, p.(Arg362Ter) and p.(Arg704SerfsTer7) respectively, leading to the production of a truncated protein. This information is now given line 164-169

> *196-200/Figure 4 - As serum starved cells/basal body (B) are not mentioned in the main text, as is, Fig 4A would be sufficient/is relevant to the text. Please make the text reflect the contents of the whole figure, or re/move to supplement.*

We agree with the reviewer that the full description of the figure should be in the text. We added two sentences to describe figure 4B see lines 217-218.

> *224 - spermatozoa (plural) fits better here, not spermatozoon*

OK changed accordingly

> *236 - According to the figure legend, 6B is only showing data from the epididymal sperm, not postnatal time points; should be referencing 6C. Alignment of Marker label*

As indicated above, the figure has been completely remade, including new results. See new figure 6. Figure 6C was suppressed. The corresponding text was changed accordingly see lines 249-266

> *255-256 - Referenced figure 7B3, however, 7B3 only shows tubulin staining, so no CCDC146 can be observed. Did authors mean to reference fig 7B as a whole?*

Sorry for this mistake. We agree and the text is now figure 8B6 (figure 7 and 8 were switched)

> *305 - "of tubules" - I presume it is meant to be microtubules?*

Yes; The text was changed as suggested

> *317-321 - a diagram of HTCA would be useful here*

We have added a reference where HTCA diagram is available see line 363. Moreover, a TEM view of HTCA is presented figure 12A

> *322/Fig 11A - an arrow denoting the damage might be useful, as A1 and A3 look similar. The size of the marker bar is missing. Please update the information on figure legend.*

Concerning, the comparison between A1 and A3, the take home message is that there is a great variability in the morphological damages. This point is now underlined in the corresponding text. We updated the size of the marker bar as suggested (200 nm). See line 365-367

> *323 - Please mark where capitulum is in the figure*

Capitulum was changed for nucleus

> *Since Fig 11B2 is not referenced in the main text, it does not seem to add anything to the data, and could be removed/moved to supplement.*

We added a sentence to describe figure 11B2 line 370

> *342-343 - manchette in step I is not seen clearly - the figure needs to be annotated better. However, DPY19L2 is absent in step I in the KO, but the main text does not reflect that - why is that?*

We do not understand the remark of the reviewer “manchette in step I is not seen clearly”. The figure shows clearly the manchette (red signal) in both WT and KO (Figure 13 D1/D2).

For steps 13-15 WT spermatids, the size of the manchette decreases and become undetectable. In KO spermatids, the shrinkage of the manchette is hampered and in contrast continue to expand (Figure 13D2). We also provide a new Figure 13-figure supplement 1 for other illustrations of very long manchettes and a statistical analysis. In the meantime, the acrosome is strongly remodeled, as shown in figure 16-new, with detached acrosome (panel H). This morphological defect may induce a loss of the DPY19L2 staining (Figure 13 D2 stage I-III). This explanation is now inserted in the text line 396399

> *Figure 15B and 15C only show KO, corresponding images from the WT should be present for comparison.*

WT images are now provided in Figure 1-figure supplement 1 new

> *Figure 12 - Figure 12 - JM?.*

JM was removed. It does not mean anything

> *Figure 12C and Supplementary Fig 10 - structures need to be labelled, as it is unclear what is where*

Done

> *338 - text mentions step III, but only sperm from step VII are shown in Figure 13*

As suggested by reviewer 3, we changed stage by step. The text was modified to take into account this remark see lines 388-396

> *360 - This is likely supposed to say Supp Figure 11E-G, not 13??*

Yes, it is a mistake. Corrected

> *388 Typo "in a in a".*

Yes, it is a mistake. Corrected

> *820 - Fig 3 legend - in KO spermatid nuclei were elongated - could this be labelled by arrows? I am not convinced this phenotype is that different from the WT.*

In fact, the nuclei of elongating KO spermatids are elongated and also very thin, a shape not observed in the WT; We have added arrow heads and modified the text to indicate this point line 200.

> *836 - Figure 5 legend says that in yellow is centrin, but that is not true for 5A, where the figure shows labelling for y-tubulin (presumably, according to the figure itself).*

We have modified the text of the legend to take into account the remark

> *837- 5A supposedly corresponds to synchronized HEK293T cells, but the reasoning behind using synchronized cells is not mentioned at all in the main text; furthermore, how this synchronization is achieved is not explained in materials and methods (serum starvation? Thymidine block?).*

Yes, figure 5A was obtained with synchronized cells. We have added one paragraph in the MM section. For cell synchronization experiments, cells underwent S-phase blockade with thymidine (5 mM, SigmaAldrich) for 17 h followed by incubation in a control culture medium for 5 h, then a second blockade at the G2-M transition with nocodazole (200 nM, Sigma-

Aldrich) for 12 h. Cells were then fixed with cold methanol at different times for IF labelling. See line 224 for changes made in the result section and lines 700-704 for changes made in the MM section.

> *845- figure legend says that the RT-PCR was done on CCDC146-HA tagged mice, but the main text does not reflect that.*

We made changes and the description of the KI is now presented before (line 240) the RT-PCR experiment (line 257).

> *949 - it is likely supposed to say A2, not B1 (B1 does not exist in Fig 15)*

Yes, it is a mistake. Corrected

> *971 - Appendix Fig 3 legend - I believe that the description for B and C are swapped.*

Yes, it is a mistake. Corrected

> *Furthermore, some questions to address in A would be: Which cross sections were from which animal/points? How many per animal? Were they always in the same location?*

Yes, we have a protocol for arranging and orienting all testes in the same way during the paraffin embedding phase. The cross-sections are therefore not taken at random, and we can compare sections from the same part of the testis. The number of animals was already indicated in the figure legend (see line 1128)

> ***Reviewer #3 (Recommendations For The Authors):***
>
> 1. *There are a number of grammatical and orthographical errors in the text. Careful proofreading should be performed.*

We have sent the manuscript to a professional proofreader

> 1. *The author should also check for redundancies between the introduction and the discussion.*

The discussion has modified to take into account reviewers' remarks. Nevertheless, we did our best to avoid redundancies between introduction and discussion.

> 1. *Can the authors provide a rationale why they have chosen to tag their gene with an HA tag for localisation? One would rather think of fluorescent proteins or a Halo tag.*

Because the functional domains of the protein are unknown, adding a fluorescent protein of 24 KDa may interfere with both the localization and the function of CCDC146. For this reason, we choose a small tag of only 1.1 KDa, to limit as such as possible the risk of interfering with the structure of the protein. This rational is now indicated in the manuscript lines 251-254. It is worth to note, that the tagged-strain shows no sperm defect, demonstrating that the HA-tag does not interfere with CCDC146 function.

> 1. *In the abstract, line 53, "provide evidence" is not the right term for something that is just suggestive. The term "suggests" would be more appropriate.*

The text was modified to take into account this remark

> 1. *Line 74: "genetic deficiency" sounds strange here, do the authors mean simply "mutation"?*

Infertility may be due to several genetic deficiency such as chromosomal defects (XXY (Klinefelter syndrome)), microdeletion of the Y chromosome or mutations in a single gene. Therefore, mutation is too restrictive. Nevertheless, we modified the sentence which is now “…or a genetic disorder including chromosomal or single gene deficiencies”

> 1. *Lines 163-164: the authors describe the mutations (premature stop mutations) and say that they could either lead to complete absence of the gene product, or the expression of a truncated protein. Did they test this, for example, with some immuno blot analyses?*

As stated above, unfortunately, we were unable to verify the presence of RNA-decay in these patients for lack of biological material.

> 1. *Line 184 and Fig 2E: the sperm head morphologies should be quantitatively assessed.*

We provide now a full statistical analysis of the observed defects: see new panel in Figure 2 F

> 1. *Fig 3: The annotation should be more precise - KO certainly means CDCC146-KO. The colours of the IH panels is different, which attracts attention but is clearly a colour-adjustment artefact. Colours should be adjusted for the panels to look comparable. It would be also helpful to add arrowheads into the figure to point at the phenotypes that are highlighted in the text.*

We have added Ccdc146 KO in all figures. We have added arrow heads to point out the spermatids showing a thin and elongated nucleus. Concerning adjustment of colors, we attempted to make images of panel B comparable. See new figure 3.

> 1. *Fig 6A: the authors use RT PCR to determine expression dynamics of their gene of interested, and use actin (apparently) as control. However, actin and CDCC146 expression levels follow the same trend. How is the interpreted?*

The reviewer did not understand the figure. The orange bars do not correspond to actin expression and the grey bars to Ccdc146 expression but both bars represent the mRNA expression levels of Ccdc146 relative to Actb (orange) and Hprt (grey) expression in CCDC146-HA mouse pups’ testes. We tested two housekeeping genes as reference to be sure that our results were not distorted by an unstable expression of a housekeeping gene. We did not see significant difference between both house keeping genes. Actin was not used.

> 1. *In line 235, the authors suggest posttranslational modifications of their protein as potential cause for a slightly different migration in SDS PAGE as predicted from the theoretical molecular weight. This is not necessarily the case, some proteins do migrate just differently as predicted.*

We have changed the text accordingly and now provide alternative explanation for the slightly different migration. See lines 258-259

> 1. *The annotation of Fig 6 panels is problematic. First, why do the authors write "Laemmli" as description of the gel? It would be more helpful to write what is loaded on the gel, such as "sperm". Second, in panels B and C it would be helpful to add the antibodies used. It is not clear why there is a signal in the WT lane of panel B, but not in the HA lane (supposing an anti-HA antibody is used: why has WT a specific HA band?). In panel C, it is not clear why the blot that has so beautifully shown a single band in panel B suddenly gives such a bad labelling. Can the authors explain this? Also, they cut off the blot, likely because to too much background, but this is bad practice as full blots should be shown. In the current state, the panel C does not allow any clear conclusion. To make it conclusive, it must be repeated.*

Several mistakes were present in this figure. This figure was recomposed. The WB on testicular extract was suppressed and we now present a new WB allowing to compare the presence of CCDC146 in the flagella and head fractions from WT and HA-CCDC146 sperm. Using an anti-HA Ab, we demonstrate that in epididymal sperm the protein is localized in the flagella only. See new figure 6. The corresponding text was changed accordingly.

> 1. *The authors have raised an HA-knockin mouse for CDCC146, which they explained by the unavailability of specific antibodies. However, in Fig 7, they use a CDCC146 antibody. Can they clarify?*

The commercial Ab work for HUMAN CCDC146 but not for MOUSE CCDC146. We have added few words to make the situation clearer, we have added the following information “the commercial Ab works for human CCDC146 only”. See line 240

> 1. *In Fig 7A (line 258), the authors hypothesise that they stain mitochondria - why not test this directly by co-staining with mitochondria markers?*

We chose another solution to resolve this question:

To avoid the issue of the non-specificity of secondary antibodies, we performed a new set of IF experiments using an HA Tag Alexa Fluor® 488-conjugated Antibody (anti-HA-AF488-C Ab) on WT and HA-CCDC146 sperm. These results are now presented in figure 7 panel A (new). The specificity of the signal obtained with the anti-HA-AF488-C Ab on mouse spermatozoa was evaluated by performing a statistical study of the density of dots in the principal piece of the flagellum from HA-CCDC146 and WT sperm. These results are now presented in figure 7 panel B (new). This study was carried out by analyzing 58 WT spermatozoa and 65 CCDC146 spermatozoa coming from 3 WT and 3 KI males. We found a highly significant difference, with a p-value <0.0001, showing that the signal obtained on spermatozoa expressing the tagged protein is highly specific. We have added a paragraph in the MM section to describe the process of image analysis. We finally present new images obtained by ExM showing no staining in the midpiece (figure 7C new). Altogether, these results demonstrate unequivocally the presence of the protein in the whole flagellum.

> 1. *It seems that in both, Fig 7 and 8, the authors use expansion microscopy to localise CDCC146 in sperm tails. However, the staining differs substantially between the two figures. How is this explained?*

In figure 8 we used the commercial Ab in human sperm, whereas in figure 7 we used the anti-HA Abs in mouse sperm. Because the antibodies do not target the same part of the CCDC146 protein (the tag is placed at the N-terminus of the protein, and the HPA020082 Ab targets the last 130 amino acids of the Cter), their accessibility to the antigenic site could be different. However, it is important to note that both antibodies target the flagellum. This explanation is now inserted see lines 304-312

> 1. *Fig 8D and line 274: the authors do a fractionation, but only show the flagella fraction. Why?*
>
> *Showing all fractions of their experiment would have underpinned the specific enrichment of CDCC146 in the flagella fraction, which is what they aim to show. Actually, given the absence of control proteins, the fact that the band in the flagellar fraction appears to be weaker than in total sperm, one could even conclude that there is more CDCC146 in another (not analysed) fraction of this experiment. Thus, the experiment as it stands is incomplete and does not, as the authors claim, confirm the flagellar localisation of the protein.*

We agree with the reviewer's remark. We provide now new results showing both flagella and nuclei fractions in new figure 6A. This experiment is presented lines 253-256

> 1. *Line 283, Fig 9D,F: The description of the microtubules in this experiment is not easy to understand. Do the authors mean to say that the labelling shows that the protein is associated with doublet microtubules, but not with the two central microtubules? They should try to find a clearer way to explain their result.*

As suggested by reviewer 2, we have changed the figure to make it clearer. The text was changed accordingly. See new figure 9 and new corresponding legend lines 1006.

> 1. *Fig 9G - how often could the authors observe this? Why is the axoneme frayed? Does this happen randomly, or did the authors apply a specific treatment?*

Yes, it happens randomly during the fixation process.

> 1. *Line 300 and Fig 10A - the authors talk about the 90-kDa band, but do say anything about what they think this band is representing.*

We have now added the following sentence lines 340-342: "This band may correspond to proteolytic fragment of CCDC146, the solubilization of microtubules by sarkosyl may have made CCDC146 more accessible to endogenous proteases."

> 1. *Fig 11A, lines 321-322: the authors write that the connecting piece is severely damaged. This is not obvious for somebody who does not work in sperm. Perhaps the authors could add some arrow heads to point out the defects, and briefly describe them in the text.*

We realized from your remark that our message was not clear. In fact, there is a great variability in the morphological damages of the HTCA. For instance, the HTCA of Ccdc146 KO sperm presented in figure 10A2 is quite normal, whereas that in figure 10A4 is completely distorted. This point is now underlined in the corresponding text. See lines 367-369

We also added the size of the marker bar (200 nm), which were missing in the figure's legend.

> 1. *Line 323: it will be important to name which tubulin antibody has been used to identify centrioles, as they are heavily posttranslationally modified.*

The different types of anti-tubulin Abs are described in the corresponding figure's legend

> 1. *Fig 11B - phenotypes must be quantified to make these observations meaningful.*

We agree that a quantification would improve the message. However, testicular sperm are obtained by enzymatic separation of spermatogenic cells and the number of testicular sperm are very low. Moreover, not all sperm are stained. Taking these two points into account, it seems to us that quantification could be difficult to analyze. For this reason, the quantification was not done; however, it is important to note that these defects were not observed in WT sperm, demonstrating that these defects are cased by the lack of CCDC146. We have added a sentence to underline this point; See lines 374-375

> 1. *Line 329: Figure 12AB - is this a typo - should it read Figure 12B?*

We have split the panel A in A1 and A2 and changed the text accordingly. See line 378

> 1. *Why are there not wildtype controls in Fig 12B, C?*

We provide now as Figure 12-figure supplement 1, a control image for fig 12B. For figure 12C, the emergence of the flagellum from the distal centriole in WT is already shown in Fig 12A1

> 1. *Fig 13: the authors write that the manchette is "clearly longer and wider than in WT cells" (lines 342-343). How can they claim this without quantitative data?*

We now provide a statistical analysis of the length of the manchette. See figure 13-figure supplement 1A. We also provide a new a new image illustrating the length of the manchette in Ccdc146 KO spermatids; See Figure 13-figure supplement 1B.